aselsan COMMUNICATION & INFORMATION TECHNOLOGIES

STC-8255

Digital Tachograph



Preparation Manual

DTVU-PRE

(76BE-8255-0001)

Revision A

October 2021

aselsan COMMUNICATION & INFORMATION TECHNOLOGIES

Content

This book contains the usage information of the below equipment.

APN	:	5820-8255-1000S
Description	:	ASELSAN 8255 DIGITAL TACHOGRAPH DEVICE

About Manual

Manual APN	:	76BE-8255-0001
Manual Description	:	8255 Digital Tachograph Preparation Manual
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Declaration of Conformity

This declaration, officially register the confirmation of the product named as

Definition	:	Aselsan Digital Tachograph Vehicle Unit
Model	:	STC-8255
TOE Version	:	v1.0
TOE SW Version	:	v1.0

According to the Council Regulation (EEC) No. 3821/85 (Annex-1B) on recording equipment in road transport (EU) dated 20 December 1985 and its latest amendments.

Product is evaluated according to the following European Union regulations:

1. Regulation (EC) No. 2135/98
2. Regulation (EC) No. 1360/2002
3. Regulation (EC) No. 561/2006
4. Regulation (EC) No. 1266/2009
5. AETR Agreement.

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WARNINGS AND REMARKS

Warnings and Remarks

In this section the below listed warnings in which the described rules required to be followed during maintenance and repair process of the device are given.

Definition	Document APN	Page Code	Revision
Personal Security Warnings	7610-0000-0001	503	A
Device Security Warnings	7610-0000-0002	503	A
Maintenance Warnings	7610-0000-0003	503	A
Mounting Warnings	7610-0000-0004	503	A

Personal Security Warnings

Procedures to Be Applied on the Personnel Injured By Electrical Shock

- Do not try to pull out the wounded person.
- If possible, cut electric current via circuit breaker or fuse.
- If it is not possible to cut electric current, then pull or lift the wounded person by using a dry stick, rope or any other nonconductive material.
- Call for help as soon as possible.
- After cutting the electricity contact of the wounded person, lay him on a flat surface and do artificial respiration.
- Do not let the person drink anything until he recovers.

Electrical shock

1. Protect yourself with dry insulation material, dry leather, deadwood, dry rubber, etc.
2. Turn off the circuit breaker or pull the patient from conducting line.
3. Do not touch the patient unless electrical circuit is cut.
4. Remove foreign materials like chewing gum from the patient's mouth.
5. Do artificial respiration immediately.
6. Call the doctor.
7. Keep him warm with blanket, etc.

Device Safety Warnings



WHEN THE DEVICE IS OPENED OR UNITS WITHIN THE DEVICE ARE CHANGED BY UNAUTHORIZED PEOPLE, WARRANTY BECOMES INVALID.



USE THE DEVICE ALWAYS WITH THE ACCESSORIES AND PERIPHERALS APPROVED BY ASELSAN. OTHERWISE, RESULTING DEFECTS AND MALFUNCTIONS WILL BE CONSIDERED AS USER'S FAULT.



USE OF THE EQUIVALENT UNITS/PARTS/MATERIALS NOT MANUFACTURED BY ASELSAN MAY CAUSE FOLLOWINGS:

- DETERIORATION OF TECHNICAL SPECIFICATIONS OF THE DEVICE DEFINED WITH THE DESIGN AND ITS OPERATIONAL PERFORMANCE UNDER DIFFERENT AMBIENT CONDITIONS
- INJURY OF THE USER
- DAMAGE TO THE DEVICE AND THE PLATFORM ON WHICH IT'S USED

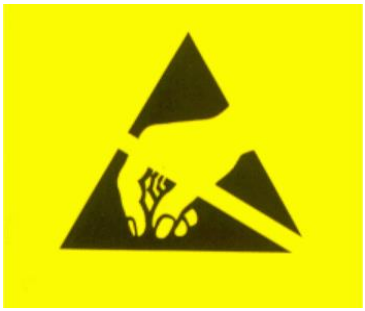


BEFORE USING THE DEVICE, PLEASE BE SURE TO READ THE MANUALS AND BOOKS RELATED TO THE DEVICE.

Maintenance Warnings

- Use your device according to the methods written in operational manual.
- Be careful not to damage the device while carrying, using and storing it.
- Keep your device away from dust, humidity, dense sunlight, excessive heat sources and liquids
- Protect your device against impact, falling and shocks.
- During extreme colds, note that the connectors and cables may become harden and more fragile.
- Every part on your device or every unit and accessory completing it is definitely required. Do not lose any of them.
- Use a soft and damp cloth to clean your device. Do not use detergent and chemical solvents.

Assembly Warning



Electrostatic Discharge (ESD)

- TO FEEL–3500 Volts or more
- TO HEAR–4500 Volts or more
- TO SEE SPARKING -5000 Volts or more is needed.

Devices contain parts that are ESD sensitive. For this reason, environment mentioned below must be provided during assembly and disassembly procedures.

Many electronic components may be damaged by 'Electrostatic Discharge'.

Voltages lower than those that you can feel, hear or see may damage components partially or completely.

Table on which assembly, disassembly and repair procedures will be performed must be covered with an ESD-protective material and grounded.

Personnel who will perform assembly and disassembly procedures must wear a smock that does not generate electricity and always use a grounding bracelet.

Grounding bracelet connects you directly to the earth and avoids static charge generation on you.

Grounding bracelet must be put on always before commencing the work. Be sure that it has connected appropriately and placed securely at your wrist. Removing your bracelet must be the last thing you do before leaving the work area.

Take the cards and electronic parts out of their protective packages only within statically protected areas.

Cards and parts are safe when they are in statically protected bags. Bags with their mouths opened do not protect the cards. This care must be provided on all cards, even on defective cards to be repaired later. By this way, intact components on defective cards would have been protected. If parts returned for fault analysis are unprotected, then they may show ESD damage signs.

Keep the non-conducting materials away from statically protected areas. Electronic materials may fail due to the static charges generated by some insulators. Thus, all materials made of plastic, nylon, foam and cellophane must not be placed within statically protected areas.

Also, your dress, tie and neckwear may cause static charge generation.

Safety

Before activating, installing and calibrating the STC-8255 digital tachograph, please read this preparation (workshop) manual.



General Precautions

ALWAYS use only ASELSAN authorized accessories. (Download devices, calibration tools, cables and etc.).

Other unauthorized accessories have the risk of fire hazard, explosion, personal injury or damage to the STC-8255 digital tachograph.



Risk of Accident!

For traffic safety reasons, please do not attempt to insert or withdraw the workshop card(s) while the vehicle is moving!



Changes or modifications to the STC-8255 digital tachograph may void its compliance with government laws/rules and make it illegal to use.

Avoid using the STC-8255 digital tachograph at temperatures below -25°C or above 80°C.

Avoid storing the STC-8255 digital tachograph at temperatures below -40°C or above 85°C



If the STC-8255 digital tachograph is stored more than 10 years on the shelf without installing to the vehicle, the battery in the tachograph is discharged and the Real Time Clock (RTC) data will be erased because of it. In this case, please re-charge the internal battery as mentioned in this manual (section 9.8) and set the RTC again.



Please read this preparation instructions carefully before installing the STC-8255 digital tachograph into a vehicle.

Legal Requirements

The use of digital tachographs is administrated by the latest consolidated valid version of the EC regulation (EEC) No. 3821/85 Annex-IB together with EC regulations (EC) no. 561/2006. These regulations assign a number of responsibilities to the workshop personnel.

It is illegal to misrepresent, deactivate or delete recordings of the digital tachographs and the tachograph cards. Also, it is forbidden to falsify printouts. Modifications of the digital tachograph or its electronic signals or data links, especially if made with the intention of cheating, is forbidden and shall be punishable according to the legal stipulations.

Workshop Responsibilities

Workshops are responsible for installation, activation, calibration, periodic inspection, maintenance, and repair and decommissioning of the digital tachograph systems according to the digital tachograph regulation (EEC No. 3821/85).

The workshops have to carry out inspection when the following conditions occur.

- After every installation.
- After any repairs carried out on the tachograph or the recording equipment system.
- After any change in the characteristic coefficient (w factor) or vehicle constant (k factor) for the vehicle.
- After any change in the effective tyre circumferences of the drive wheels resulting from changing the tyre size.
- If the Vehicle Registration Number (VRN) has changed.
- If the tachograph's UTC time differs from the current UTC time by more than 20 minutes.

Workshop personnel has to affix an installation plaque after each installation or after the calibration parameters changed during inspection. The installation plaque must have a seal or it is affixed in such a way that removing destroys details of the plaque.

1 INTRODUCTION

1.1 General

The ASELSAN STC-8255 digital tachograph (hereinafter abbreviated as STC-8255) is a secure recording device which records the driving times, work periods, availability periods and rest periods of the driver and co-driver. Additionally, STC-8255 automatically records the speed and distance covered by the vehicle. The securely recorded vehicle and driver related data by the digital tachograph serves as a proof for the controlling bodies (e.g. police). The digital tachograph STC-8255 complies with the technical specifications as specified in the Commission regulation (EC) No. 1360/2002 of 13 July 2002 and Commission regulation (EC) No. 1266/2009 of 16 December 2009, the 7th and 10th adaptations of Council regulation (EEC) No. 3821/85 to the technical progress of Recording Equipment in Road Transport respectively.

This preparation manual is intended to serve as a guidance document especially for workshop personnel who is responsible for installation, calibration, maintenance, and periodic inspection and decommissioning. This manual describes the detailed usage of the digital tachograph STC-8255 as specified in the regulation. The primary concern of this document is Calibration Mode of operation. Read the corresponding sections carefully and familiarize yourself with the individual functions and menu settings, installation, inspection, maintenance and software update operations of the STC-8255 digital tachograph. However, reading the user manual (knowledge of operational mode) is necessary in case of driving a tachograph installed vehicle for calibration or testing purposes.

This workshop manual also provides guidance to the personnel of vehicle manufacturers who install the STC-8255 digital tachograph into the vehicle during production and perform some basic configurations.

1.2 Notational Conventions

The following notational conventions are used in this document.



Warning Message
A warning message indicates potential risk of injury or accident



Attention!
The text nearby this symbol contains critical information to prevent damage to the device, to avoid loss of data or to obey legal obligations.



The hint sign will give you some guidance, information or clues which is beneficial for you if it is taken into account.

Symbols

* The asterisk mark indicates a special feature or option.
➔ The reference sign indicates that the related issue will be explained detailed in the given pages.

Steps

1. The numbered steps describe actions for the given step by step instructions.
2. Subsequent actions are numbered.

1.3 Tachograph System Overview

The tachograph system components consist of the followings as shown in the Figure 1: Tachograph System .

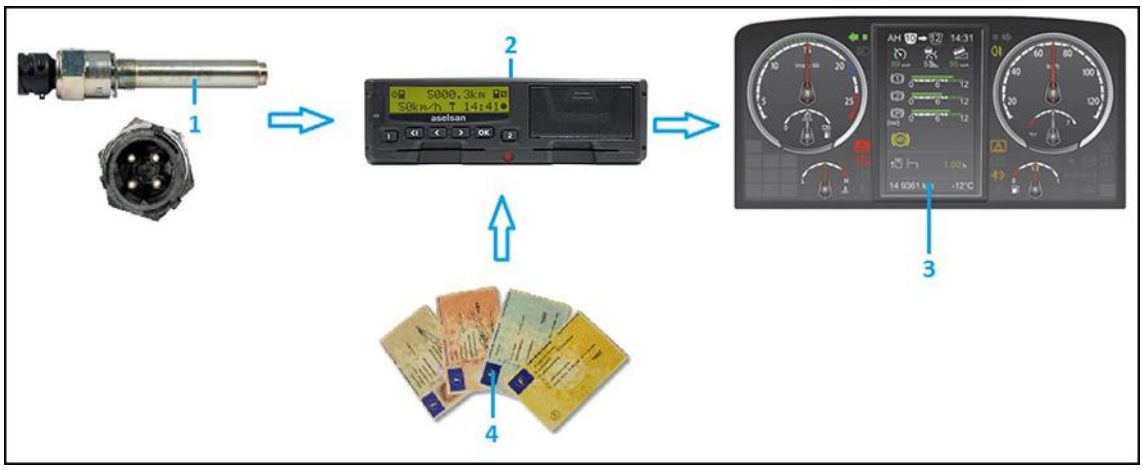


Figure 1: Tachograph System Components

1. Motion sensor,
2. Digital tachograph (STC-8255),
3. Display in instrument cluster,
4. Various type of tachograph cards.

1.3.1 Motion Sensor

Secure motion sensor is used to provide the digital tachograph (STC-8255) with speed signal pulses from the vehicle gearbox. To ensure the validity of the motion sensor signal, the speed pulses sent to the tachograph is validated with a separate encrypted signal. Invalidated speed signal information which is obtained from authentication data leaves evidence in case of tampering which will be recorded by the tachograph.

1.3.2 Digital Tachograph

The tachograph records and stores several information related to the vehicle and the drivers using it. These data are:

- Driver and workshop card activities,
- Warning (events) and malfunctions (faults) related to the tachograph and cards,
- Vehicle information, detailed speed, odometer values,
- Calibration, time adjustments and control data,
- Security breach attempts, etc.

1.3.3 Display in Instrument Cluster

The display in the instrument cluster can be used to display information arising from the digital tachograph such as odometer, trip distance, speed or other warning messages via the CAN-Bus of the vehicle.

1.3.4 Tachograph Cards

There are four different types of tachograph cards which are marked with four different colors.

Driver card (white) – operational mode
Workshop card (red) – calibration mode
Company card (yellow) – company mode
Control card (blue) – control mode

The mode of operation of the digital tachograph is set subject to which type of card is inserted and is specified by a symbol on the display.

1.3.4.1 Driver Cards

Driver's card is used to identify the driver to the digital tachograph. Driver cards are personal and can be used for driving purposes. Driver cards store driver related data such as activities, identification of the driver etc. Since there is a limit of data storage of the driver cards, the data in the driver cards should be backed up and stored at least once in 28 days to a secured media. The validity of the driver cards is 5 years from issue date.

1.3.4.2 Company Cards

Company cards identify the respective commercial-vehicle company. A company card is used to display, print and download the data of the digital tachograph. By using company card, the vehicle or the company owner can lock the data which is company specific and cannot be accessed from other companies (cards). The company is also responsible for storing the downloaded data from the tachograph regularly and has to make these data available to the control authorities if necessary. The validity of the company cards is 5 years from issue date.

1.3.4.3 Control Cards

Control card is used to identify the control bodies such as police. By using control card, all of the data recorded in the digital tachograph can be displayed, printed out or downloaded. The validity of the control cards is 2 years from issue date.

1.3.4.4 Workshop Cards

Workshop card identifies the appropriate expert of an approved or certified workshop. Workshop cards are the essential part of the workshops and most of the functions (activation, calibration) carried out by the workshops require the use of workshop cards. Cards are obtained by a workshop by applying to the relevant authorities. In contrast to company and control cards, the workshop cards can be used for test drive purposes. The validity of the workshop cards is one year from issue date.



If the workshop card is lost or damaged or if the PIN is forgotten, please inform the relevant authorities immediately and apply for the new one.

2 WORKSHOP INTERACTION

2.1 Front Panel

The STC-8255 digital tachograph has the following display and control elements.



Figure 2: STC-8255 Front View

1. Alphanumeric display
2. Service (Calibration and Data Download) interface (6-Pin)
3. Keypad **#1** for the driver (driver#1)
   Short push for adjusting the driver activity,
   Long push for driver card withdrawal.
4. Backspace key ⌫ (backspace & clear or cancel)
5. Card slot **#1** for the driver's tachograph card
6. Control key ‹ (backward / previous entry)
7. Control key › (forward / next entry)
8. OK Key (call menu, confirm)
9. Sealing plug
10. Keypad **#2** for the co-driver (driver#2)
    Short push for adjusting the co-driver activity,
    Long push for co-driver card withdrawal.
11. Card slot **#2** for the co-driver's tachograph card
12. Printer drawer (for inserting a new roll of paper see "section 0

14. Paper Roll")

2.2 Back Panel

At the back side of the STC-8255, there is an ISO-16844 rear connector, a rubber alignment and a GNSS fakra connector (depending on the configuration). ISO-16844 connector is used for connecting the power and the peripherals to the STC-8255. The pin assignment diagram is given in Table 2,

Table 3,

Table 4 and Table 5. Rubber alignment is used to fix the STC-8255 in the installation frame in order to reduce the vibrations. Lastly, the GNSS fakra connector is used for external GNSS antenna connection if the internal GNSS antenna performance is poor. GNSS is optional and depends on the STC-8255 configuration.

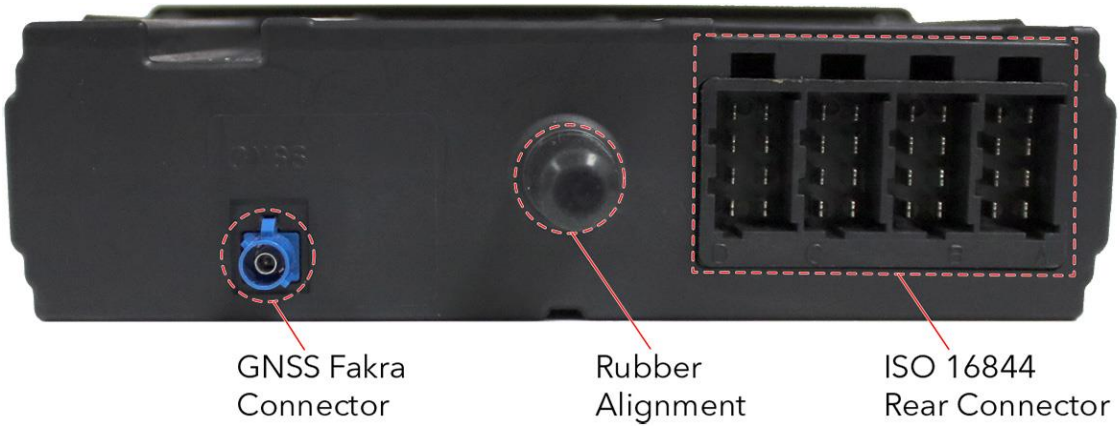


Figure 3: STC-8255 Rear View

2.3 Display

Display used in the STC-8255 is a Liquid Crystal Display (LCD) with backlight. Display has 2 lines with 16 characters shown at each line. Backlight color choice is optional and can be selected prior to ordering and only be assembled at the factory during production.

The display may be used in different ways:

Basically, the display is used to show information to the user. The display is used to navigate in a menu system together with the four buttons (not the numbered ones). The display is also used as a feedback when entering information into the tachograph or changing settings.

2.3.1 Display Modes

If the tachograph is not in sleep mode and no need to display warnings or no other menu is selected, pressing the Control keys ‹ or › changes the default display into following screens.

1. Default display (Speed and odometer),
2. Driver 1 and driver 2 DAGS times at driving state,
3. Driver 1 DAGS times,
4. Driver 2 DAGS times,
5. Driver 1 weekly DAGS times,
6. Driver 1 daily DAGS times,
7. DAGS extended information,
8. Driver and co-driver activities and local time,
9. UTC time.

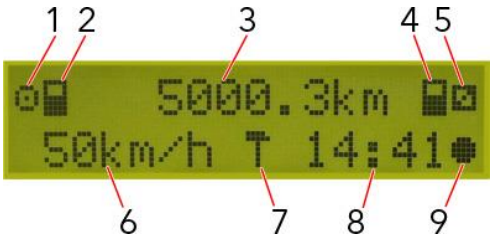
While the vehicle is moving, only the displays #1, 2, 8 and 9 are available for selection.

When the vehicle is stationary, only the displays #1, 3, 4, 8 and 9 are available for selection.

When the vehicle is stationary and the valid card is inserted into the slot-1, all displays #1, 2, 3 4, 5, 6, 7, 8 and 9, are available for selection. If reading process of the driver card is not completed, on the display #3, 5, 6 and 7, "-" symbol is shown instead of durations.

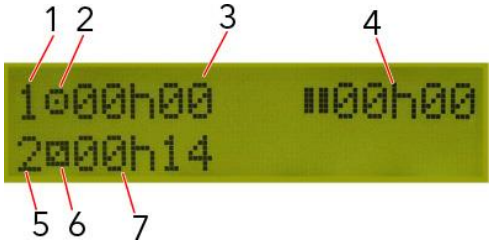
2.3.1.1 Default Display (Speed & Odometer) (#1)

Normally, (when the ignition is OFF) the tachograph is in sleep (power safe) mode and the display does not show any information in the display window, but with the touch of any button or turning ignition ON will wake it up (both illumination and text).



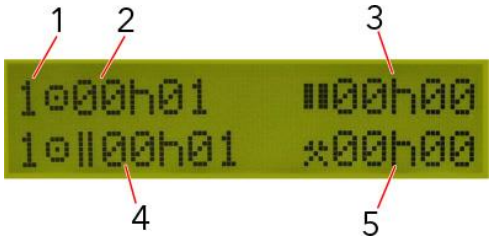
1 Driver 1 activity
2 Driver 1 card symbol
3 Total odometer value
4 Driver 2 card symbol
5 Driver 2 activity
6 Speed
7 Mode of operation
8 Time (UTC time if "●" symbol is not present)
9 Local time symbol "●".

2.3.1.2 Driver 1 and Driver 2 DAGS Times at Driving State (#2)



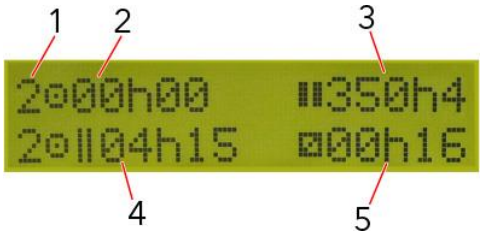
1 Indicates the driver information
2 Driver 1 activity (driving)
3 Driver 1 driving time since last break
4 Valid rest/break time in break periods of at least 15 min. followed by 30 min.
5 Indicates the driver 2 information
6 Driver 2 activity (available)
7 Driver 2 current activity time

2.3.1.3 Driver 1 DAGS Times (#3)



1 Indicates the driver information
2 Drive time since last break/rest
3 Cumulative break/rest time
4 Cumulative driving time during the current and previous week
5 Current activity time

2.3.1.4 Driver 2 DAGS Times (#4)



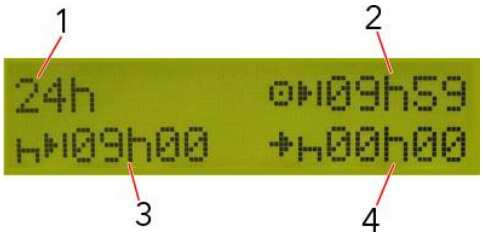
1 Indicates the driver 2 information
2 Driving time since last break/rest
3 Cumulative break/rest time
4 Cumulative driving time during the current and previous week
5 Current activity time

2.3.1.5 Driver 1 Weekly DAGS Times (#5)



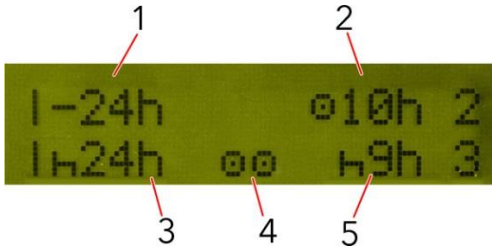
1 Weekly pictogram icon
2 Driver 1 weekly remaining driving time
3 Driver 1 weekly remaining rest time
4 Driver 1 remaining time to next weekly rest period beginning

2.3.1.6 Driver 1 Daily DAGS Times (#6)



1 Daily pictogram icon
2 Driver 1 daily remaining driving time
3 Driver 1 daily remaining rest time
4 Driver 1 remaining time to daily shift period ending

2.3.1.7 DAGS Extended Information (#7)



1 Weekly and daily pictogram icon
2 Number of remaining daily extended driving time
3 Weekly rest limit
4 Crew driving
5 Number of remaining daily reduced rest time

Note That!

DAGS informations mentioned above are in accordance with Regulation (EC) no. 561/2006. For detailed information about rules on driving times, breaks and rest periods for drivers, please refer to Regulation (EC) no. 561/2006.

2.3.1.8 Driver and Co-driver Activities and Local Time (#8)



In this display mode, following information is shown (first line) from left to right:

- Symbol for driver activity (driving is shown in this case).
- Card symbol indicates that card is inserted to the card slot #1.
- Current local time.
- Symbol for co-driver activity (available is shown in this case).

In the second line the following information is displayed:

- Current date.
- Symbol for the current mode of operation (in this case 'Operational' mode is shown).

2.3.1.9 UTC Time (#9)



In this display mode, following information is shown (first line) from left to right:

- Pictogram for UTC time.
- Current date.

In the second line the following information is displayed:

- Current UTC time.
- Time offset set for the local time.

2.3.2 Special Displays

2.3.2.1 Activation status

If the STC-8255 has not been activated then the symbol "⊟" will be shown in the display. The STC-8255 will not accept any tachograph cards except the workshop cards while this symbol "⊟" appears on the display.



In order to use the STC-8255 digital tachograph, ensure that it is installed, activated and calibrated according to the vehicle specific parameters at the certified workshops.

2.3.2.2 Out of scope

When the vehicle is driving in countries where tachograph legislation does not exist "**Out of Scope**" mode could be activated. When it is activated from the menu the display shows the **"OUT"** symbol on the screen.



➔ *Please refer to "Out of Scope Condition" on the "Operation manual".*

2.3.2.3 Ferry transfer / Train crossing



While the vehicle is located on a ferry or on a train, the Ferry transfer/Train crossing mode is selected by the user, then the display shows the symbol " ⛴ ". User action is required to set this functionality from the menu.

➔ *Please refer to "Ferry or Train Condition" on the "Operation manual".*

2.4 Keypad

OK OK key : Calls the menu, confirms a selection and acknowledges a message or warning.

<I	Backspace key	:	Moves back in a menu, clears or cancels the entry, returns to the default display when pressed repeatedly.
<	Backward key	:	Moves right in a menu to select the desired function or selection. Decreases a value. Toggles options.
>	Forward key	:	Moves left in a menu to select the desired function or selection. Increases a value. Toggles options.
1	Keypad #1	:	Short push for adjusting the driver activity, Long push for driver card withdrawal.
2	Keypad #2	:	Short push for adjusting the co-driver activity, Long push for co-driver card withdrawal.

2.5 Card Slots

There are two card slots where you insert your tachograph cards.



Figure 4: STC-8255 Card Slots

The card slot for the driver is on the left and card slot for the co-driver is on the right. Each card must be inserted all the way to the stop. The cards are locked electro-mechanically and normally they can only be removed with the help of the electromechanical release mechanism triggered by the user action (long pressing the **keypad #1** and **keypad #2**) when the vehicle is stationary.

2.6 Printer

The printer is used to provide printed record of the information stored in the tachograph.



Figure 5: STC-8255 Printer

Inside the printer drawer there is a printer mechanism that holds printer and a roll of paper. The printed paper will come out from a small opening at the bottom of the printer drawer.

2.7 Paper Roll

In order to print the stored information in the tachograph a valid type approved paper rolls (recommended by Aselsan Inc.) should be inserted into the printer drawer. To replace the new paper roll, please obey the following instructions.

1. Press on the upper part of the printer drawer lid as shown in the Figure 6.



Figure 6: STC-8255 Printer Drawer Lid (Closed)

2. The bottom side of the printer drawer will be raised as shown in the Figure 7.



Figure 7: STC-8255 Printer Drawer Lid (Opened)

3. Carefully pull out the printer drawer by grabbing from its lower part until it is entirely detached from the tachograph body as shown in the Figure 8. Check if any plastic piece exists in the drawer from the old paper roll. If so, remove this plastic piece from the paper holder.



Figure 8: STC-8255 Printer Drawer (Pulled Outside)

4. Insert new roll of paper as shown in the figures below. Keep the feeding part of the paper to be on the upper side as shown in Figure 9 and forward it to pass the lower edge of the drawer as shown in Figure 10.



Figure 9: STC-8255 Replacing Paper Roll-1



Figure 10: STC-8255 Replacing Paper Roll-2

5. Hold the drawer to be the same level with the printer slot and push it gently into the tachograph to close the drawer as shown in the Figure 11, Figure 12 and Figure 13.



Figure 11: STC-8255 Replacing Paper Roll-3



Figure 12: STC-8255 Replacing Paper Roll-4



Figure 13: STC-8255 Replacing Paper Roll-5

6. Pull the paper upwards and tear it off.

2.8 Service (Calibration and Data Download) Interface (6-Pin)

The service interface (6-Pin) which is located on the front panel of the STC-8255 digital tachograph is covered with a silicon cover.



Figure 14: STC-8255 6-Pin Interface

The port has two independent components; a calibration interface and a download interface.

The calibration interface serves the calibration, configuration and diagnostics of the STC-8255 digital tachograph.

The download interface is used to download the data stored in mass memory or on the inserted driver or co-driver smartcard.

2.9 Vehicle Dashboard Integration

Some vehicles can display some of the tachograph information on an integrated display. Here are the examples on the type of information that can be displayed:

- Speed,
- Total Odometer,
- Travelled (trip) Distance,
- Date and Time,
- Accumulated Daily Driving Time,
- Accumulated Driving Time,
- Messages, Warnings, System Events, Faults.

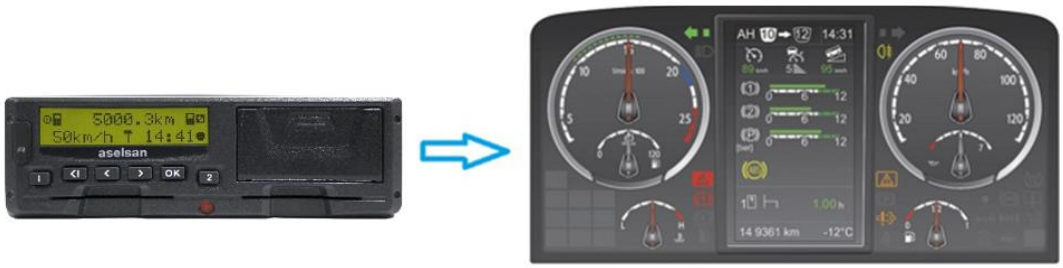


Figure 15: Vehicle Dashboard Integration

2.10 Acoustic Warnings

STC-8255 generates acoustic alerts (beep tone) to warn user about some warning situations (events) or irrelevant user entries. Acoustic alerts can be heard on these events:

1. Overspeed,
2. DAGS warning,
3. Pre-warnings,
4. Irrelevant user entries,
5. Buzzer test.

2.11 Pictograms

The following table lists the details of the pictograms in the STC-8255. For the pictogram combinations please refer to “Pictogram Combinations” on page 98.

Table 1: List of Pictograms

Symbol	Meaning / Description
	Company, company mode
	Controller/inspector, control mode
	Driver, driving, continuous driving time, operational mode
	Workshop/test station, calibration mode
	Manufacturer, tachograph not activated
	Available, current availability period
	Rest, current rest period
	Work, current work period
	Break, cumulative break time
	Unknown
	Driver card slot
	Co-driver card slot
	Smartcard
	Clock
	Display
	External storage
	Power supply
	Printer/printout
	Motion sensor
	Tyre size
	Vehicle/vehicle unit (tachograph)
OUT	Out of Scope (digital tachograph not required)
	Ferry/train crossing
	Events
	Faults
	Start of daily work period
	End of daily work period
	Location
	Manual entry of driver activities

Symbol	Meaning / Description
	Security
	Speed
	Time
Σ	Total/summary
24h	Daily
|	Weekly
||	Two weeks
+	From or to

INTENTIONALLY LEFT BLANK.

3 MODE OF OPERATIONS

3.1 Operational Mode

By inserting a valid driver card, the STC-8255 digital tachograph enters into the operational mode. It is explained in the Operational Manual in detail, therefore will not be explained again in this manual.

➔ Please refer to the "Operation Manual" for further information of operational mode.

3.2 Company Mode

By inserting a valid company card, the STC-8255 digital tachograph enters into the company mode. It is explained in the Operational Manual in detail, therefore will not be explained again in this manual.

➔ Please refer to the "Operation Manual" for further information of company mode.

3.3 Control Mode

By inserting a valid control card, the STC-8255 digital tachograph enters into the control mode. It is explained in the Operational Manual in detail, therefore will not be explained again in this manual.

➔ Please refer to the "Operation Manual" for further information of control mode.

3.4 Calibration Mode

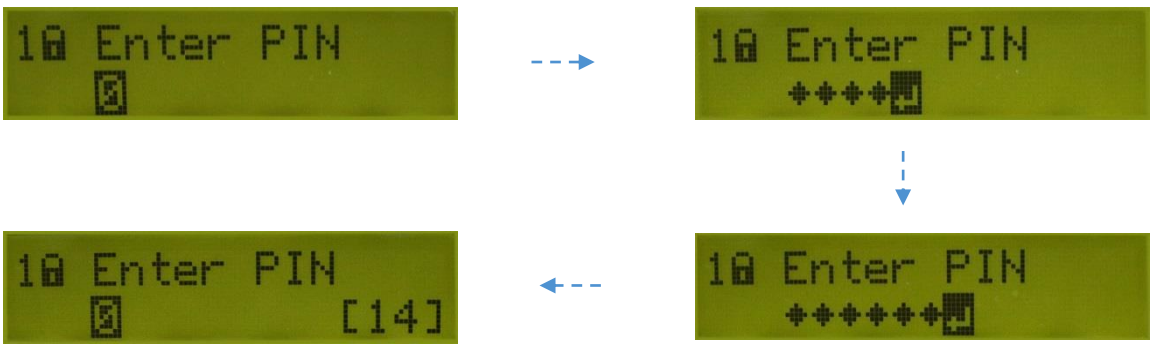
By inserting a workshop card and entering its respective PIN, the STC-8255 digital tachograph enters into the calibration mode. In this mode, workshop staff can be able to activate, calibrate, configure vehicle specific parameters and download mass storage data of the STC-8255. By entering the calibration mode, the user has gained access rights to 'Service Menu', which is located under the main menu.

3.4.1 PIN Entering

After insertion of a valid workshop card into one of the slots, the STC-8255 requests the entry of the respective PIN. The PIN may be 4 to 8 characters long and may consist of numbers (0-9), upper-case letters (A-Z), lower case letters (a-z) or special characters (@#$).

To enter the PIN:

1. Select the desired character by using control ‹ / › keys. Then press **OK** to assign the character. After assigning the cursor moves to the next character and previously assigned character will be shown as "*". Repeat this procedure until all of characters in the PIN is entered. After 4 characters are entered, the carriage return symbol appears. Press **OK** to confirm the PIN or if it contains more than 4 characters, continue on selecting the desired characters by using control ‹ / › keys.



2. To correct the entered character, use backspace ⌫ key to move back to previous cursor. The entered character becomes visible and select the desired character as described above. Press **OK** until the cursor

stops on the carriage return symbol. Pressing **OK** again on the carriage return symbol confirms the whole PIN entry. The card will be read and the PIN will be checked.

3. If PIN entry is not validated, the PIN entry procedure starts again from beginning showing the number of remaining attempts.

3.4.2 Calibration

Vehicle specific parameters will be programmed into the STC-8255 data memory in the calibration procedure. Vehicle parameters include vehicle identification (VIN, VRN and Registering Member State) and vehicle characteristics (w, k, l, tyre size, speed limiting, device setting (if applicable), current UTC time, current odometer value). The front 6-Pin interface is used for calibration purposes in the Calibration Mode.

- The calibration procedure will be explained in section 6.2.5 Calibration.

➔ Please refer to that section for more explanations.



Calibration of the STC-8255 digital tachograph requires calibration equipment (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B, Appendix-8). Aselsan recommends to use Digital Tachograph Calibration Device (STKC-8250) to calibrate the STC-8255.

4 MENU STRUCTURE

STC-8255 menu structure is shown in Figure 16 and Figure 17.

Depending on the mode of the tachograph (which is set when certain tachograph cards are inserted), STC-8255 menu items that can be accessed from the main menu are changed.

The menu items which are appended "*" symbol before the following number in the figures require some prerequisites to be accessed.

*1) Only available in the active tachograph device.

*2) Only available with insertion of driver or workshop card to the driver slot.

*3) Only available with insertion of driver or workshop card to the co-driver slot.

*4) Only available when language change by card and manual is selected from "General configuration".

*5) Only available in the active tachograph device. OUT Begin is only available when out of scope is close, OUT End is only available when out of scope is open.

*6) Only available in the active tachograph device and when out of scope option is close.

*7) Only available if backlight is selected to be arranged by menu.

*8) Only available with existence of status inputs D1-D2 as a hardware configuration.

*9) Only available with existence of GNSS module as a hardware configuration.

*10) Only available in the Company Mode.

*11) Only available (once) in the Company Mode if VRN is not entered to the tachograph.

*12) Only available in the Calibration Mode.

*13) Only available if baud rate is enabled under the CAN A or CAN C Config menu.

*14) Only available if ISO is enabled from the CAN Protocols menu.

*15) Only available if UDS is enabled from the CAN Protocols menu.

*16) Only available with the existence of Auto-R feature as a hardware configuration.

*17) Only available if UDS and RDDW. in Sleep are enabled from related menus.

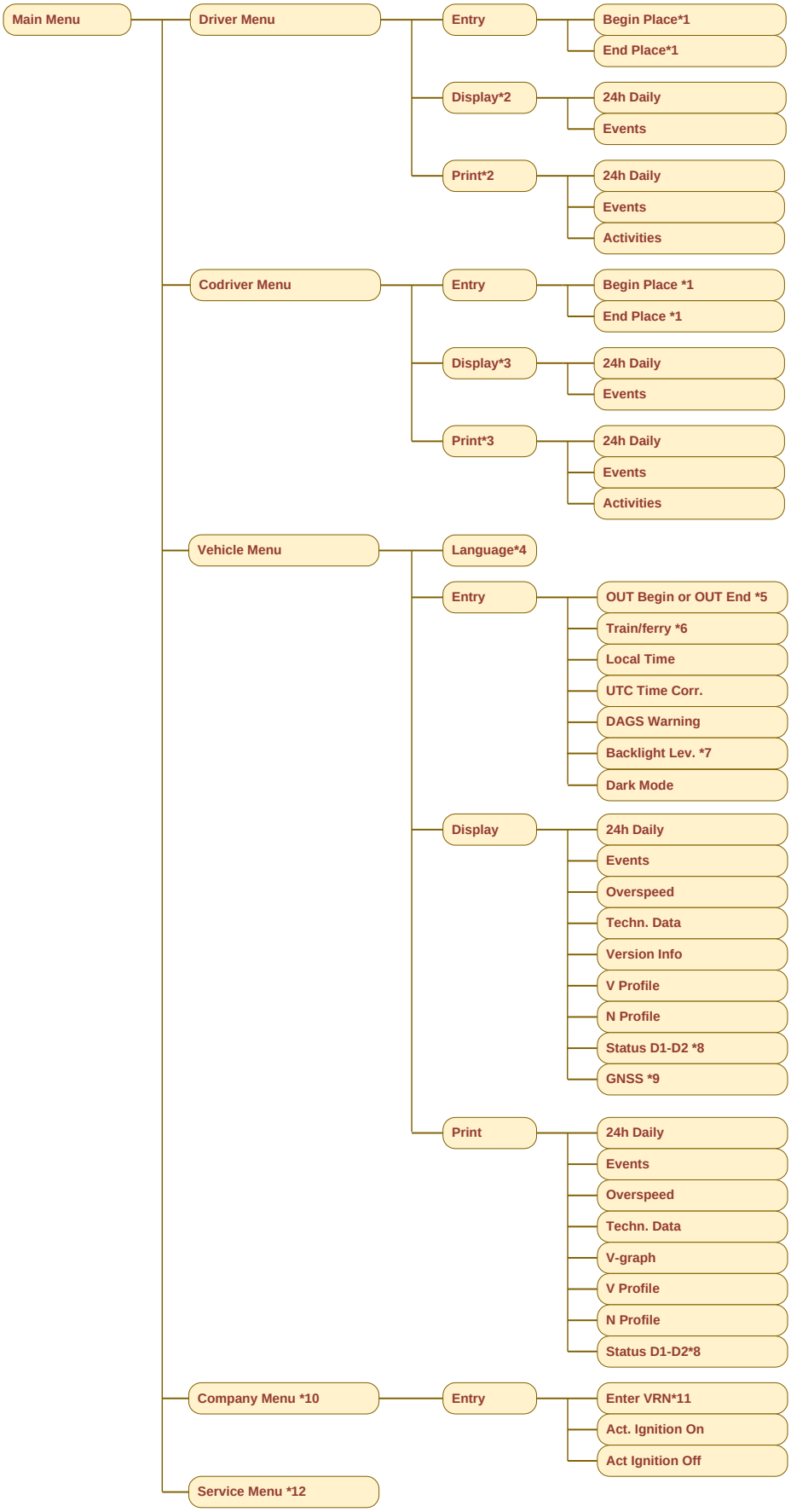


Figure 16: Menu Structure-1

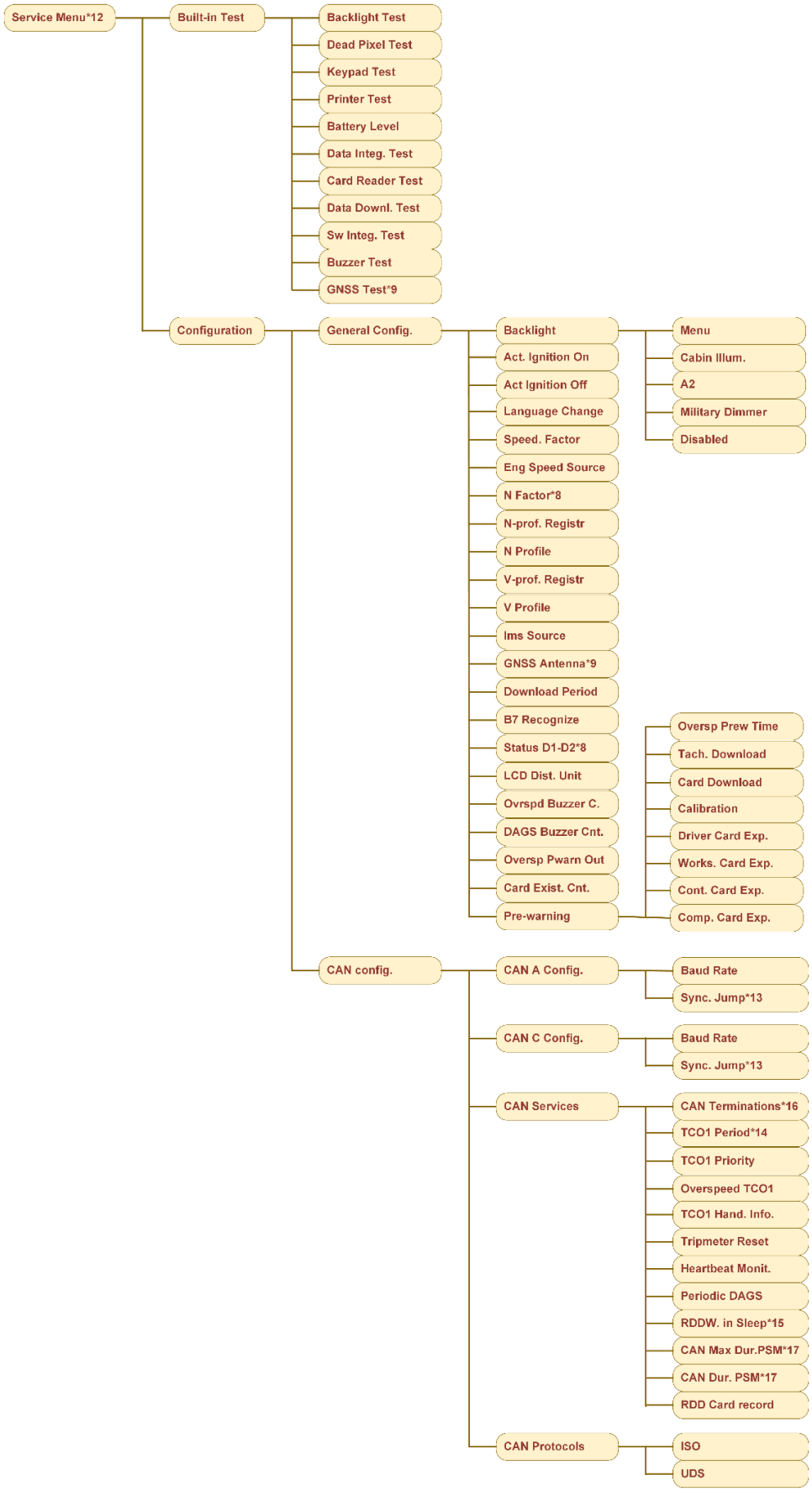


Figure 17: Menu Structure-2

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5 GENERAL INSTRUCTIONS PRIOR TO INSTALLATION

The following sections describe the preparations required before installing the STC-8255 digital tachograph into the vehicle. Before the digital tachograph can actually be installed in a vehicle it is very crucial to perform a careful (visual) inspection of the components present.

In order to verify the approved and the certified versions of software and hardware of the STC-8255, please check whether the correct versions of the software and hardware information are installed or not, by following the explanations given in section 9.12 Software & Hardware Information in detail. Software and hardware versions must be identical with the ones defined in section -2.1 of the Security Target (DTVU-ST) document (i.e. ST public version).

5.1 Receipt of Device

Upon receipt of the STC-8255, please check that the packaging of the tachograph is intact. Check that the tachograph is not damaged during shipping.

Please also check that;

- The sealing label, sealing plug and sealing cap on the tachograph are undamaged and intact (please refer to section-5.3 Checking before Installation")
- If the current approved and certified software is installed on the STC-8255.

The STC-8255 software must be approved by the manufacturer (Aselsan Inc.) and certified by the official EU approval bodies. The certified hardware and software versions are available at the following link (exact link will be given later).

Aselsan's web site (https://www.aselsan.com.tr/en/capabilities/public-safety-communicatio-systems/digital-tachograph-systems).

5.2 Storage

Store the STC-8255 digital tachograph on its original packaging and protect it from over temperature and moisture. The tachograph must not be exposed to any mechanical load or force that could make physical damage to it.

Storage Temperature: -40 °C to +85 °C

5.3 Checking before Installation

This section describes the visual check procedures prior to installation of the STC-8255 digital tachograph. The visual checks verify that the tachograph to be installed is a genuine Aselsan Inc. digital tachograph; namely, STC-8255.

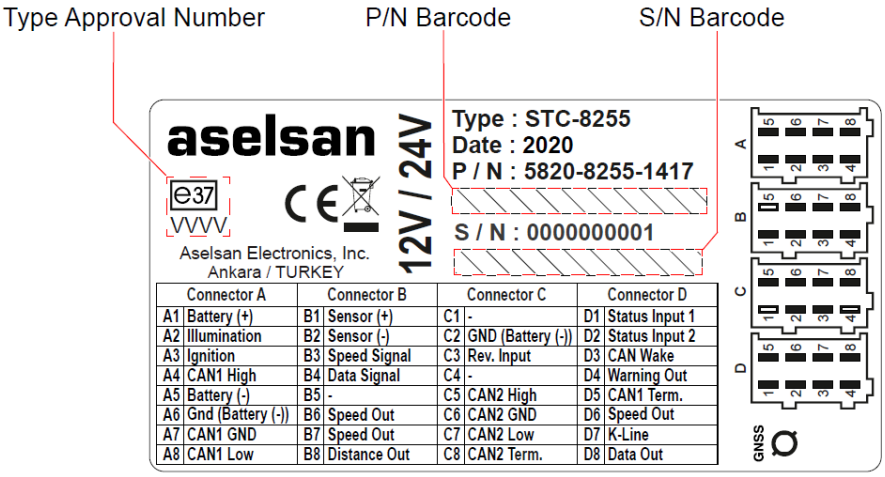
The following points must be checked:

- The descriptive label (which is transparent) must be properly affixed to the tachograph and the type approval number on the label shows the correct Aselsan tachograph type approval number "e37-VVVV" (will be assigned after type approval certificate is granted).
- The sealing cap must be intact and not interfered with. Please refer to next section "☐
- Details of the Descriptive Label and Visual " for further information about tamper seals.
- The sealing label (tamper evident security label with hologram) must be present and genuine. This label is located at the upper side of the device. Please refer to next section "☐
- Details of the Descriptive Label and Visual " for further information about hologram.
- Check that the sealing plug which is a tamper-evident breakable mechanical part covering the locking joint of VU enclosure is present and intact. See the section ☐ for details.
- Check that any evidence of tamper with the sealing labels, sealing plug or sealing cap exists.

- Make sure that there is no damage i.e. any drill holes in the exterior mechanical case of the tachograph.
- Check that any additional seals or labels does not exist on the case as they might cover drilled holes.
- Make sure that the tachograph does not come into contact with water, other liquids or any chemicals.
- Make sure that the tachograph is not exposed to direct sunlight for extended periods.
- Check also that the vehicle use is NOT intended for the transport of dangerous goods. If it is, then do not install the STC-8255 since it has not ADR compliance (Ex-proof).

5.4 Details of the Descriptive Label and Visual Checks

Figure 18 shows the details of STC-8255 descriptive label which is affixed on top of STC-8255.



Connector A		Connector B		Connector C		Connector D	
A1	Battery (+)	B1	Sensor (+)	C1	-	D1	Status Input 1
A2	Illumination	B2	Sensor (-)	C2	GND (Battery (-))	D2	Status Input 2
A3	Ignition	B3	Speed Signal	C3	Rev. Input	D3	CAN Wake
A4	CAN1 High	B4	Data Signal	C4	-	D4	Warning Out
A5	Battery (-)	B5	-	C5	CAN2 High	D5	CAN1 Term.
A6	Gnd (Battery (-))	B6	Speed Out	C6	CAN2 GND	D6	Speed Out
A7	CAN1 GND	B7	Speed Out	C7	CAN2 Low	D7	K-Line
A8	CAN1 Low	B8	Distance Out	C8	CAN2 Term.	D8	Data Out

Figure 18: Showing the Descriptive Label of STC-8255.

The descriptive label has the following information written on it.

- Name and address of the manufacturer: Aselsan Electronics, Inc. Ankara / Turkey.
- Manufacturer's model or type number: STC-8255
- Manufacturer part number (P/N): 5820-8255-1417 (depends on the device configuration and can be a variant of 5820-8255-1xxx)
- Product serial number (S/N): 0000000001 (ten character)
- Manufacture date of the Equipment: 2020 (YYYY)
- Barcodes of the serial number and product number
- Type approval number e37-0008
- Equipment's operating voltage (12V/24V)
- Pin connection diagram of the tachograph.

5.4.1 Visual Checks

The VU mechanical structure is so designed such that physical tampering attempts can be easily detected through visual inspection of the plastic enclosure, the sealing (security) label, the sealing plug and the sealing cap. Workshops or fitters have to make visual inspection before installing the tachographs into the vehicle.

1. The sealing cap (shown in the Figure 19) which covers one of the screws connecting front panel assembly and the mechanical enclosure. It is not possible to open the VU without removing or breaking this tiny piece of round plastic. To provide its genuineness Aselsan brand name is cold pressed on this round plastic to avoid imitation.



Figure 19: Showing the Sealing Cap Covering the STC-8255 Screw

2. The sealing label (shown in the Figure 20) is a tamper-evident label. In order to provide genuineness and authenticity of the label, it also contains a hologram and a unique serial number which is also embedded into the VU software. This sealing (security) label cannot be removed without being destructed if one attempts to open the VU enclosure. The workshops are responsible for checking if the sealing label on the VU is genuine and intact.



Figure 20: Showing the Sealing (Security) Label on the Top of the STC-8255

The sealing plug (shown in Figure 21) is a tamper-evident breakable mechanical part covering the locking joint of the VU enclosure. It is not possible to reach this locking point without breaking this sealing plug. To provide its genuineness, Aselsan brand name is cold pressed on this plastic part to avoid imitation. The workshops are responsible for checking if the sealing plug on the VU is genuine and intact.



Figure 21: Showing the Sealing Plug of the STC-8255

6 INSTALLATION OPERATIONS

By inserting a workshop card and entering its respective PIN, the STC-8255 digital tachograph enters into the Calibration mode. In this mode, workshop staff can be able to activate, pair motion sensor, calibrate and configure vehicle specific parameters and download mass storage data of the STC-8255.

The complete installation procedure consists of the following:

- Installation of the motion sensor and the sensor cable.
- Making compulsory electrical power and required signal connections.
- Mounting the tachograph.
- Activating the tachograph i.e. pairing with motion sensor.
- Calibrating and programming the tachograph.
- Sealing the tachograph system (especially for the motion sensor).
- Completing and fixing an installation plaque.

6.1 Tools and Equipment Needed

Following tools and equipment could be needed during installation process.

- Aselsan's digital tachograph calibration (programming) device which is STKC-8250 (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B, Appendix-8).
- Digital tachograph data download device (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B, Appendix-7).
- Valid (not expired) workshop card.
- Digital tachograph removal tool.
- Connector housing and connector crimps.
- Crimp nippers (for fixing the cable to crimps).
- Lead seal pliers (for pressing workshop ID into seals).

6.2 Installation of Digital Tachograph

This section describes the main steps of installation of the STC-8255 digital tachograph into a vehicle and putting into operation.

6.2.1 Motion Sensor

The motion sensor that will be used with the Aselsan STC-8255 digital tachograph must conform to EU regulations; namely, it has to be type approved according to the EEC No: 3821/85.

➔ Please refer to the motion sensor manufacturer's installation instructions before installing it.

Install the motion sensor directly to the vehicle gearbox as specified in the related documentation. Aselsan STC-8255 digital tachograph can only work with the type approved motion sensor. Any existing motion sensor that was previously installed to use with an analogue tachograph must be replaced before progressing.

6.2.2 Cabling

6.2.2.1 Sensor Cable

After the correct type of motion sensor has been installed, it can be connected to the motion sensor cable. The other end of the motion sensor cable is connected to the connector-B on the back side of the STC-8255. If the cable is going to be installed for the first time to the vehicle, please refer to the pin assignments of the connectors (B) given in section 6.2.2.4 Pin Assignments. Please also give attention to

the following criteria if you are installing the cable assembly for the first time. The cross-sectional area of the sensor cable which is going to be installed should be at least 4 x 0.5 mm^2.

- The cable assembly must be laid into the vehicle in adequate length and wherever possible it must be routed alongside other cables to reduce risk of damage.
- Connect the connectors tightly to avoid loose connections.
- Do not tie the cable to any moving parts.
- Make sure that the cable length is long enough because it is possible to remove the tachograph without disconnecting the sensor cable.
- Do not let the cable tensile too much.
- Ensure that the sensor cable is not cut or stretched by the mounting bracket while connecting the rear connector to the tachograph.
- Make sure that the cable is routed away from the intense heat sources like exhaust manifolds or turbocharger.

If all of the conditions mentioned above have been satisfied then the cable assembly can be appropriately secured by using cable ties.

If you are using existing cable assembly for the installation, please check that any branch connection does not exist and exclude it if exists.

6.2.2.2 Power Connections

The power cable must be connected to the tachograph from Connector-A. Please make sure that the same criteria mentioned above are also applied while assembling the power cables.

If the cable is going to be installed for the first time to the vehicle, please refer to the pin assignments of the connectors (A) given in section 6.2.2.4 Pin Assignments.

Power and motion sensor connections of the digital tachograph should be done according to the following figure.

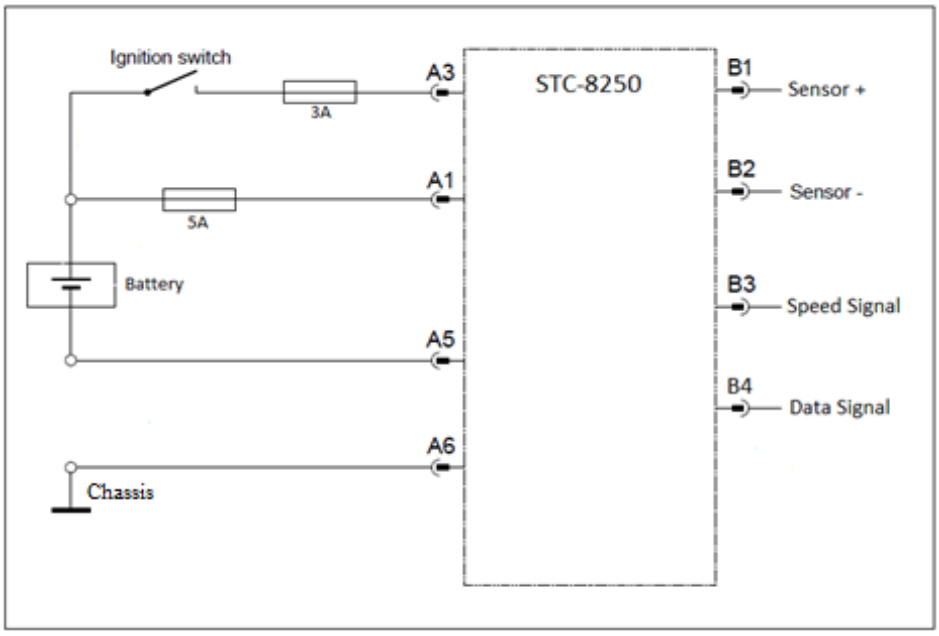


Figure 22: Wiring Diagram of STC-8255 in Vehicle

6.2.2.3 Connector Panel

STC-8255 digital tachograph is connected to the vehicle's electronic system via this rear connector panel at the back side of the device. The connector pins and functions are compatible with the ISO-16844 connector format. Pin assignments of the connector panel are described in the following section.

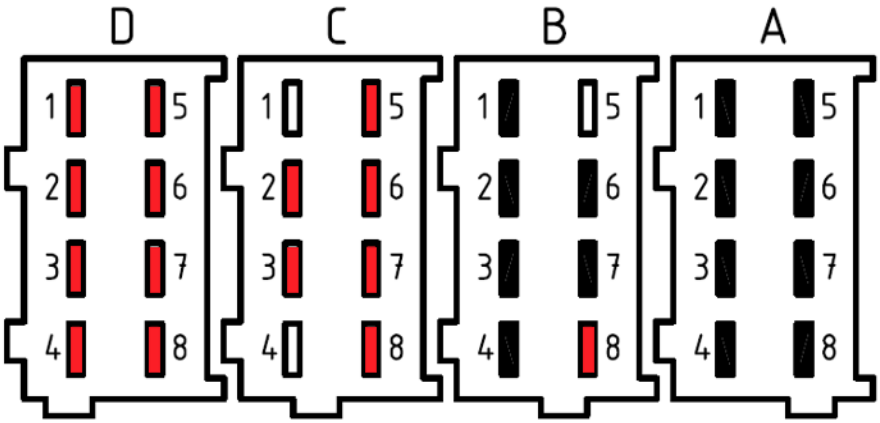


Figure 23: Showing the Rear Connector Panel Pin Allocation

The black colored pins (A1 to A8 and B1 to B7 except B5) are standard pins, the red colored pins (B8, C2, C3, C5, C6, C7, C8, D1, D2, D3, D4, D5, D6, D7 and D8) are optional pins according to the ISO16844 that are used in the STC-8255 connector panel. The white color pins are also optional according to ISO-16844 but not used and they have no functionality in STC-8255.

➔ For the mated connectors, please refer to the manufacturer's (TE Connectivity) datasheets for the related products.

6.2.2.4 Pin Assignments

Table 2: Pin assignment for connector-A

Pin	Direction	Name	Description
A1	-	Battery (+)	Permanent power supply.
A2	In	Illumination	Illumination control input.
A3	-	Ignition	Power supply line connected when ignition ON.
A4	Bi	CAN1_High	CAN-Bus High signal line for CAN-A.
A5	-	Battery (-)	Return line for the permanent power supply.
A6	-	GND (Battery (-))	Vehicle ground, chassis.
A7	-	CAN1_GND	Can-Bus ground for CAN-A.
A8	Bi	CAN1_Low	CAN-Bus Low signal line for CAN-A.

Table 3: Pin assignment for connector-B

Pin	Direction	Name	Description
B1	-	Sensor (+)	Motion sensor power supply (+).
B2	-	Sensor (-)	Motion sensor power supply (-).
B3	In	Speed Signal	Real-time speed signal from motion sensor.
B4	Bi	Data Signal	Encrypted data channel for the motion sensor.
B5	n/a	-	Not used.
B6	Out	Speed Out	Output for speed pulses.
B7	Out	Speed Out	Output for speed pulses.
B8	Out	Distance Out	4 pulse/m distance output.

Table 4: Pin assignment for connector-C

Pin	Direction	Name	Description
C1	n/a	-	Not used.
C2	-	GND (Battery (-))	Vehicle ground, chassis.
C3	In	Revolution Input	Motor revolution input pulse signal.
C4	n/a	-	Not used.
C5	Bi	CAN2_High	Second CAN-Bus High signal line for CAN-C.
C6	-	CAN2_GND	Can-Bus ground for CAN-C.
C7	Bi	CAN2_Low	Second CAN-Bus Low signal line for CAN-C.
C8	-	CAN2 Termination	CAN2 termination pin, (connect a wire to C7 to enable the 120 Ohm termination resistance on CAN2, depends on the configuration).

Table 5: Pin assignment for connector-D

Pin	Direction	Name	Description
D1	In	Status Input 1	Digital input for status input 1.
D2	In	Status Input 2	Digital input for status input 2.
D3	In	CAN Wake	Can wake up input signal.
D4	Out	Warning Out	Digital output for external alarm.
D5	-	CAN1 Termination	CAN1 termination pin, (connect a wire to A8 to enable the 120 Ohm termination resistance on CAN1, depends on the configuration).
D6	Out	Speed Out	Digital pulse output for speedometer.
D7	Bi	K-Line	K-Line data communication.
D8	Out	Data Out	Serial communication (information) interface.

6.2.3 Mounting the Tachograph

Aselsan STC-8255 digital tachograph conforms to the DIN ISO-7736 standard (which describes the dimensions of a car radio) and it is convenient to install Aselsan STC-8255 to 1-DIN slot easily.

Please read the following instructions before mounting the tachograph.

The tachograph must be installed in such a position that the driver can see and read the display, access and operate the tachograph safely from driver's seat.

- Open the DIN ISO-7736 bay in the driver's cab which is intended for installation. If there is no standard bay present or free, install an appropriate mounting bay or a frame which conforms to the ISO-7736 standard.
- Pull the connectors from the back of the radio cage and plug it into the tachograph according to the rear connector panel given in Figure 23. Make sure that, the wires are long enough to allow the unit to be moved in and out of the slot when connected.
- Insert the tachograph into the slot until you hear the clips click into place.
- Ensure that the wires will not be damaged when the tachograph is inserted to its slot. Leave enough space for the wires not to be damaged.
- If you have some difficulty while inserting the tachograph into the existing installation frame, then you should try with the new installation frame that is approved by Aselsan Inc. Please contact Aselsan Inc. to obtain these installation frames.
- Support the tachograph at the rear end of the unit from the rubber alignment.

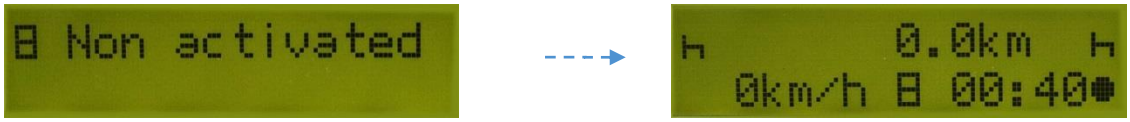


Do not install the tachograph on a dashboard or a place that is exposed to direct sunlight.

6.2.4 Activation and Motion Sensor Pairing

The STC-8255 digital tachograph is supplied in the "not-activated" state. In this state, the tachograph is not fully functional and no data will be stored in to the STC-8255.

After all of the connections explained in the previous sections are made carefully and the tachograph is installed in its slot STC-8255 can be powered on. The first screen that appears on the display is the warning message (shown below) indicating that the tachograph is not activated. After the warning message displayed, the display turns to the non-activated display format (explained in section 2.3.2.1 Activation status) which is shown below.



After a valid (not expired) workshop card is inserted into the tachograph and the PIN is entered (explained in section 3.4.1PIN Entering), the pairing and activation process starts automatically in calibration mode of operation.

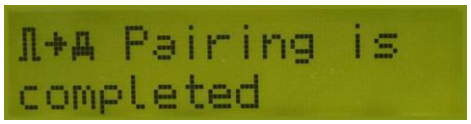
The paring process can also be initiated from the STKC-8250 calibration device during the calibration process.

The STC-8255 digital tachograph does not accept any tachograph cards except the valid workshop card until it is activated.

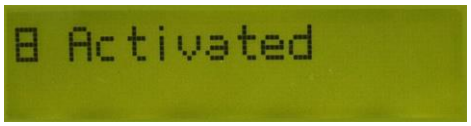
After insertion of a workshop card, the activation and pairing process starts automatically and the following screen is displayed.



When the pairing process is completed and the following screen is displayed, press **OK** to confirm.



Since a valid workshop card is inserted into the tachograph the STC-8255 is activated and when you remove the workshop card it will give the following message meaning the activation process is successful.



You can continue the calibration procedure before ejecting the workshop card. Then you can get this message after the calibration process.

If activation or pairing fails you get the following display messages.



Then;

- ➔ Remove the workshop card,
- ➔ Check the system connections and re-insert the workshop card and retry the process until the pairing/activation is successful.

6.2.5 Calibration

Calibration of a tachograph is compulsory after the installation process, during its regular inspections or after any repair that requires a tachograph system to be disconnected or seal to be broken.

After the physical installation of the tachograph is complete, the following vehicle parameters are required to be programmed into the tachograph's internal memory.

Calibration of the STC-8255 digital tachograph requires calibration equipment (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B, Appendix-8). Aselsan recommends to use Digital Tachograph Calibration Device (STKC-8250) to calibrate the STC-8255.

The front 6-Pin interface is used for calibration purposes in the Calibration Mode.

Vehicle parameters include;

- Vehicle identification number (VIN), vehicle registering member state (RMS) and the vehicle registration number (VRN),
- The current UTC date and time,
- The current odometer value,
- Vehicle characteristic coefficient (w-factor) and tachograph constant value (k-factor),
- Tyre size and the effective circumference of the wheels (L-factor),
- Maximum permitted speed,
- Next calibration due date (usually 2 years from current date).

When a calibration is carried out, the STC-8255 digital tachograph keeps a record of this calibration. Please see next chapter for details of this record.

The vehicle calibration parameters (e.g., w, k and l values) should be determined by using approved methods, for instances by use of a rolling road.

- Please refer to the calibration equipment's related documentation for further information of these methods.

A similar calibration record is also generated for the workshop cards and written to the inserted active workshop card (the card in the slot 1 if both slots are occupied with). The contents of the workshop card's calibration records consist of the followings:

- The calibration purpose (will be explained in next section),
- The VIN, vehicle registration nation and VRN numbers,
- The w, k and l values,
- Tyre size,
- Authorized speed limit,
- The old and new odometer values,
- The old and new time values,
- Next calibration due date,
- The tachograph part number, serial number, year of manufacture, and manufacturer's code,
- The motion sensor type, serial number, year of manufacture and manufacturer's code.

If just the time is updated during the calibration procedure and no other parameters are changed then this record is not stored as a calibration record but a time adjustment record.

When the calibration process is completed the calibration device will give a confirmation message about the success of the calibration. If the calibration procedure is failed, then remove the card and check the connections or faults before re-inserting the workshop card. Then, re-connect and try to calibrate the tachograph with the calibration device once again.

By using the calibration device (STKC-8250) a several tests can be performed. Speed test, odometer and clock test are the most important of these tests. Connecting the device from the 6-Pin interface the tachograph could be triggered for the speed testing and these driving activities are ignored and not recorded to the STC-8255. Meanwhile, the STKC-8250 calibration device test the real time clock deviations which should be less than ±2 seconds for a day. Besides, some other parameter configurations rather than the mentioned ones above can also be programmed by means of the calibration devices. These parameters are also configurable from the service menu of the tachograph when a valid workshop card is inserted.

6.2.6 Checking Calibration Data

After a calibration process is completed, a check should be made to confirm that the records stored to the tachograph are correct. To perform this, download the tachograph memory and check the downloaded data or take a technical data printout to confirm the correctness of the stored data.

When you take a technical printout, check the following content:

- Calibration purpose,
  - Activation (recording of calibration parameter known at the moment of activation)
  - First installation (first calibration of the VU after its activation)
  - Installation (first calibration of the VU in the current vehicle)
  - Periodic inspection
- The workshop name, address, card number and expiry date,
- The VIN and VRN,
- The w, k, and l parameters,
- The tyre size and authorized speed limit,
- The odometer values,
- The UTC date and time,
- The date of next calibration.

→Please refer to the “Section 10.12 Technical Data Printout”.

6.2.7 Sealing

Sealing is the most crucial tamper-proofing method in the tachograph system. All of the electrical and mechanical parts (motion sensor, cables) which can be removed should be sealed.

To ensure the integrity of the signal coming from the motion sensor through vehicle gearbox to the tachograph, the motion sensor has to be sealed at its connection to the gearbox. Sealing of the motion sensor may be accomplished either by using of the sealing pliers or traditional seals (same methods as the analogue tachographs). The actual sealing method must be authorized by the relevant authorities. If re-sealing is necessary it must be performed by the authorized workshops.

The installation plaque must also be sealed, if it is not damaged while being removed.

If a broken or a damaged seal is detected, the workshop must;

- Inspect, calibrate and re-seal the tachograph system.
- Prepare a report about the cause of the damaged or broken seal and inform the relevant authorities about the situation.

6.2.8 Test Drive

After installation and calibration procedures, a test drive or roller test bench is necessary (according to the legal requirements) to confirm that calibrated tachograph is functioning properly. The test drive includes the following controls:

Checking of the tachograph if it displays the speed correctly, its odometer value is coherent with the vehicle's odometer values and its displayed time is correct. Also checking the proper operation of the speedometer if it is connected.

6.2.9 Installation Plaque

After successful completion of the calibration and testing, an installation plaque must be affixed to the vehicle.

The installation plaque must be secure from manipulation, i.e. it must be impossible to modify the data on it without destroying the label or without leaving visible traces on it. The installation plaque must be affixed to the vehicle in such a way that, it can be clearly visible and easily accessible by the inspection authorities at any time. In general, it is mounted on the vehicle's "B" pillar or the doorframe on the driver's side of the vehicle.

The installation plaque must include the followings:

- Name and address of the authorized workshop,
- w = ... Imp/km (characteristic coefficient of the vehicle),
- k = ... Imp/km (constant of the vehicle),
- l = ... mm, actual tyre circumferences measured during calibration,
- Tyre size according to the written on the tyre itself,
- Date of calibration,
- Vehicle Identification Number (VIN).

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7 GENERAL SETTINGS

7.1 Language

The STC-8255 supports 17 (seventeen) languages at the moment. The language shown by the display depends on the tachograph card which is inserted to the driver card slot. The language is automatically adjusted upon card insertion or withdrawal. However, it can be selected manually from the vehicle menu.

To perform manual language selection:

1. Press **OK** to enter main menu. Select **Vehicle menu>Language** by using control ‹ / › keys. Press **OK** to enter the menu. Select the desired language you want to set. Press **OK** to activate the current language selection. After confirmation STC-8255 returns one level back.



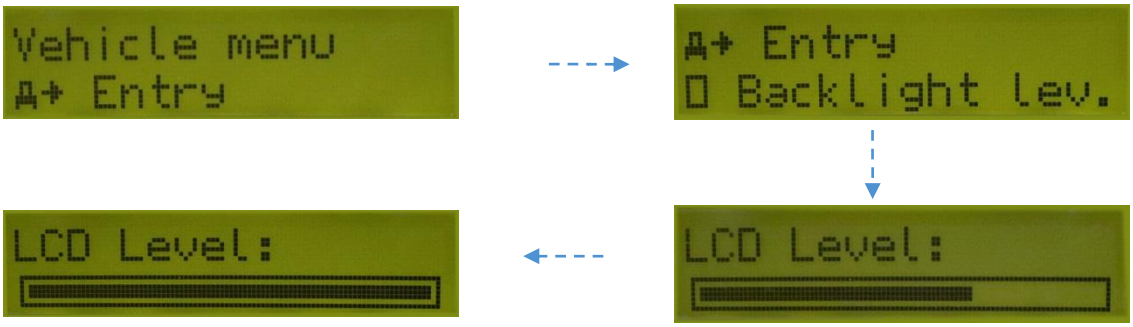
7.2 Display Backlight Level

STC-8255 Backlight level can be adjusted manually from this menu if the backlight source is selected as ***"Menu"*** from ***Service menu>General Configurations>Backlight*** settings. Backlight level can also be adjusted from the A2 illumination pin or from the CAN-Bus of the vehicle.

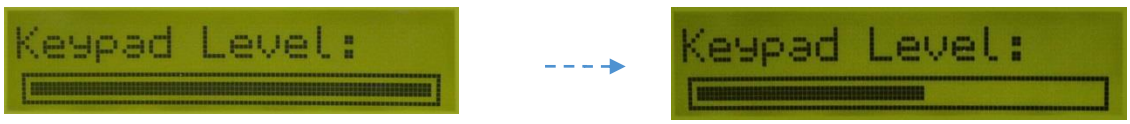
➔ Please refer to "Section 8.2.1.1 Backlight" for more information.

To adjust display backlight level:

1. Press **OK** to enter main menu. Select **Vehicle menu>Entry** by using control ‹ / › keys. Press **OK** to enter the menu. Select **Entry>Backlight lev.** by using control ‹ / › keys. Press **OK** to enter the menu. The current LCD backlight level is displayed. Adjust the LCD backlight level to the desired level by using control ‹ / › keys.



After adjusting the desired LCD backlight level, press **OK** to adjust the Keypad backlight level to the desired level by using control ‹ / › keys. Press **OK** to save current settings. After confirmation, STC-8255 returns one level back.

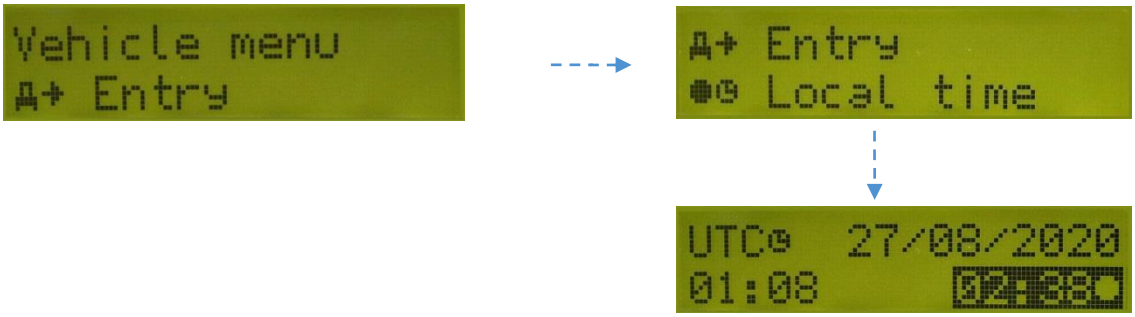


7.3 Local Time

In STC-8255 standard (default) display view, local time is displayed. It may be adjusted according to the current time in a specific country and can be adjusted manually in steps of ±30 minutes from the menu.

To adjust the local time, follow the instructions below:

1. Press **OK** to enter main menu. Select **Vehicle menu>Entry** by using control ‹ / › keys. Press **OK** to enter the menu. Select **Entry>Local time** by using control ‹ / › keys. Press **OK** to enter the menu. The current UTC and Local time is displayed. Adjust the local time to the desired value in steps of ±30 minutes by using control ‹ / › keys. Press **OK** to save current settings. After confirmation STC-8255 returns one level back.



Please refer to section - 9.15

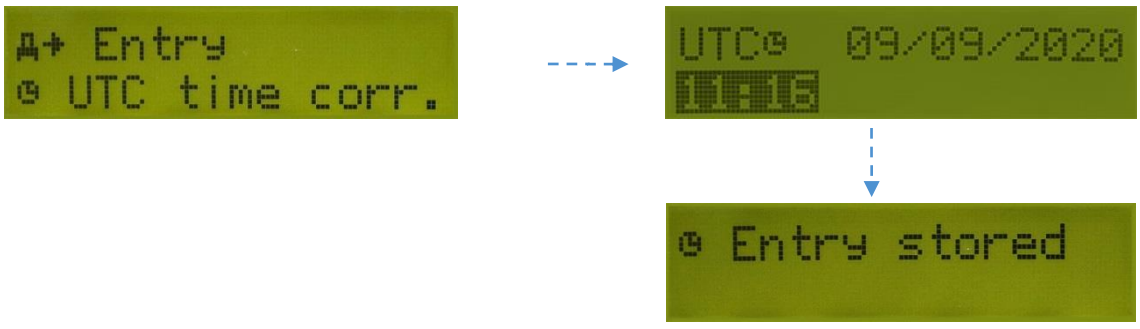
- National / Regional Abbreviations and the European Time Zones.

7.4 UTC Correction

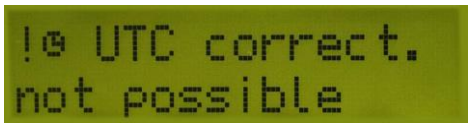
All of the tachograph activities are registered to the memory in UTC time (Universal Time Coordinated). You can correct the UTC time by ±1 minute per week. Greater deviations can only be changed by an authorized workshops with calibration process.

To make UTC correction from the menu, follow the steps:

- Press **OK** to enter main menu. Select **Vehicle menu>Entry** by using control ‹ / › keys. Press **OK** to enter the menu. Select **Entry>UTC time corr.** by using control ‹ / › keys. Press **OK** to enter the menu. The current UTC time is displayed. Adjust the UTC time to the desired value in steps of ±1 minute by using control ‹ / › keys. Press **OK** to save current settings. After confirmation STC-8255 returns one level back.



- If you try to correct the UTC time more than once within 7 days, you will get the following message on the display.



Note that: If you try to correct the UTC time while the STC-8255 is in Calibration Mode you will not get this warning message on the display.

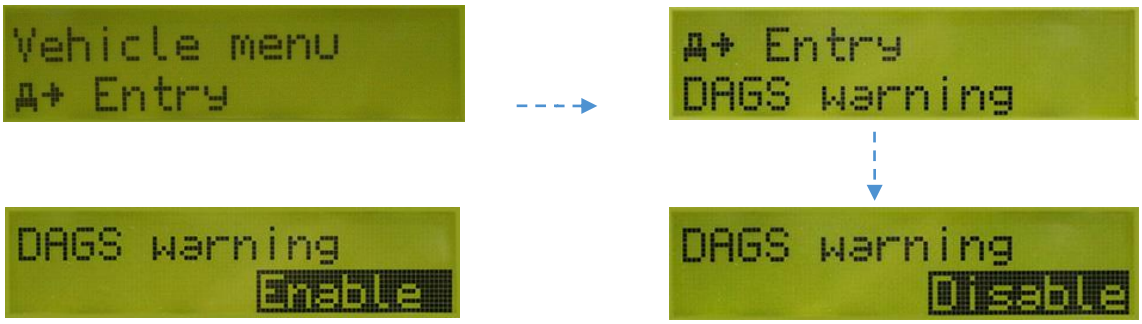
7.5 DAGS Warning

Driver Activity Guidance System (DAGS) warning, reminds the driver to give a break after continuous driving time of 4h15min, 4h30min, daily, weekly and fortnightly driving and upcoming resting times. DAGS Warning of the STC-8255 can be enabled or disabled manually from this menu. If the DAGS buzzer control is enabled from ***Service>General Configurations>DAGS buzzer cnt.*** settings, it gives an audible alert to the user when DAGS warning system is enabled.

➔ Please refer to "Section 8.2.1.19 DAGS Buzzer Control" for adjusting buzzer control.

To adjust ***Vehicle menu>Entry>DAGS warning***, do as follows:

1. Press **OK** to enter main menu. Select ***Vehicle menu>Entry*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***DAGS warning*** by using control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the DAGS warning to the desired status by using control ‹ / › keys. Press **OK** to save current settings. After confirmation STC-8255 returns one level back.

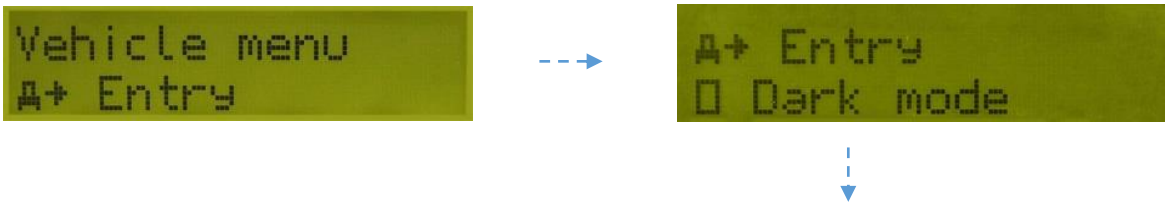


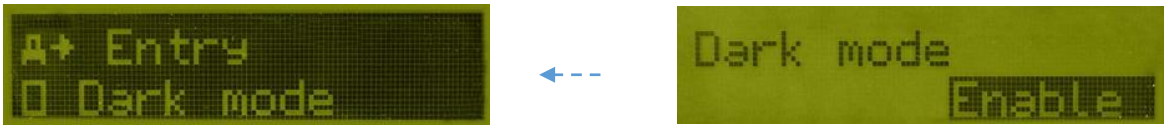
7.6 Dark Mode

The STC-8255's contrast of LCD can be altered by using Dark mode menu.

To enable ***Vehicle menu>Entry>Dark Mode***, do as follows:

1. Press **OK** to enter main menu. Select ***Vehicle menu>Entry*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***Dark Mode*** by using control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the desired mode by using control ‹ / › keys. Press **OK** to save current settings. After confirmation, STC-8255 returns one level back.




A+ Entry
0 Dark mode
Dark mode
Enable

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8 SERVICE MENU (PARAMETER CONFIGURATION)

To activate ***Service menu*** in the STC-8255, insert your valid workshop card into one of the card slots. Make sure that the other slot has a driver card or no card is inserted (STC-8255 must be in Calibration Mode). By using Service Menu, workshop staff can gain access rights to ***Built-in Tests***, ***General Configurations*** and ***CAN Configurations*** of the STC-8255 and can program certain parameters without using a calibration equipment.

8.1 Built-in Tests

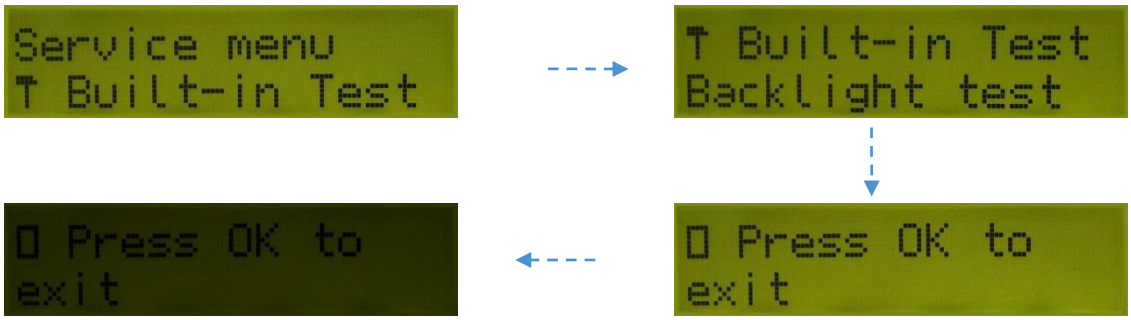
By entering the Calibration Mode, the user gains access rights to Service Menu, which is located under the Main menu. In Service Menu, a built-in test menu exists. Workshop staff can perform some built-in tests (e.g., backlight test, dead pixel test, keypad test, printing test, battery level test, data integrity test, card reader test, data download test, software integrity test, buzzer test and GNSS test depending on the configuration) by using this functionality of the STC-8255. In the following sub-sections, you will find the explanations of these built-in tests.

8.1.1 Backlight Test

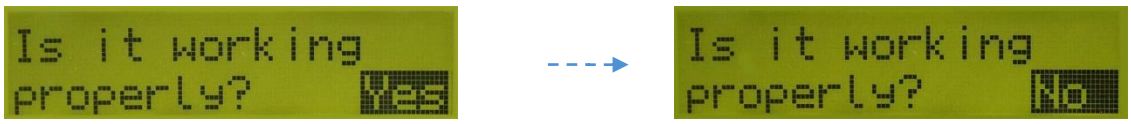
STC-8255 backlight test can be done under this menu.

To perform ***Built-in Test> Backlight test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Backlight test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, the backlight of the display turns ON and OFF successively.



If you see the backlight of the display turns ON and OFF, press **OK** to exit the menu. Then the following screen shows up.



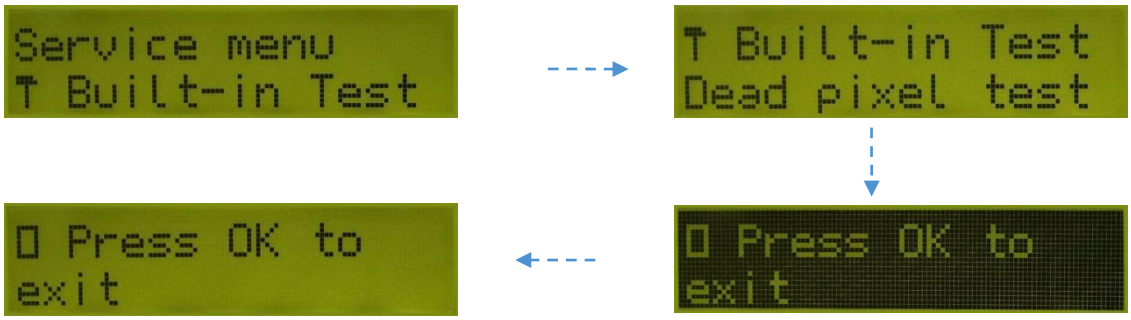
If you think backlight of the display is working properly select **Yes** and press **OK** to confirm. Otherwise select **No,** then press **OK**. If **No** is entered then STC-8255 will generate Display Error DTC. If the backlight is not working, the tachograph must be decommissioned and replaced.

8.1.2 Dead Pixel Test

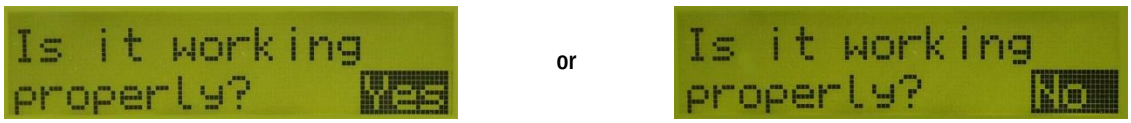
STC-8255 dead pixel test can be done from this menu.

To perform ***Built-in Test> Dead Pixel Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Dead Pixel test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, display shows positive and negative views successively.



Press **OK** to exit the menu. Then the following screen shows up.



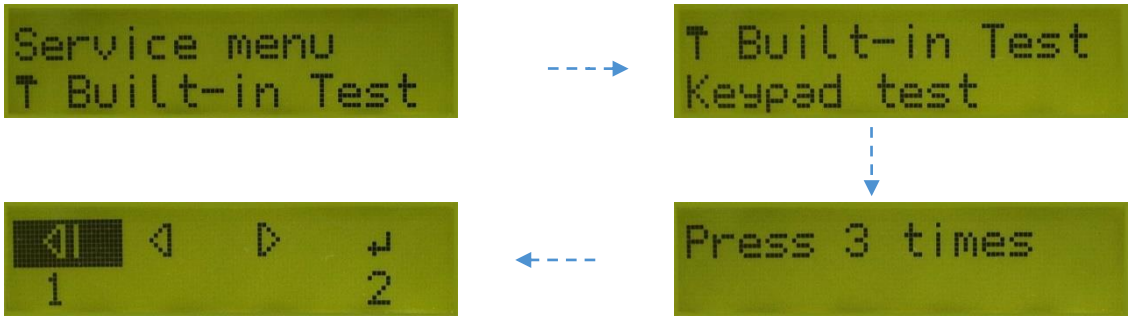
If you don't detect any dead pixels on the display, it means that display is working properly, then select **Yes** and press **OK** to confirm. Otherwise select **No,** then press **OK**. If **No** is entered then STC-8255 will generate Display Error DTC. If the display is unreadable, the tachograph must be decommissioned and replaced.

8.1.3 Keypad Test

STC-8255 keypad test can be done under this menu.

To perform ***Built-in Test> Keypad Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Keypad test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, display shows the following message and you are prompted to push each button shown on the display 3 times.



If you press each button 3 times, then the STC-8255 verifies that keypad is working properly, finishes the test by returning one level back. If you cannot press any button or forgot to press within 60 seconds,

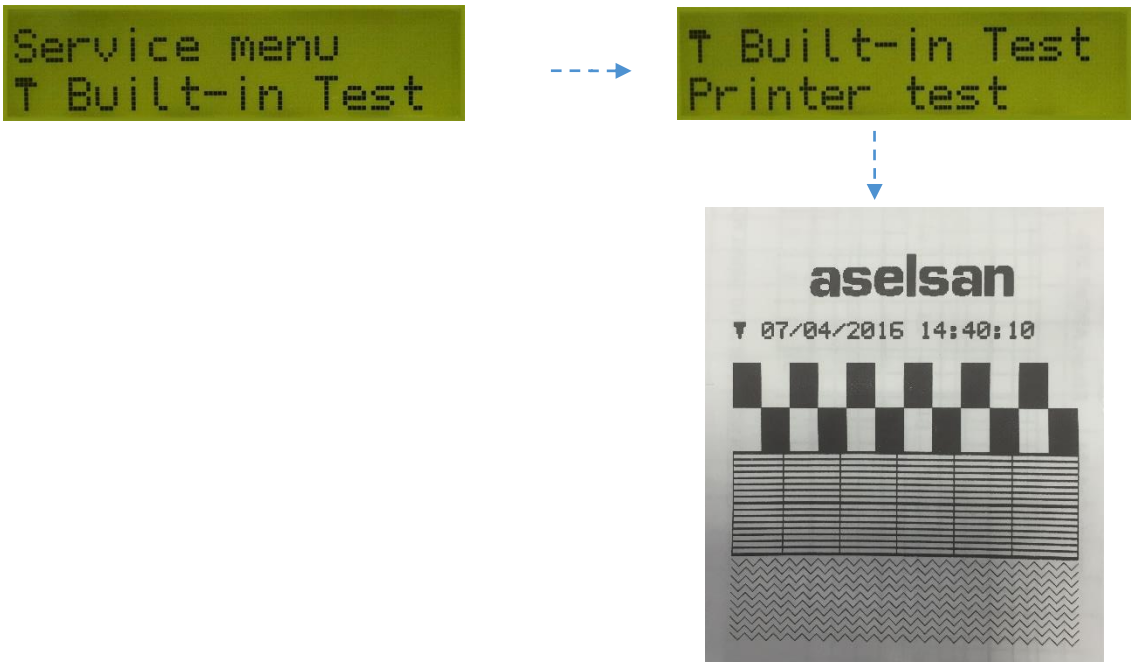
keypad test fails and STC-8255 exits the test by generating Keypad DTC Error. If the keypad is not functioning properly, the tachograph must be decommissioned and replaced.

8.1.4 Printer Test

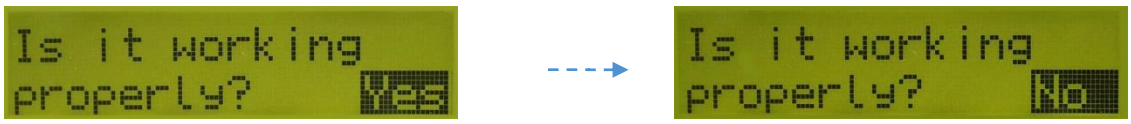
STC-8255 printer test can be done from this menu.

To perform ***Built-in Test> Printer Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Printer test*** menu by using the control ‹ / › keys. As long as you press **OK** to enter the menu, the printer starts printing test printout as shown in the following figure.



Then the following screen shows up after printing finished.



If the printer is working properly, select **Yes** and press **OK** to confirm. If the printing is not started because of out of paper, please insert new paper roll and re-perform the test. If printing fails again, select **No** and then press **OK**. If **No** is entered then STC-8255 will generate Printer Error DTC.

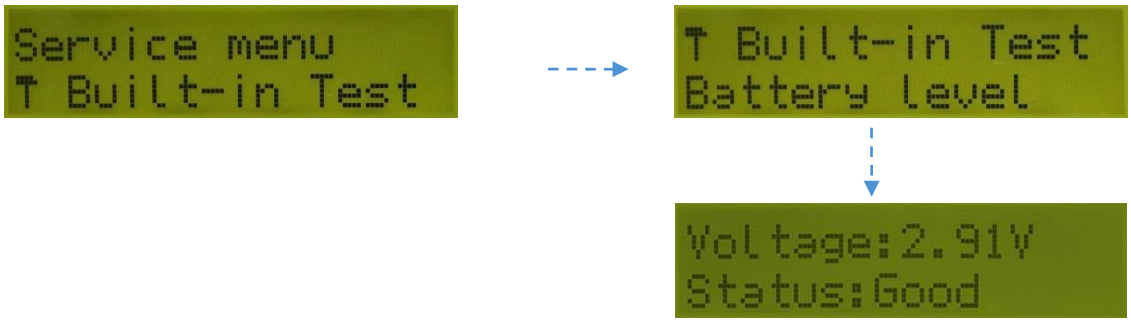
The STC-8255 digital tachograph must be decommissioned and replaced if printer is not working.

8.1.5 Battery Level

STC-8255 battery level can be seen under this menu.

To perform ***Built-in Test> Battery Level Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Battery level*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, display shows the voltage measurement of the internal battery.



Press **OK** or backspace ⌫ key after reading the battery level.

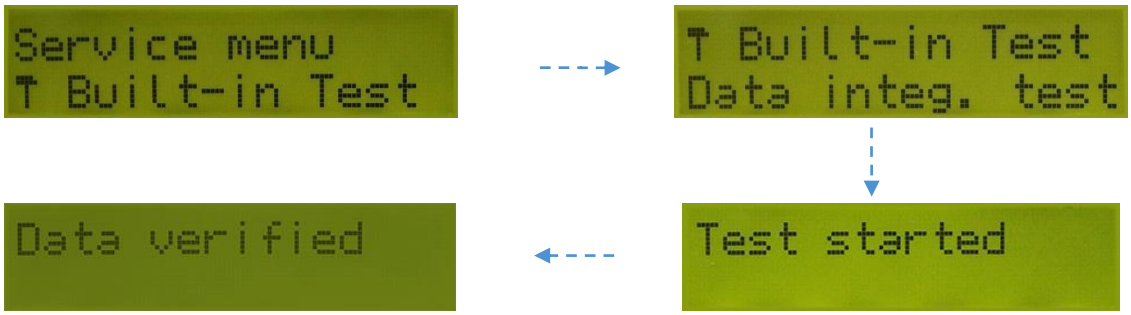
If the battery level is not good enough please consider re-charging the battery according to the procedure given in “section 9.8 Battery Re-charging Procedure of the STC-8255”.

8.1.6 Data Integrity Test

STC-8255 data integrity test can be done under this menu.

To perform ***Built-in Test> Data Integrity Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Data integrity test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, data integrity test starts and returns the success of the test.



Press **OK** button to return back.

If the data integrity test fails (data is not verified) please try to download the mass storage and consider to send the unit to the manufacturer (Aselsan Inc.).

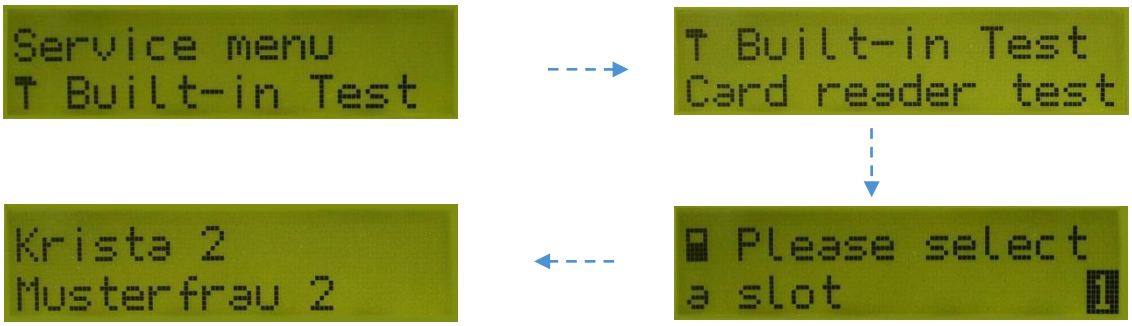
8.1.7 Card Reader Test

STC-8255 card reader test can be done under this menu.

To perform ***Built-in Test> Card Reader Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Card reader test*** menu by using the control ‹ / › keys. Press **OK** to enter

the menu. After pressing the **OK**, select the card reader slot in which the card is inserted, then press **OK** to start test.



If the card reader works properly and the inserted cards are valid, the reader reads the name and surname of the relevant card holder and display data for 10 seconds on the screen as shown above.

Otherwise, you will not see the identification information on the display. If the test fails, please answer **No** to the following question and the DTC code will be generated according to the selected card slot.

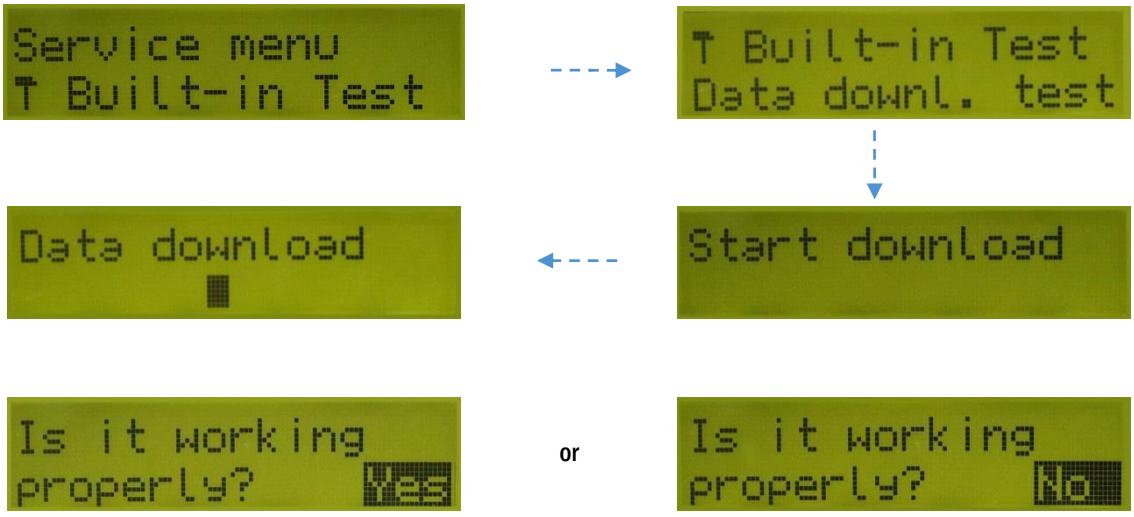


8.1.8 Data Download Test

STC-8255 data download test can be done under this menu.

To perform ***Built-in Test> Data Download Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Data download test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Connect the data download device into 6-Pin interface and start download from the download device. After download starts, you will see the data download message and flashing animation plays.



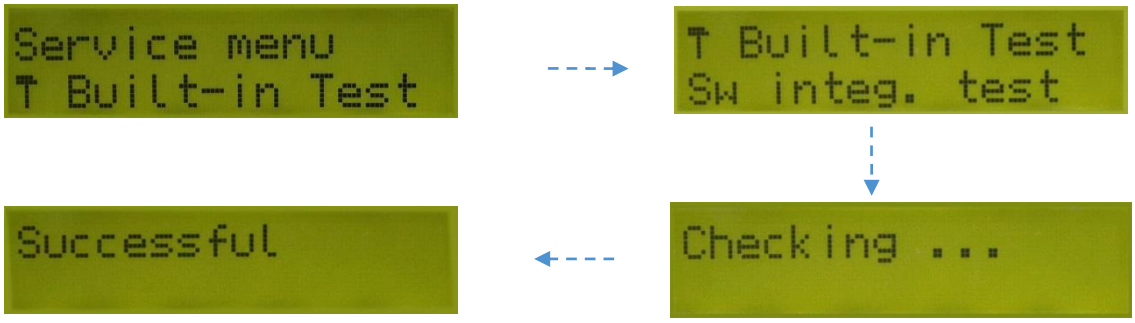
After downloading of the data finishes, you are prompted to confirm the result of the data download test. If it is completed without any errors (you can check it from the data download device), then select **Yes** and confirm with **OK.** Otherwise, select **No**, and confirm with **OK.**

8.1.9 Software Integrity Test

STC-8255 software integrity test can be done under this menu.

To perform ***Built-in Test> Software Integrity Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Software integrity test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, software integrity test starts and returns the result of the test.



If no failure occurs in the integrity of OS software then you will not get any failure messages but receive **"Successful"** message. Press **OK** button to return back.

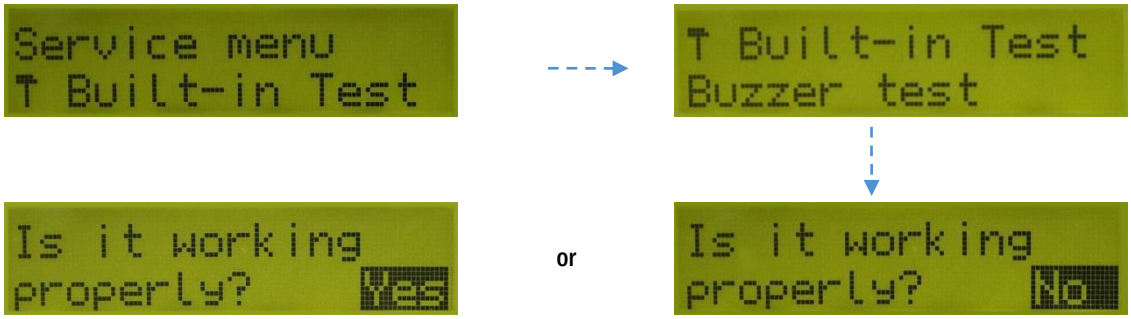
If the software integrity test fails (OS software is not verified) please try to download the mass storage and consider to send the unit to the manufacturer (Aselsan Inc.).

8.1.10 Buzzer Test

STC-8255 buzzer test can be done under this menu.

To perform ***Built-in Test> Buzzer Test***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Built-in Test*** by using control ‹ / › keys. Press **OK**, then select ***Buzzer test*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. After pressing the **OK**, buzzer test starts and buzzer generates an audible sound.



If you hear the generated buzzer sound then select **Yes** and confirm with **OK** button. Otherwise, select **No** and confirm with **OK** button.

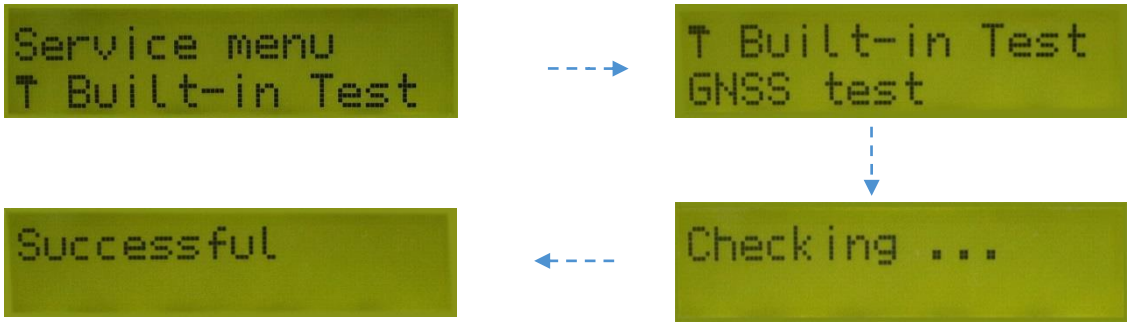
8.1.11 GNSS Test

STC-8255 GNSS test can be done under this menu.

To perform ***Built-in Test> GNSS Test***, do as follows:

1. Press **OK** to enter Main Menu. Select **Service menu>Built-in Test** by using control ‹ / › keys. Press **OK**, then select **GNSS test** menu by using the control keys. After pressing the **OK**, the internal

GNSS module is tested. If the module is OK then, "**Successful"** message is displayed as shown below.



If GNSS test fails, "Failed" message will be displayed and the DTC code will be generated accordingly. The STC-8255 digital tachograph must be decommissioned and replaced if GNSS module is not working.

8.2 Configuration

Some of the STC-8255 parameter configurations can be done under the sub-menus of this configuration menu. Configuration menu is divided into two sub-menus, one is the ***General Configurations*** which is related to the basic STC-8255 parameter settings and the other is the ***CAN Configurations*** which is related to CAN Interface specific parameters.

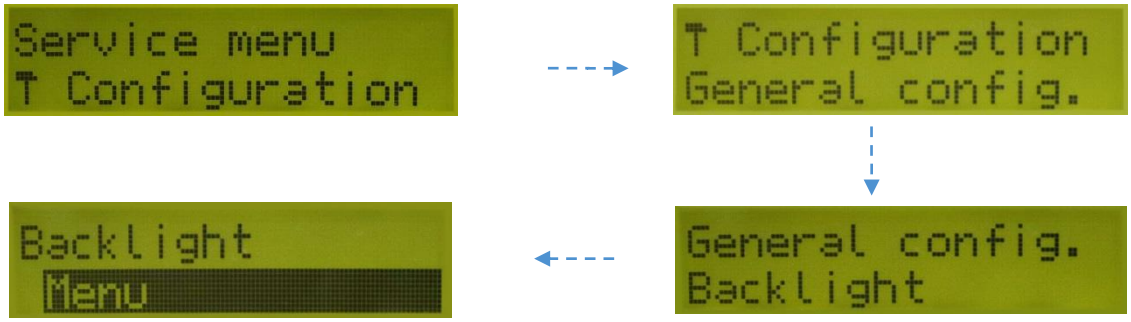
8.2.1 General Configurations

8.2.1.1 Backlight

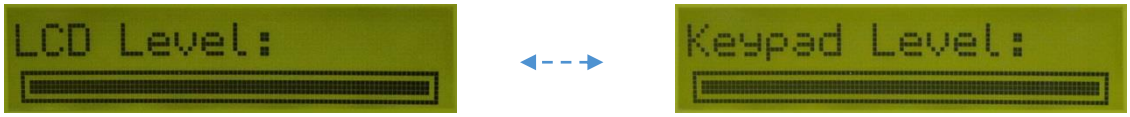
STC-8255 Backlight level setting source can be selected from this menu. Backlight level can be adjusted from Menu, from Cabin Illumination, from the Military Dimmer, from A2 Illumination Pin, or it can be disabled totally.

To select backlight source:

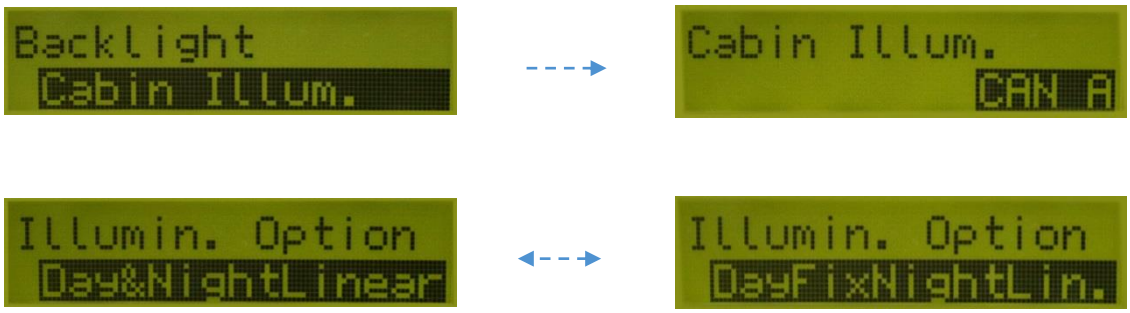
1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Backlight*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the backlight source that you desired to change.



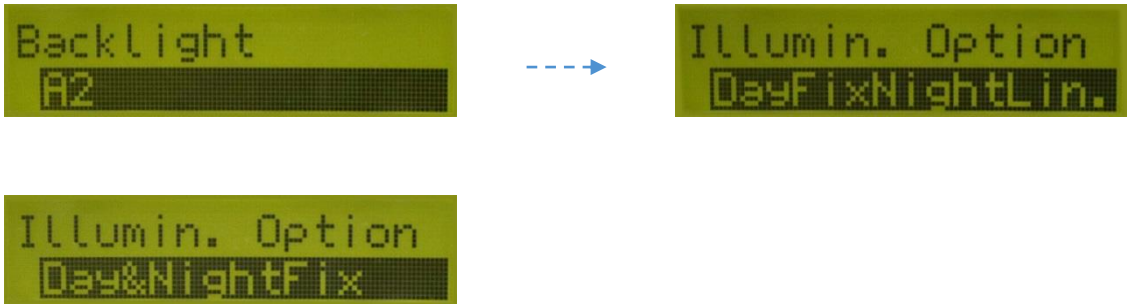
If the source is selected as ***Menu***, then the backlight level can be adjusted from that menu (***Vehicle Menu>Entry>Display Backlight Level*** explained in "section 7.2 Display Backlight Level") by manually.



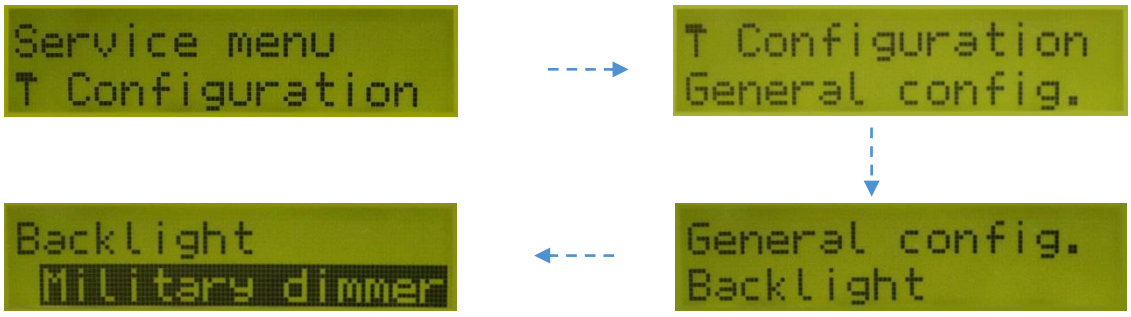
If the source is selected as ***Cabin Illumination***, then the backlight level can be adjusted from vehicle's illumination settings of instrument cluster which is connected to the CAN-Bus in the A or C connector. The Illumination options can also be selected as "Day Fix, Night Linear" or "Day & Night Linear".



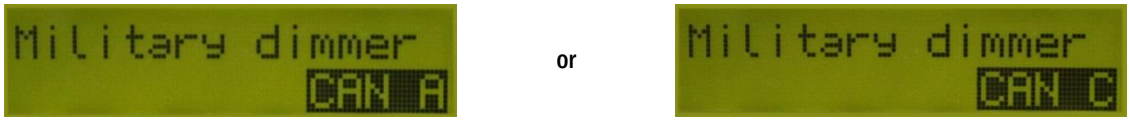
If the source is selected as ***A2***, then the backlight level can be adjusted from vehicle's illumination output which is connected to the A2 Pin in the A-connector. The Illumination options can also be selected as "Day Fix, Night Linear" or "Day & Night Fix".



STC-8255 military dimmer setting source can be selected from this menu especially in the military vehicles. Military dimmer can be adjusted from CAN-Bus of the military vehicle.



The military dimmer can be adjusted from military vehicle's instrument cluster illumination settings which is connected to the CAN-Bus in the A or C connector.



If the source is selected as ***Disabled,*** then backlight of the LCD can be switched OFF.



8.2.1.2 Activity Ignition ON

To adjust the ***Ignition ON*** activities in Calibration mode;

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select **Act. ignition on** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the activities that you desired to change automatically when the ignition on.



After igniton is **ON** again:

The activities can be set automatically to a pre-defined activity (e.g. **Work “ ”** and **Available “ ”**). The display shows the activities as shown below.



- Change the set activity according to your current task, as necessary.

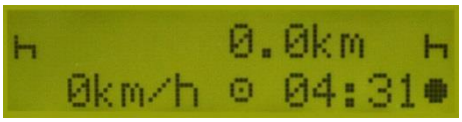
8.2.1.3 Activity Ignition OFF

To adjust the ***Ignition OFF*** activities in Calibration mode;

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select **Act. ignition off** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the activities that you desired to change automatically when the ignition off.



For example: Rest **“ ”** activities are selected both for the driver and co-driver. When the ignition becomes off, the activities of driver and co-driver will both change to pre-defined activity (Rest **“ ”**) automatically and display will be as follows:



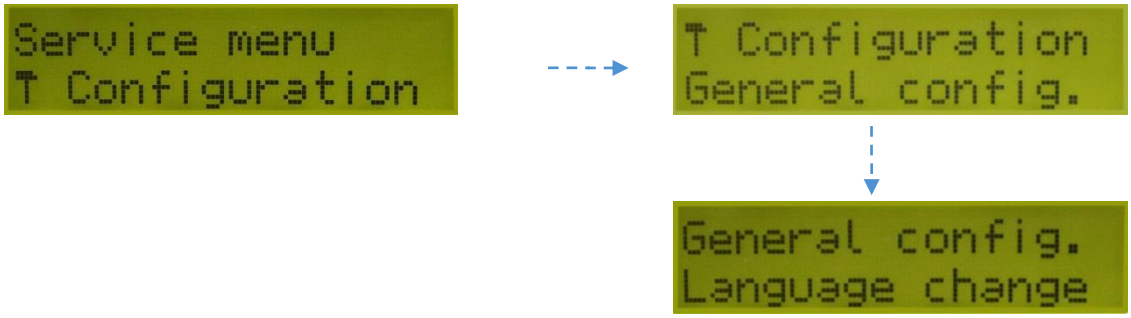
> Adjusting the activities to “?” both for driver/co-driver maintains the activities of driver and co-driver when the ignition is turned off/on again.

8.2.1.4 Language Change

The language change preferences of the STC-8255 can be adjusted from this menu.

To adjust the Language change preferences, do as follows:

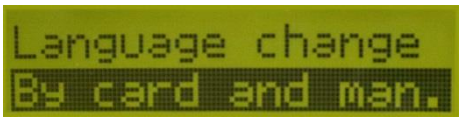
1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Language change*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the preferences that you desired to change.



The displayed language of the STC-8255 can be selected manually from the menu ***Vehicle Menu>Language*** or it is changed automatically with the inserted cards' defined language (defined in the card holder>preferred language field of the cards') properties.



If you select "only by card" then displayed language of the STC-8255 can only be adjusted automatically with the inserted cards' language properties defined in the cards' preferred language field. If two card slots are occupied with the cards, then STC-8255 automatically adjusts the displayed language according to the active card that is inserted to the first slot of the tachograph. After "only by card" selection is chosen, then displayed language cannot be changed from the Vehicle Menu>Language menu.



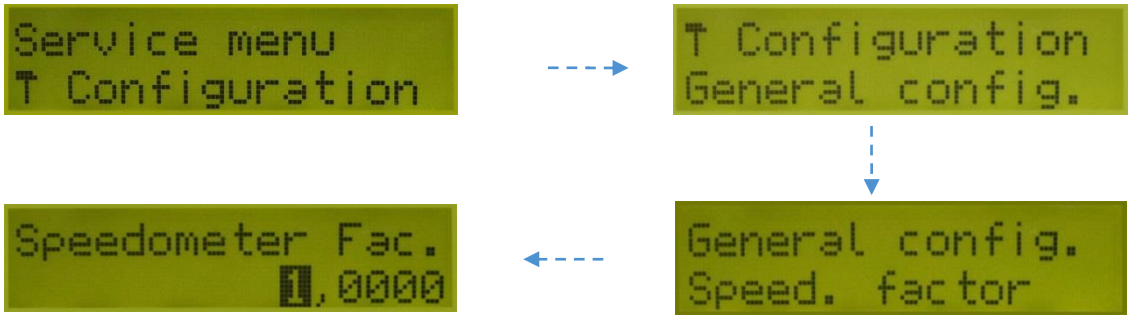
If you select ***"by card and manually"*** then displayed language of the STC-8255 can either be adjusted automatically according to the inserted card or can be selected manually from the ***Vehicle Menu>Language*** menu.

8.2.1.5 Speed. Factor

STC-8255 speedometer factor setting can be adjusted from this menu. STC-8255 has three speed outputs pins which are B6, B7 and D6. Two of these signals (B6 and B7) have the same frequency with respect to the real time speed signal input (B3). However, the third speed out signal (D6) frequency can be adjusted independently from the abovementioned signals (B6 and B7) with a multiple of this speedometer factor. The reason of this frequency adjustment is to integrate the speedometer (instrument cluster) of the vehicle with the STC-8255 digital tachograph.

To adjust the speedometer factor:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Speed. factor*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the speedometer factor that you desire by using the control ‹ / › keys. Pressing **OK** moves the cursor one by one.



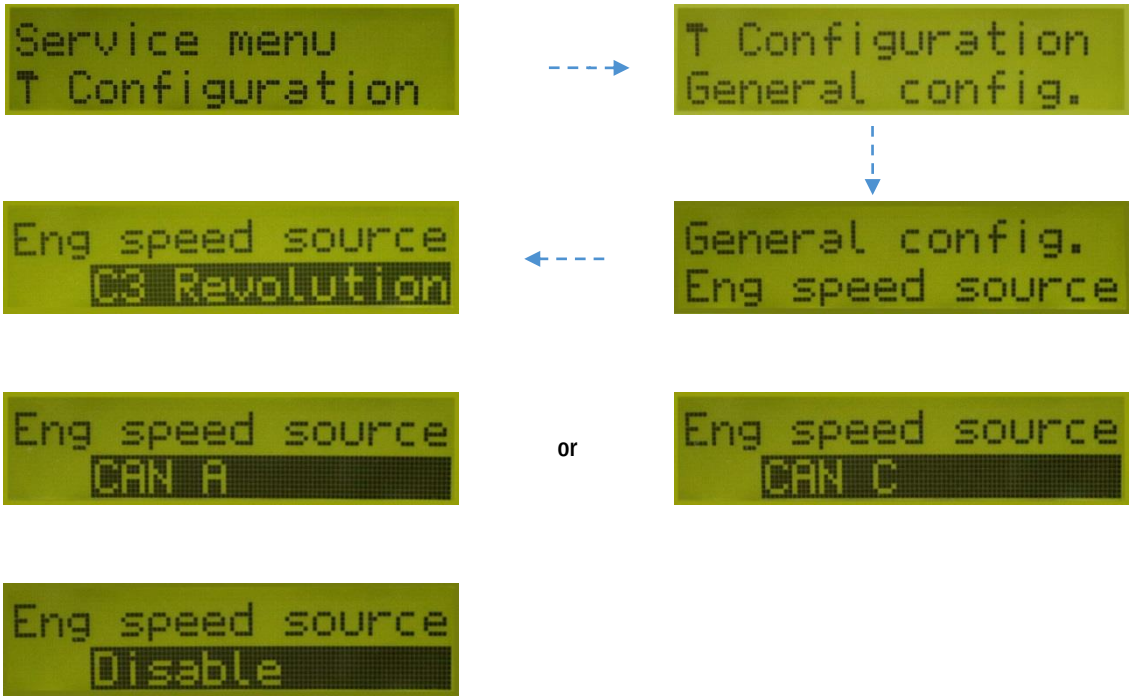
If speedometer factor is 1,0000, the D6 signal frequency is the same with the frequencies of B6 and B7 signals. If speedometer factor is 0,5000 then the D6 signal frequency becomes half of the B6 and B7 signals.

8.2.1.6 Engine Speed Source

For N-Profile records, displays and printouts of the STC-8255, the engine speed source should be selected. The engine speed source can be selected as C3 Revolution, CAN-A, CAN-C or totally disabled from this menu.

To adjust ***Eng speed source***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Eng speed source*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select the engine speed source that you desire by using the control ‹ / › keys. Engine speed source can be C3 revolution pin, CAN-A or CAN-C. Pressing **OK** will store the selection, STC-8255 returns one level back.



If the engine speed source is disabled then the N-profile registr, N-Profile and N-factor settings will take no effect.

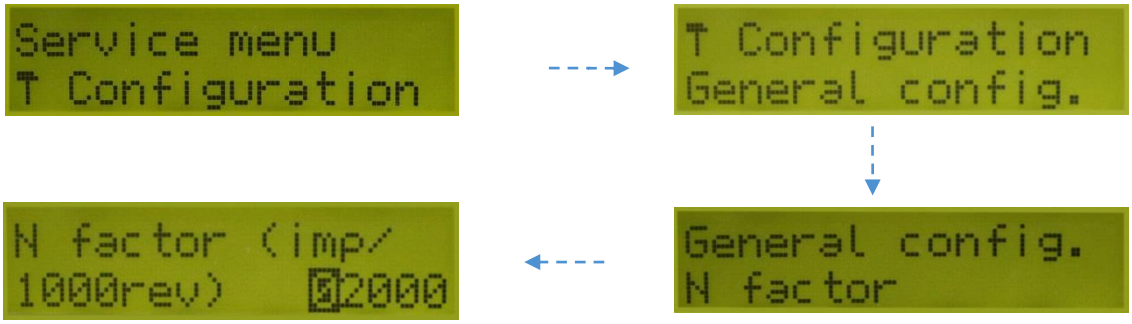
If the engine speed source is selected as CAN-A or CAN-C, STC-8255 will use the vehicle engine speed information on the CAN-Bus then the N-factor setting will take no effect.

8.2.1.7 N factor

STC-8255 N factor (rpm calibration factor) can be adjusted from this menu. STC-8255 uses this coefficient in order to calculate the motor revolution per minute (rpm) from C3 signal.

To adjust the ***N factor***:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***N factor*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the N factor that you desire by using the control ‹ / › keys. Pressing **OK** moves the cursor one by one. After completing the levels with **OK,** STC-8255 records this coefficient and returns one level back.

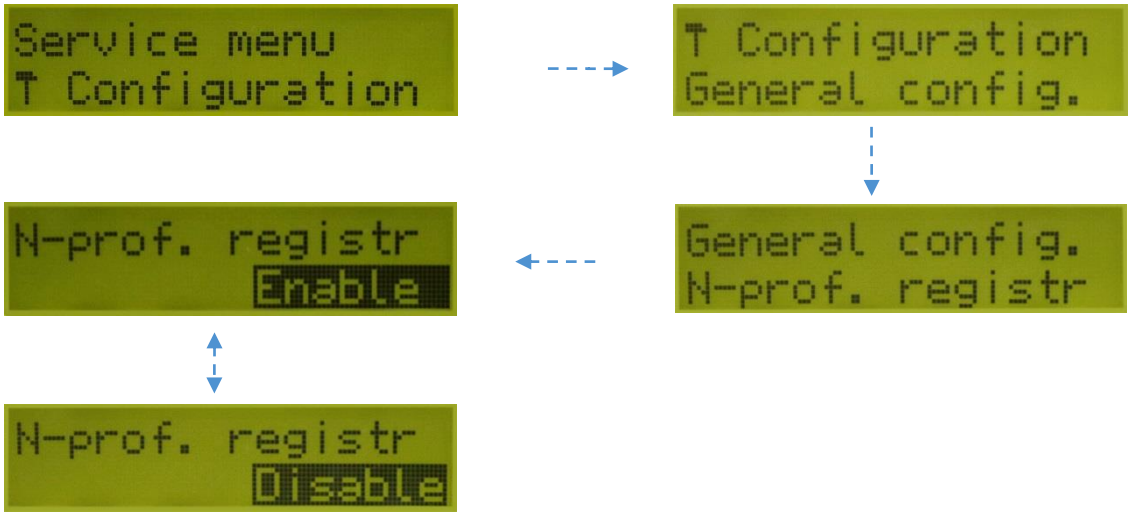


8.2.1.8 N-prof registr

STC-8255 N-profile (motor revolution (rpm) profile) registration can be adjusted from this menu. If N-profile data, which will be shown in the next chapter, is requested to be stored to the tachograph memory then N-prof. registr should be enabled from this menu.

To adjust the ***N-prof. registr***:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***N-prof. registr*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the setting by using the control ‹ / › keys. After confirmation with **OK,** STC-8255 returns one level back.

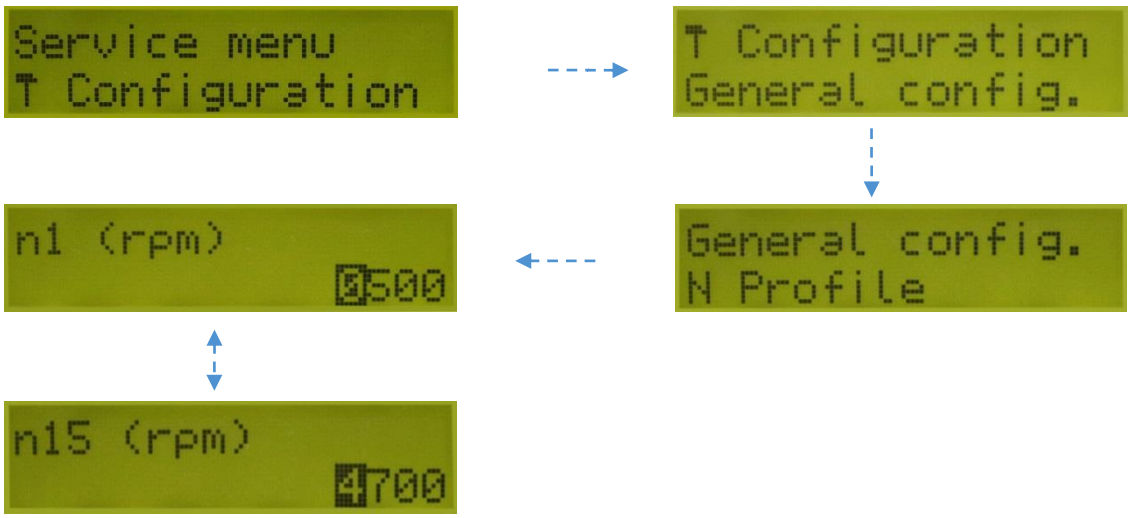


8.2.1.9 N-profile

STC-8255 N-profile (motor revolution (rpm) profile) levels can be adjusted from this menu. STC-8255 will automatically store, display or print the N-profiles (rpm levels) according to these set levels. STC-8255 records total elapsed time in minutes for each of these rpm levels. Please refer to the “section 10.7 N-Profile Printout” on page 110.

To adjust the ***N-profile*** levels:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***N-profile*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the N-profile (rpm) levels (from level n1 to n15) that you desire by using the control ‹ / › keys. Pressing **OK** moves the cursor one by one. After completing the levels with **OK**, STC-8255 returns one level back.

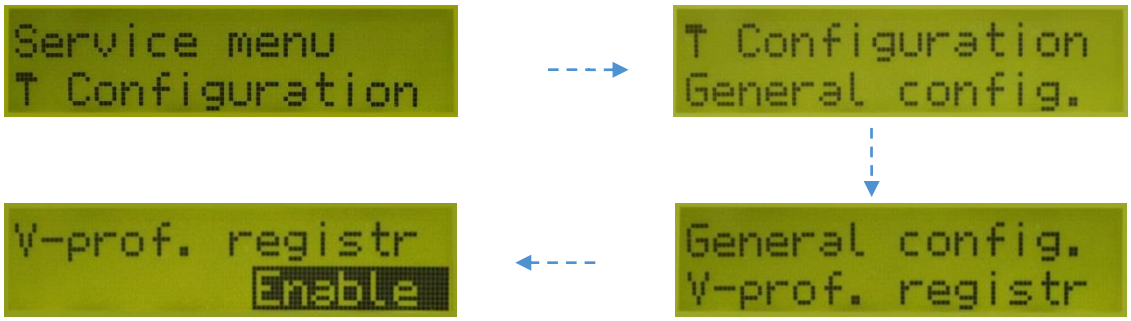


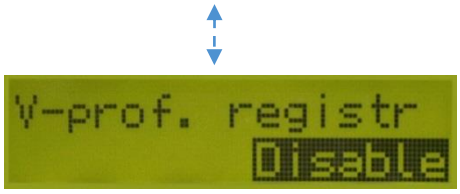
8.2.1.10 V-prof. registr

STC-8255 V-profile (speed profile) registration can be adjusted from this menu. If V-profile data, which will be shown in the next chapter, is requested to be stored to the tachograph memory then V-prof. registr should be enabled from this menu.

To adjust the ***V-prof. registr***:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***V-prof. registr*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the setting by using the control ‹ / › keys. After confirmation with **OK**, STC-8255 returns one level back.



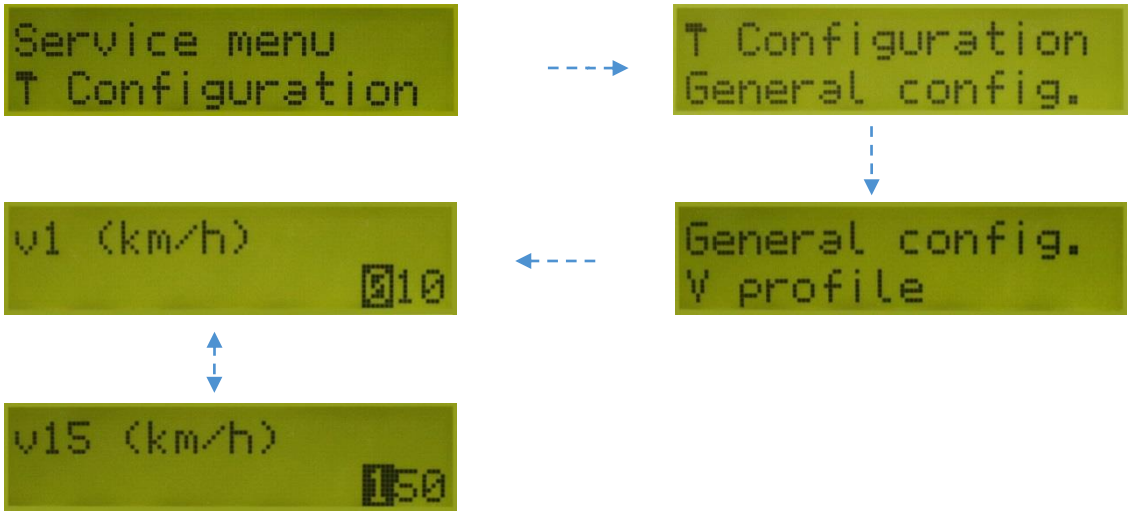


8.2.1.11 V-profile

STC-8255 V-profile (speed profile) levels can be adjusted from this menu. STC-8255 will automatically store, display or print the V-profiles (speed levels) according to these set levels. STC-8255 records total elapsed time in minutes for each of these speed levels. Please refer to the "section 10.8 V-Profile Printout" on page 111.

To adjust the ***V-profile*** levels:

1. Press **OK** to enter Main Menu. Select ***Service Menu Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***V-profile*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the V-profile (km/h) levels (from level v1 to v15) that you desire by using the control ‹ / › keys. Pressing **OK** moves the cursor one by one. After completing the levels with **OK,** STC-8255 returns one level back.



8.2.1.12 Ims Source

Independent motion source (IMS) such as GNSS (depends on the configuration) or CAN-A / CAN-C can be used to verify the speed of the primarily used motion sensor. This option enables the second motion source and the interface of the second source can be selected from this menu.

To adjust ***Ims source***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Ims source*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select the Ims source that you desire by using the control ‹ / › keys. Ims source can be GNSS, CAN-A or CAN-C. Pressing **OK** will store the selection, STC-8255 returns one level back.

If ***GNSS*** source is selected as an IMS source, then the GNSS module inside the STC-8255 (depending on the configuration) will be used.

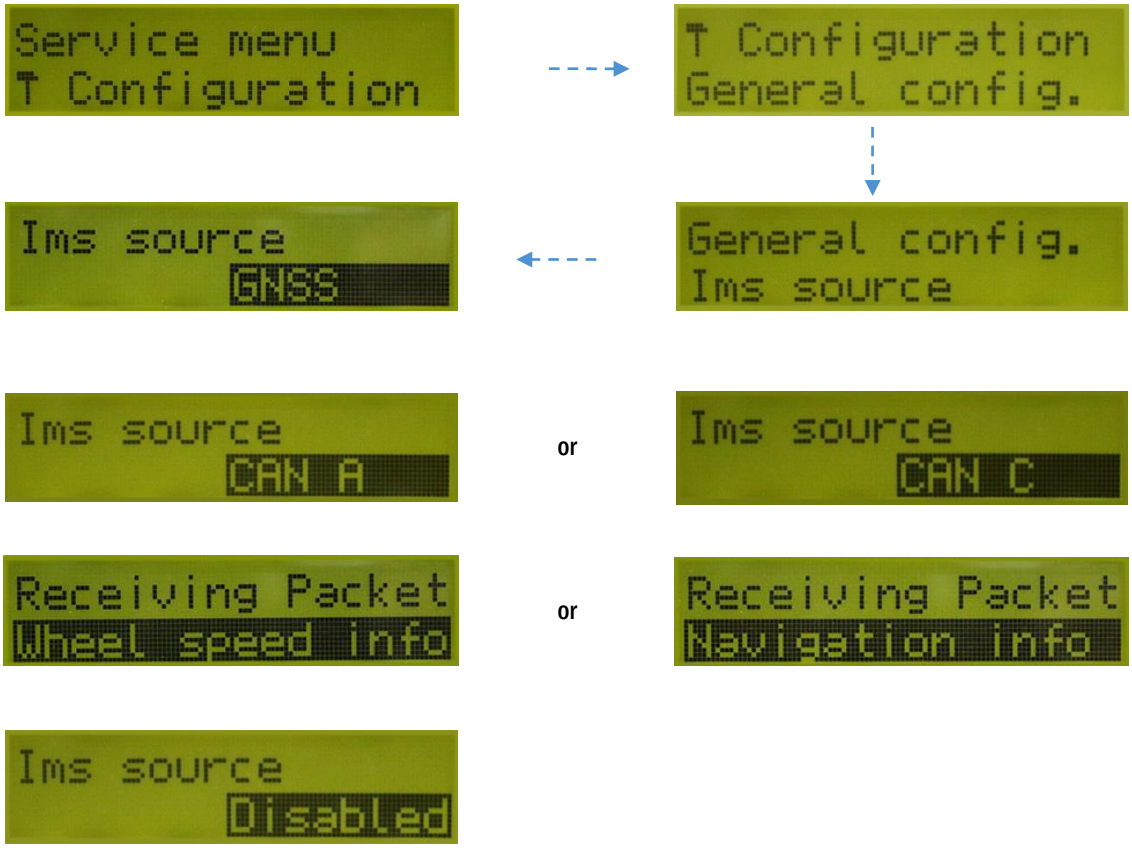
If ***CAN-A*** or ***CAN-C*** is selected, STC-8255 looks for the second source of motion from the CAN-A or CAN-C interfaces respectively. IMS CAN PGN should be entered from the **Receiving Packet** type screen if the Ims source is selected as CAN-A or CAN-C.

You can select either Wheel speed info (PGN 65215) or Navigation info (PGN 65256) as a receiving packet (IMS CAN PGN).

Wheel speed info (PGN 65215): The average speed of the two front wheels (that information is present in the vehicle which has a CAN-Bus network).

Navigation info (PGN 65256): Speed of the vehicle is calculated from a device such as a Global Navigation Satellite System (GNSS to CAN).

If it is disabled, no such (independent) source of motion will be used.



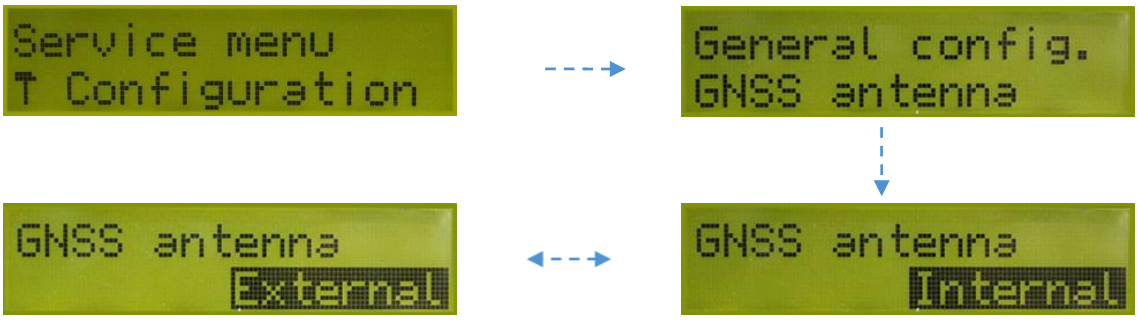
8.2.1.13 GNSS Antenna

If the IMS source of the STC-8255 is selected as GNSS, then depending on the configuration of STC-8255 internal or external antenna selection can be made from this menu. If the IMS source is chosen as CAN-A or CAN-C, then GNSS antenna selection has no effect on the speed verification function.

If the external antenna selection is made, be sure that the external antenna is connected to the Fakra-C connector at the back side of the STC-8255 and the antenna installation should be carefully performed in the vehicle.

To choose the ***GNSS antenna***, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***GNSS antenna*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select the antenna type that you desire by using the control ‹ / › keys. Pressing **OK** will store the selection, STC-8255 returns one level back.



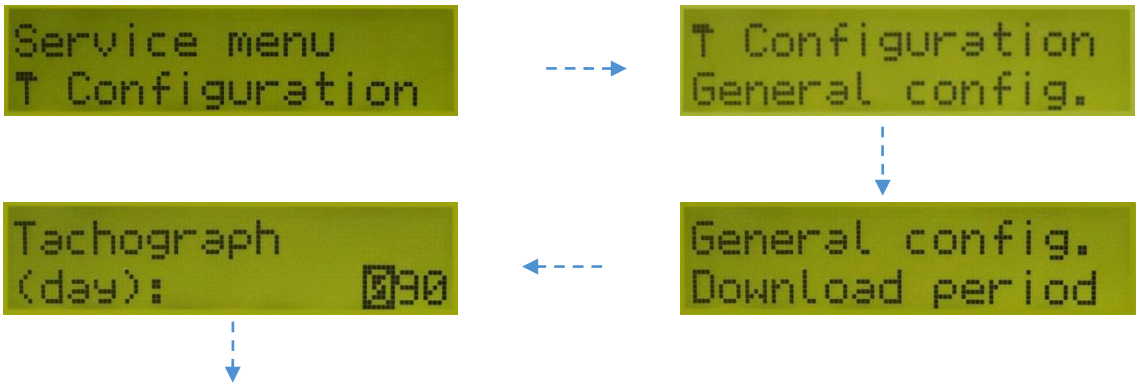
Note: The internal and external antenna selection depends on the hardware configuration of the STC-8255. Internal antenna performance characteristics vary from vehicle to vehicle. Please use external antenna if your internal antenna reception performance is poor. Please refer to **Vehicle menu>Display GNSS** for GNSS signal levels and number of satellites' signals received.

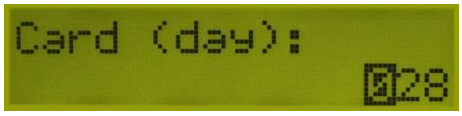


8.2.1.14 Download Period

Tachograph devices' mass memory has to be downloaded to the external tachograph data download devices at least in every 90 days for control and legal issues. Besides, the driver card has to be downloaded to external media at least in every 28 days. These compulsory tachograph and card download periods vary from each country's legal authorities. In this menu, tachograph's or cards' memory download periods can be entered. To enter the ***Download period*** menu, do as follows:

1. Press **OK** to enter Main Menu. Select **Service menu>Configuration** by using control ‹ / › keys. Press **OK** to enter the menu. Select **General Configuration** by using control ‹ / › keys. Press **OK**, then select **Download period** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Enter your country's legal tachograph and card download periods by using the control ‹ / › keys. Pressing **OK** moves cursor one by one. When entering finished the STC-8255 returns one level back.



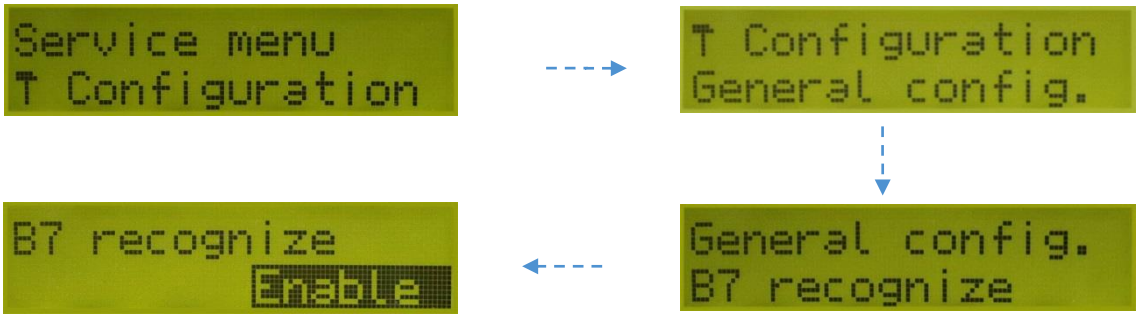


8.2.1.15 B7 Recognize

STC-8255 sends speed pulses from B7 pin on the B Connector (when the vehicle is moving) to speed pulse output recording units. STC-8255 can recognize the speed pulse output failures if this option is selected.

To enable the ***B7 Recognize***, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***B7 Recognize*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** it and press **OK** to confirm.



If B7 Recognize is enabled and there is a failure at the B7 speed pulses then the DTC code (Speed pulse output recording unit - B7) will be generated.

Please refer to “section-9.10 Diagnostic Trouble Codes (DTCs) ”.

8.2.1.16 Status D1-D2

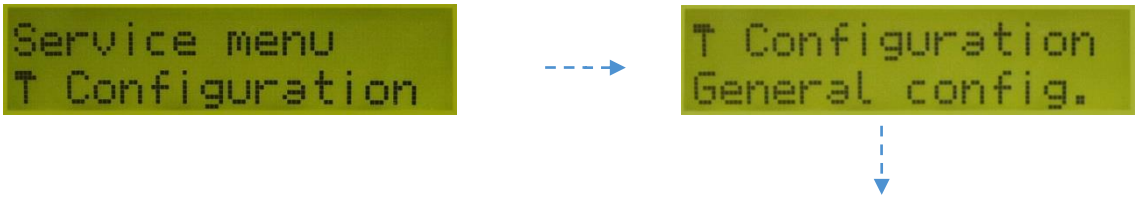
Depending on the device configuration, STC-8255 has an option to record the status of the input pins D1 and D2 to the tachograph memory. In this concept, only the input pins D1 and D2 status changes are recorded, displayed or printed from the tachograph menus.

Please refer to the related display and print menus of the status D1 & D2 on the operational manual. In addition, refer to the “section 10.9 Status D1-D2 Printout” on page 112.

In this menu, one can enable or disable the status input pins D1 and D2 in order to record, display and print the status changes to the tachograph mass memory.

To enable the ***Status D1-D2***, perform as follows:

1. Press OK to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Status D1-D2*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the status inputs D1 and D2 respectively to the desired state and press **OK** to confirm. STC-8255 returns one level back in the menu.



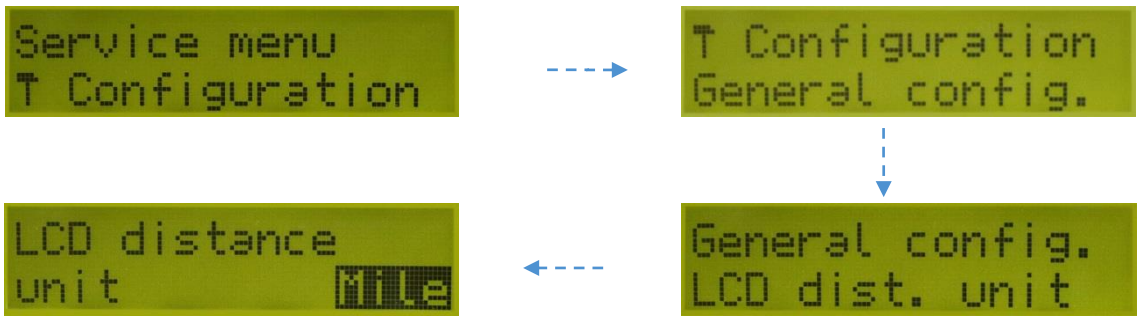


8.2.1.17 LCD Distance Unit

The unit of the distance, which appears on the display of the STC-8255, can be adjusted from this menu. The speed and the odometer values are affected when the unit is changed from km/h to mph.

To adjust the ***LCD distance unit*** to ***mile*** or ***km,*** do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***LCD Distance Unit*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the distance unit that you desire to change.



If the LCD Distance unit is changed to **Mile** then the default screen will be seen as below.



If the LCD Distance unit is changed to **km** then the default screen will be seen as below.

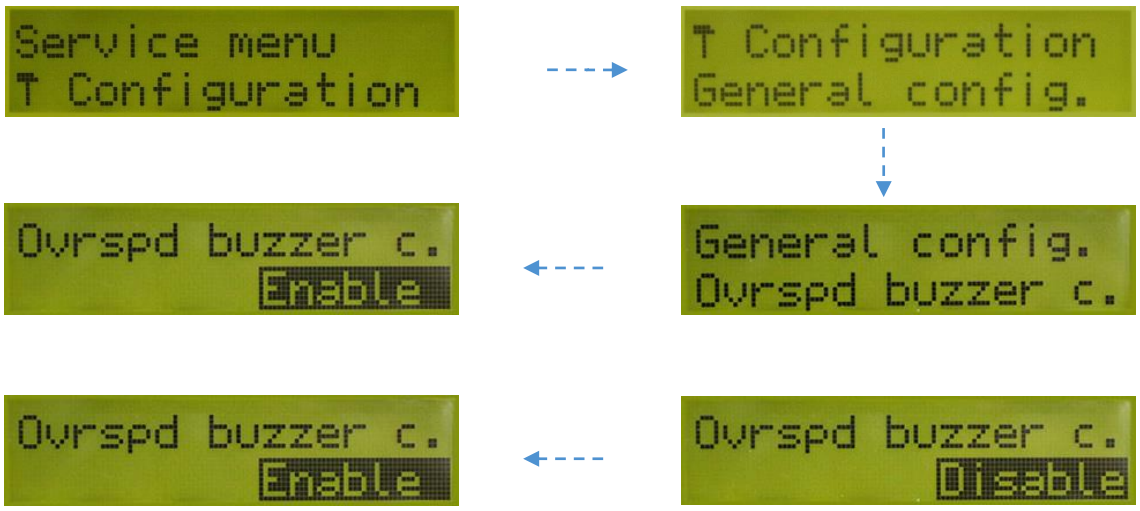


8.2.1.18 Overspeed buzzer control

After 1 minute of driving at a speed higher than the adjusted authorized speed limit, digital tachographs record this breach as an overspeed event. This event is controlled by the police officers in the road checks and if there is a breach, the driver or the company can be punished with a traffic fine. STC-8255 can warn the driver with an audible (buzzer) sound if over-speed event is recorded to the tachograph.

To enable or disable the ***Ovrspd buzzer c.***, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select **Ovrspd buzzer c.** menu by using the control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the buzzer alert to the desired state and press **OK** to confirm. STC-8255 returns one level back in the menu.

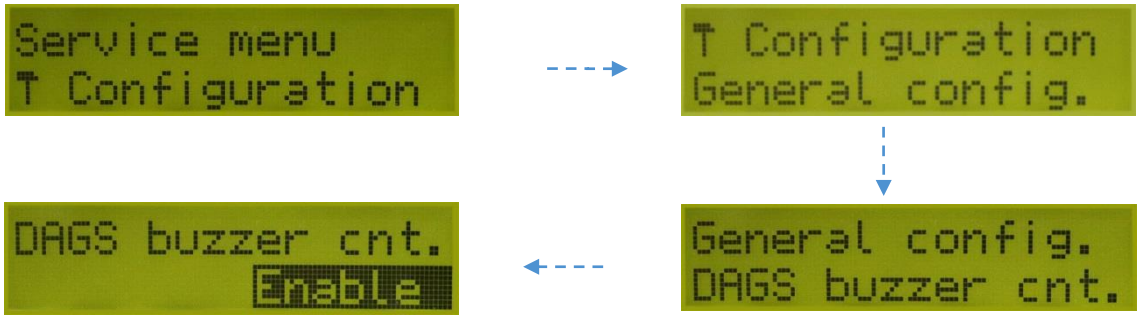


8.2.1.19 DAGS Buzzer Control

Driver Activity Guidance System (DAGS) always observes the driver activities (continues driving times (4h30m), daily, weekly, fortnightly driving and rest times) and warns the driver with an audible alert (buzzer) for giving a break for the sake of driver not to violate the working and resting hours of the social legislative (EC No: 561/2006).

To enable or disable the ***DAGS buzzer cnt.***, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select **DAGS buzzer cnt.** menu by using the control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the buzzer alert to the desired state and press **OK** to confirm. STC-8255 returns one level back in the menu.

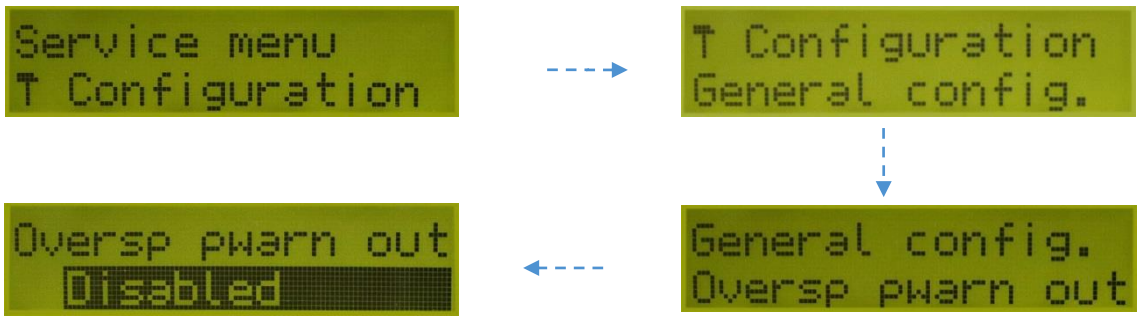


8.2.1.20 Overspeed Pre-warning Out

Overspeed pre-warning out can be configured from this menu. Pre-warning output can be selected as buzzer, display, warning out and all or can be disabled.

To open the ***Overspeed Pre-warning Out*** *menu*, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Oversp pwarn out*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select the desired pre-warning out selection then press **OK** to confirm.



If overspeed pre-warning out is selected as **Display**, then overspeed pre-warning message will be shown only on the display.

If overspeed pre-warning out is selected as **Buzzer**, then overspeed pre-warning message will be heard only from the buzzer.

If overspeed pre-warning out is selected as **Warning** output, then overspeed pre-warning output (D4 Pin) will be active.

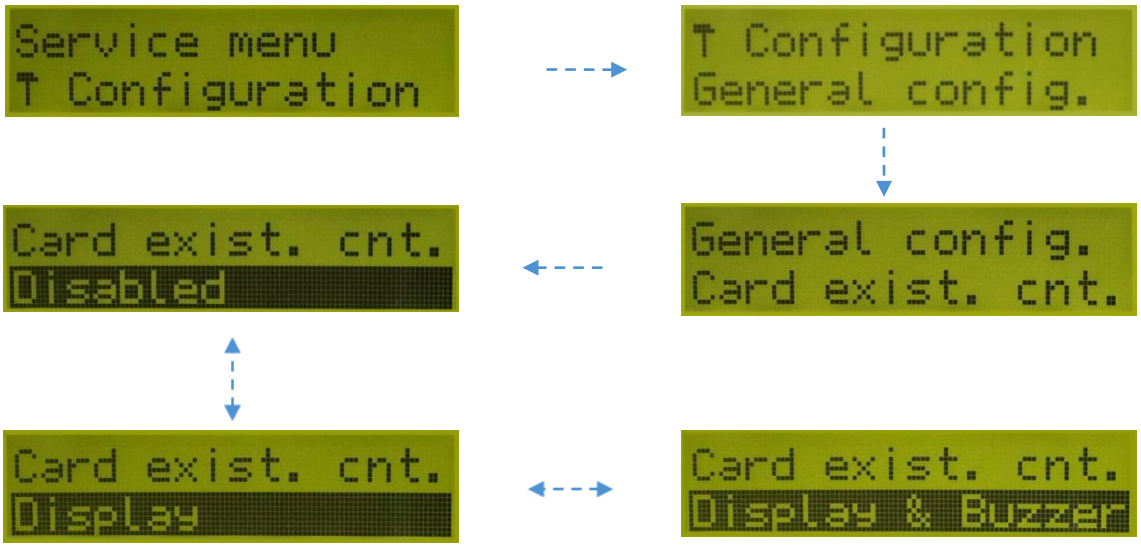
If overspeed pre-warning out is selected as **All** or **Disabled**, then all of the outputs (display, buzzer and warning output) will be enabled or disabled respectively.

8.2.1.21 Card existence control

STC-8255 warns the drivers not to forget to insert or withdraw of their cards to/from the vehicle unit in order not to create "Driving without valid card" event. Therefore, if the valid driver card is absent in the first slot of the tachograph, (or if there is a different card type rather than the driver card in the second slot) the STC-8255 warns the driver that the current situation is not suitable for driving. By the help of this feature, the driver is reminded to insert his/her valid driver card to the first slot before he/she starts his/her trip. From this STC-8255 menu, the warning type can be adjusted.

To enable or disable the ***Card exist. cnt.***, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select **Card exist. cnt.** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Adjust the alert type to the desired state and press **OK** to confirm. STC-8255 returns one level back in the menu.



If "***Disabled***" option is adjusted, then the STC-8255 will not warn the driver before his/her trip starts.

If "***Display***" option is adjusted, then the STC-8255 only warns the driver by the help of the display.

If "***Display & Buzzer***" option is adjusted, then the STC-8255 warns the driver by the help of the display and give an audible (buzzer) alert at the same time.

8.2.1.22 Pre-warning

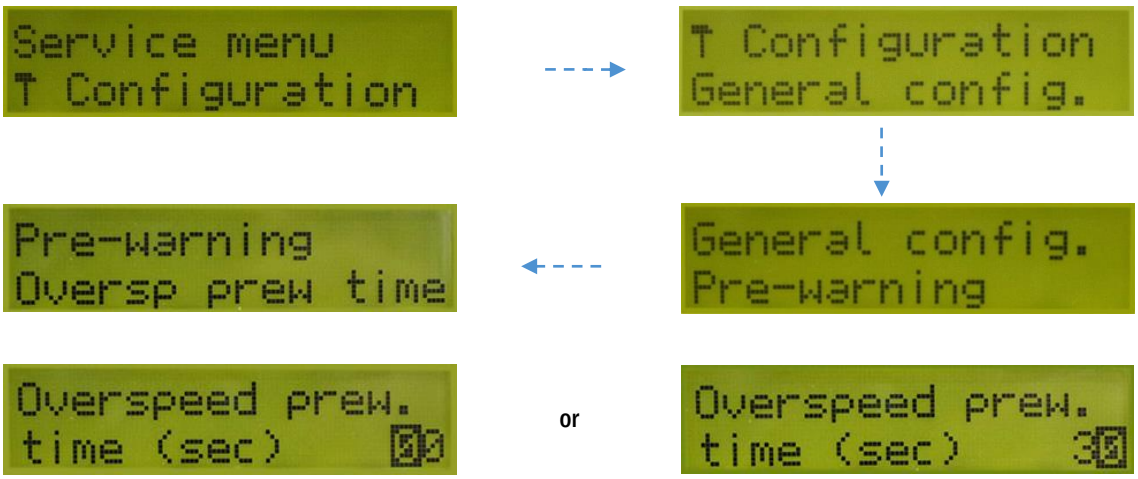
STC-8255 is able to warn its users before overspeed event occurs, before calibration due date comes or tachograph cards become expired. In order to get these Expiration Date Warning messages, workshop staff should configure the following dates in the sub-section of this menu.

8.2.1.22.1 Overspeed Pre-warning Time

After 1 minute of driving at a speed higher than the adjusted authorized speed limit, digital tachographs record this breach as an event. This event is controlled by the police officers in the road checks and if there is a breach the driver or the company can be punished with a traffic fine. To prevent this, STC-8255 can warn the driver before (pre-warning time) overspeed event is recorded the tachograph.

To open the ***Overspeed Pre-warning Time*** *menu*, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Overspeed Pre-warning Time*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select the desired pre-warning time value (0-60 seconds) then press **OK** to confirm.

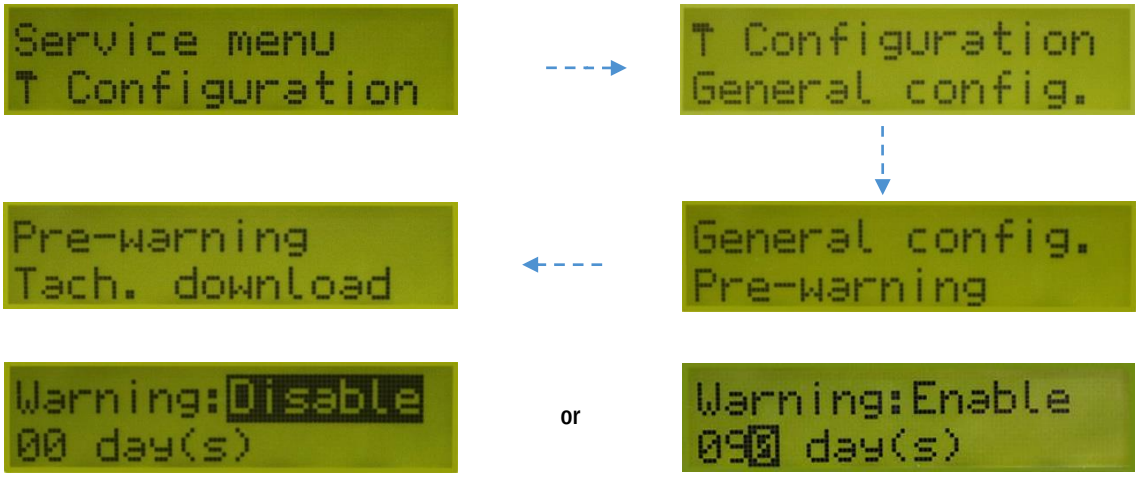


8.2.1.22.2 Tachograph Download

Tachograph devices' mass memory has to be downloaded to the external tachograph data download devices at least in every 90 days for control and legal issues. This compulsory tachograph download period varies from each country's legal authorities. By the help of this menu, a tachograph download pre-warning can be set before the due date of the download period comes.

To open the ***Tach. download Pre-warning*** menu, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Pre-warning*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Tach. download*** menu then press **OK** to enter.



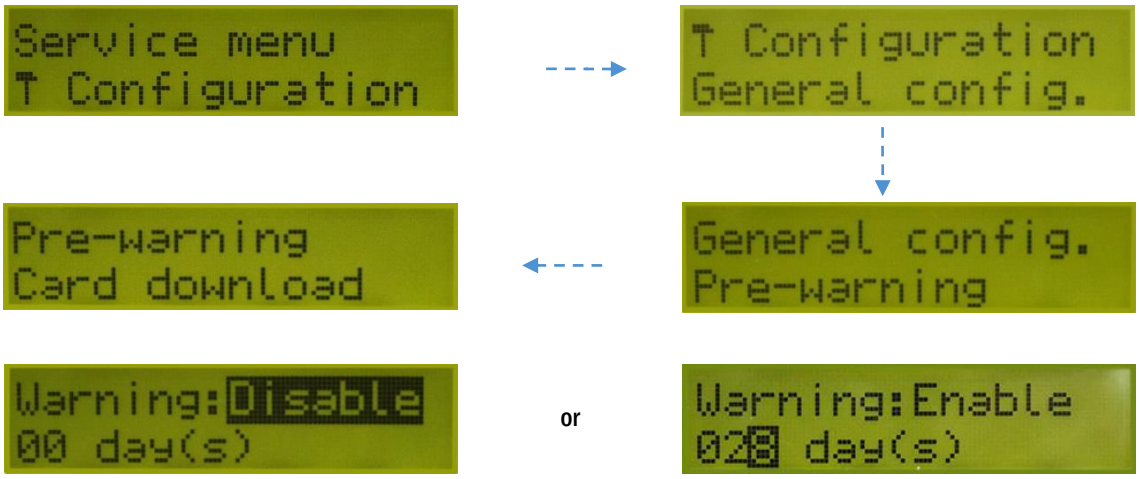
After entering the menu, you can either disable the warning messages or you can enable the warning messages up to 90 days (max. allowable time is equal to download period) before the tachograph download period due date.

8.2.1.22.3 Card Download

Tachograph driver cards have to be downloaded to the external data download devices at least in every 28 days for control and legal issues. This compulsory driver card's download period varies from each countries' legal authorities. By the help of this menu, a card download pre-warning can be set before the due date of the card download period comes.

To open the ***Card download Pre-warning*** menu, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Pre-warning*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Card download*** menu then press **OK** to enter.

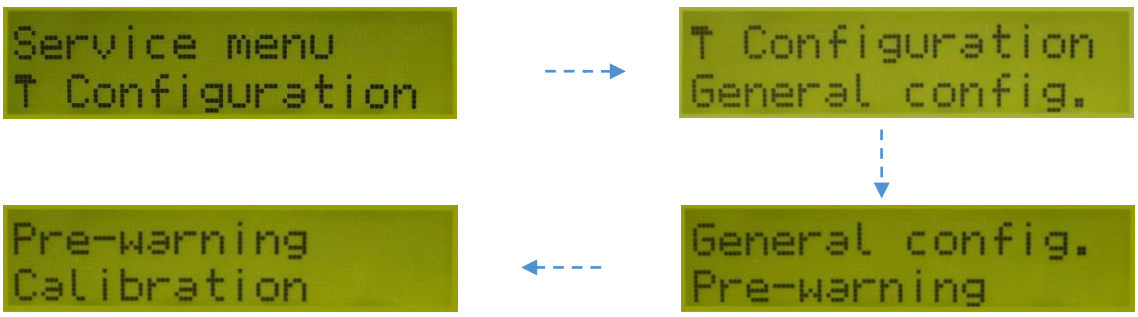


After entering the menu, you can either disable the warning messages or you can enable the warning messages up to 28 days before the card download period due date.

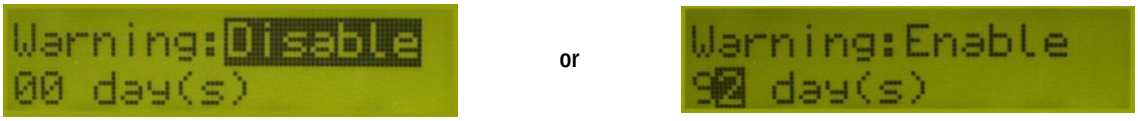
8.2.1.22.4 Calibration due Date Warning

To open the ***Warning Expiration Date Menu*** of Calibration, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Pre-warning*** menu by using the control ‹ / › keys. Press **OK**, then select ***Calibration*** menu by using the control ‹ / › keys.



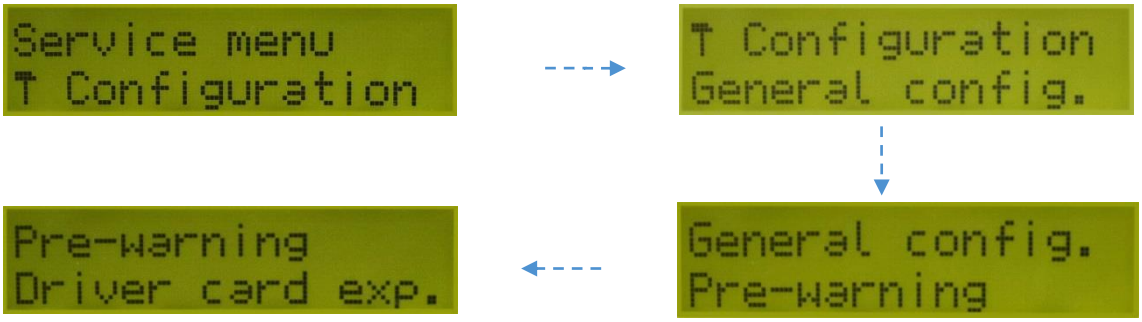
After entering the menu, you can either disable the expiration warning messages or you can enable the warning messages up to 250 days before the calibration due date.



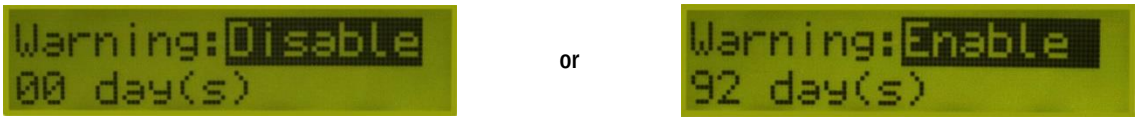
8.2.1.22.5 Driver Card Expiration Date Warning

To open the **Driver card** ***exp.*** date pre-warning menu, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ◄ / ► keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ◄ / ► keys. Press **OK**, then select **Pre-w*arning*** menu by using the control ◄ / ► keys. Press **OK** to enter the menu. Select ***Driver card exp.*** menu then press **OK** to enter.



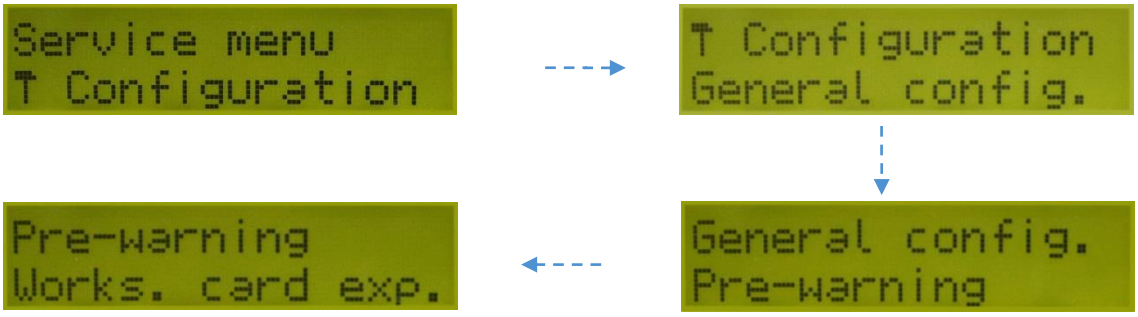
After entering the menu, you can either disable the expiration warning messages or you can enable the warning messages up to 250 days before the driver card's expiration date.



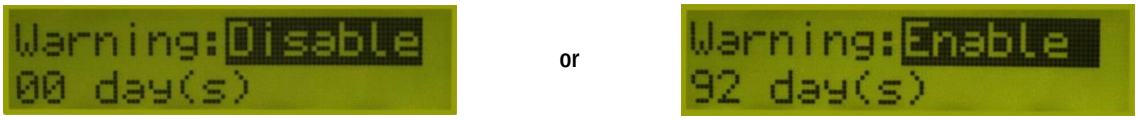
8.2.1.22.6 Workshop Card Expiration Date Warning

To open the **Works. card** ***exp.*** date pre-warning menu, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ◄ / ► keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ◄ / ► keys. Press **OK**, then select **Pre-w*arning*** menu by using the control ◄ / ► keys. Press **OK** to enter the menu. Select ***Works. card exp.*** menu then press **OK** to enter.



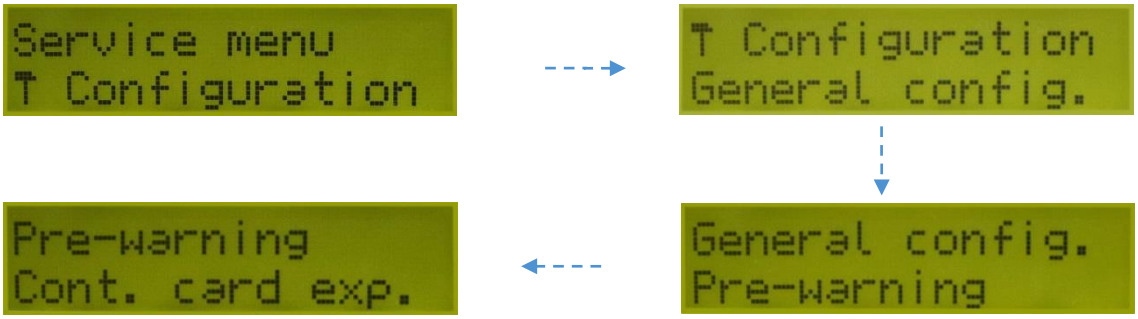
After entering the menu, you can either disable the expiration warning messages or you can enable the warning messages up to 250 days before the workshop card's expiration date.



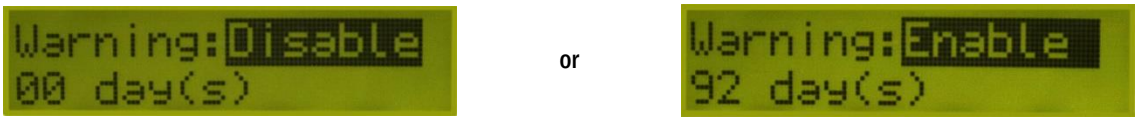
8.2.1.22.7 Control Card Expiration Date Warning

To open the **Cont. card *exp.*** date pre-warning menu, do as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Pre-warning*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Cont. card exp.*** menu then press **OK** to enter.



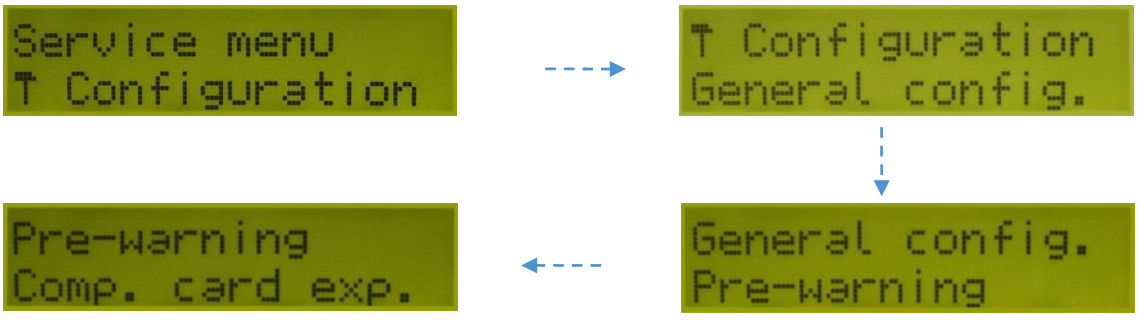
After entering the menu, you can either disable the expiration warning messages or you can enable the warning messages up to 250 days before the control card's expiration date.



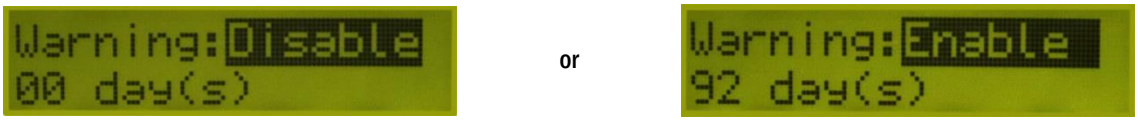
8.2.1.22.8 Company Card Expiration Date Warning

To open the **Comp. card *exp.*** date pre-warning menu, do as follows:

Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***General Configuration*** by using control ‹ / › keys. Press **OK**, then select ***Pre-warning*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Comp. card exp.*** menu then press **OK** to enter.



After entering the menu, you can either disable the expiration warning messages or you can enable the warning messages up to 250 days before the company card's expiration date.



8.2.2 CAN Configurations

STC-8255 has two configurable CAN Interfaces (CAN-A and CAN-C). These CAN interfaces are used to connect the STC-8255 to vehicle's CAN-Bus or separate CAN-Bus devices such as FMS (Fleet Management Systems) or remote download units. CAN interfaces are located (in A and C connectors) at the backside of the STC-8255. CAN parameters are vehicle specific parameters and may differ from vehicle to vehicle. Therefore, these CAN settings should be configured according to the vehicle's CAN configuration specification document (will be given separately to the workshops by vehicle manufacturers or provided by ASELSAN Inc.).

8.2.2.1 CAN-A Configurations

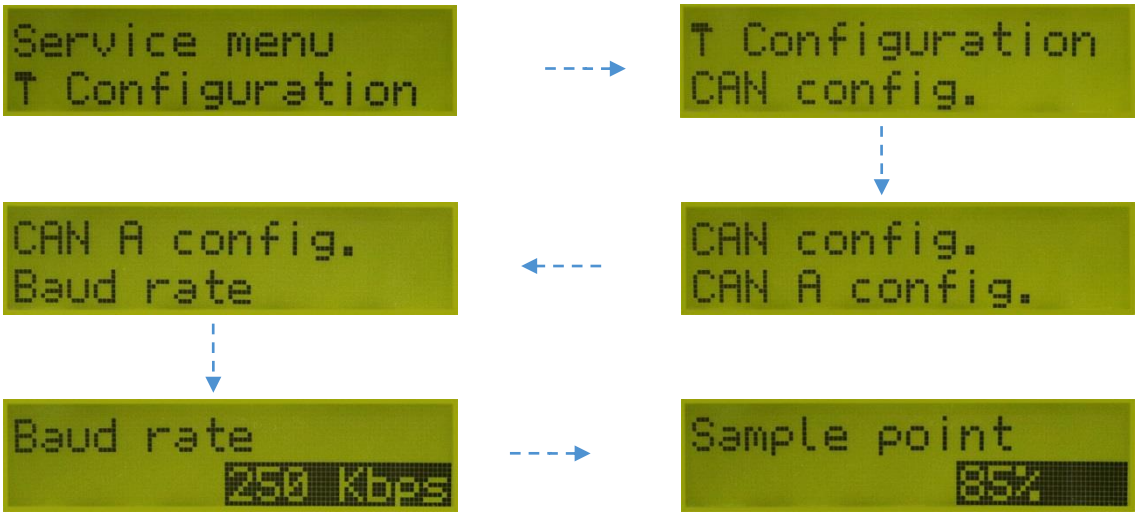
CAN-A configurations can be done from this menu. Please read the "Vehicle specific CAN parameter documentation" (will be provided later) for further information.

CAN-A configuration parameters are ***Baud rate, Sample*** **point** and ***Synchronous jump*** and will be explained in the following sections.

8.2.2.1.1 Baud Rate

To open the ***CAN-A Configurations>Baud Rate*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN Configuration*** by using control ‹ / › keys. Press **OK**, then select ***CAN-A Configurations*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Baud Rate*** and press **OK** to enter the sub menu. Select the desired baud rate value then press **OK** to confirm. After confirmation ***Sample Point*** should be selected by using control ‹ / › keys and press **OK** to confirm.



Baud rate can be selected as 125Kbps, 250 Kbps, 500Kbps, 1 Mbps or it can be disabled from this menu.

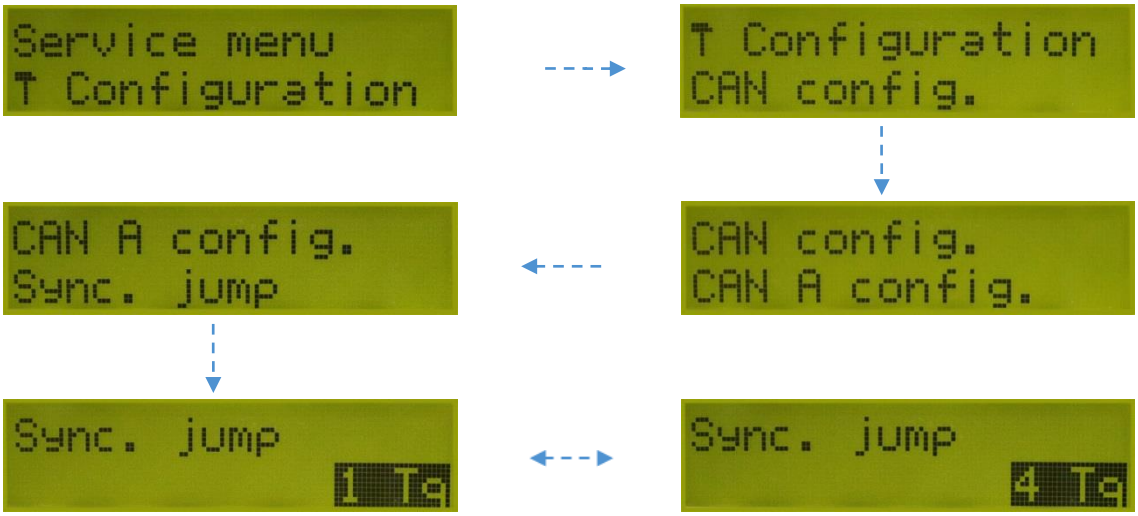
If the baud rate is disabled in the above steps, then the following sub-menus (sample point and synchronous jump menus) will not be activated.

8.2.2.1.2 Synchronous Jump

To open the ***CAN-A Configurations> Synchronous*** Jump menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN Configuration*** by using control ‹ / › keys. Press **OK**, then select ***CAN-A Configurations*** menu by using the control ‹ / › keys. Press **OK** to enter the menu.

Select ***Synchronous Jump*** and press **OK** to enter the sub menu. Select the desired sample rate (1 Tq to 4 Tq) then press **OK** to confirm.



8.2.2.2 CAN-C Configurations

CAN-C configurations can be done from this menu. Please read the "Vehicle specific CAN parameter documentation" (will be provided later) for further information. CAN-C configurations are very similar to the CAN-A configurations and will not be explained again.

→Please refer to "section 8.2.2.1 CAN-A Configurations".

8.2.2.3 CAN Services

Some of the services related to CAN-A or CAN-C can be configured from the sub-menus of this menu.

If enabled, some parameters or parameter groups (TCO1 – PGN 65132) will be transmitted cyclically by the STC-8255. TCO1 has 8-byte message format structure and includes working states, driver 1 states, driver 2 states, tachograph status, tachograph output shaft speed and tachograph vehicle speed parameters. These parameters will be configured in the following sub-sections.

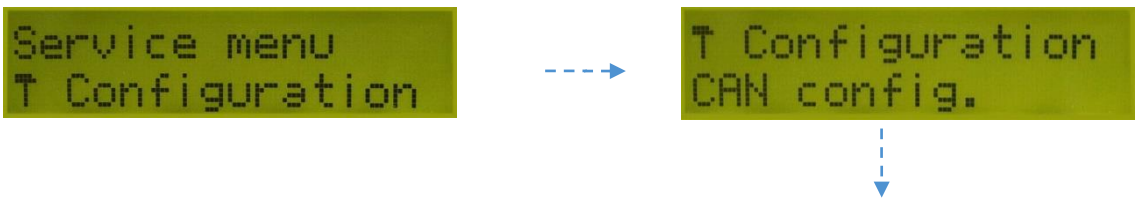
→Please refer to SAE-J 1939-71 standard for further information on TCO1 and other PGN's.

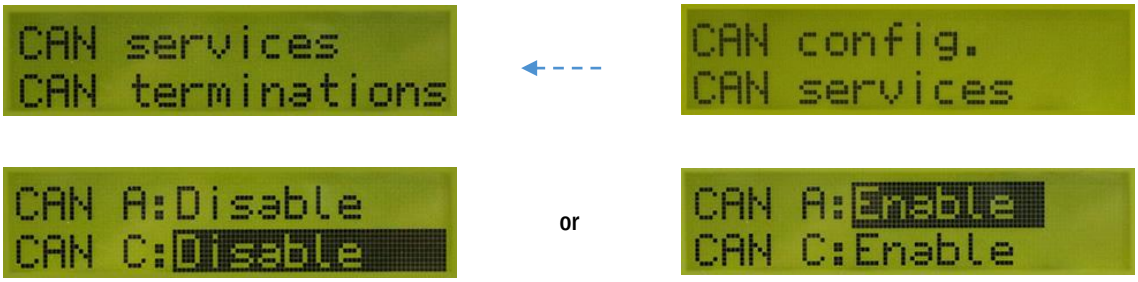
8.2.2.3.1 CAN Terminations

Depending on the hardware configuration of STC-8255, CAN termination resistances can be adjusted from this menu. The termination resistance of the CAN-A and/or CAN-C can be enabled or disabled independently.

To open the ***CAN Services > CAN terminations*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN Configuration*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN terminations*** and press ***OK*** to enter the sub menu. Select the desired termination resistance (***Enable*** or ***Disable***) on each CAN-A or CAN-C separately then press **OK** to confirm.





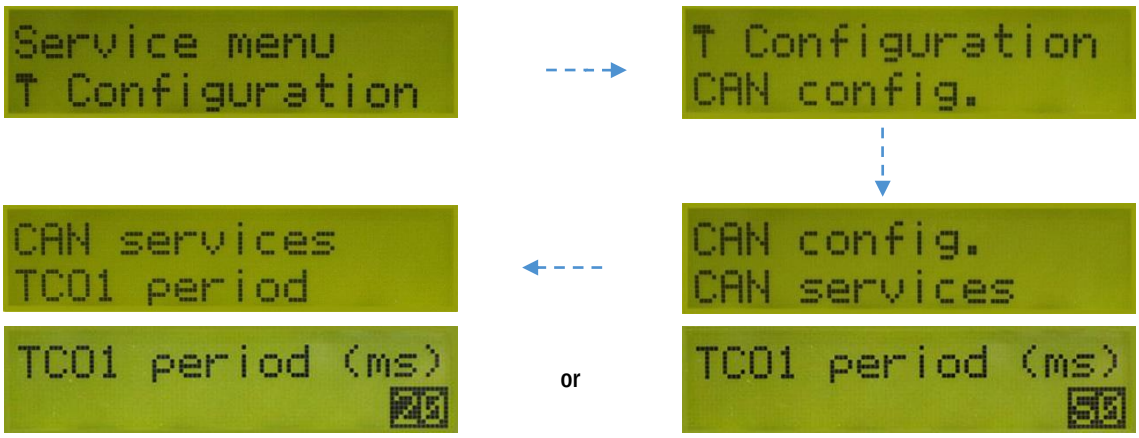
If you enable the CAN termination resistance, you will measure 120 Ohm resistance between CAN-high and CAN-low pins.

8.2.2.3.2 TCO1 Period

TCO1 messages are periodic messages and it is sent from the STC-8255 cyclically. The period of this message can be set 20ms or 50ms from this menu.

To open the ***CAN Services > TCO1 Period*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***TCO1 period*** and press **OK** to enter the sub menu. Select the desired period (***20 ms*** or ***50 ms***) then press **OK** to confirm.

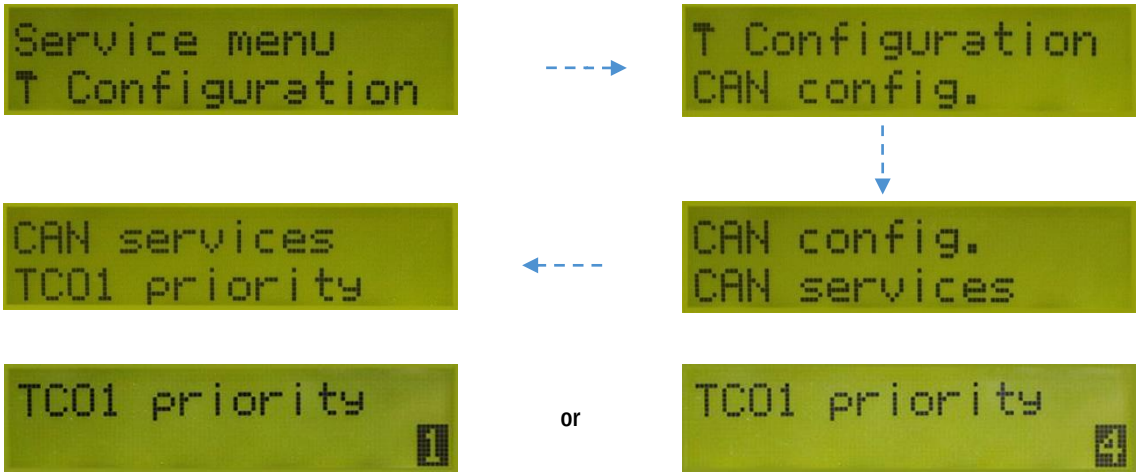


8.2.2.3.3 TCO1 Priority

TCO1 message priority in the vehicle CAN-Bus can be set in this menu. TCO1 priority indicates the importance of the messages in the Vehicle Can network. The lower values mean the priority is higher. The default TCO1 priority should be set to 3.

To open the ***CAN Services > TCO1 Priority*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***TCO1 priority*** and press ***OK*** to enter the sub menu. Select the desired value (***0*** to ***7***) then press **OK** to confirm.



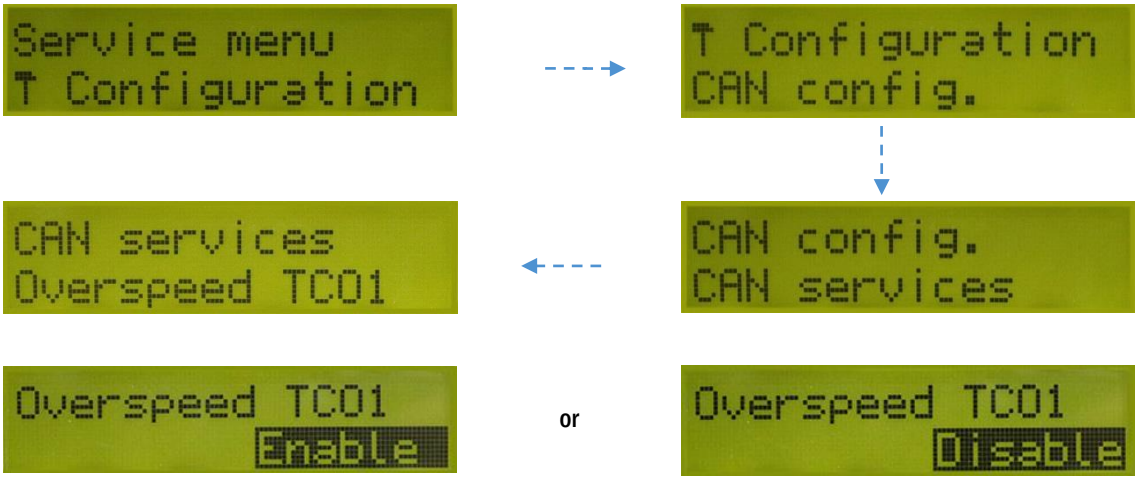
Remark: Do not set the priority value to a lower value if you are not sure what to do.

8.2.2.3.4 Overspeed TCO1

Overspeed information is transmitted within the TCO1 packets. This information indicates whether the vehicle is exceeding the legal speed limit set in the tachograph.

To open the ***CAN Services > Overspeed TCO1*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Overspeed TCO1*** and press **OK** to enter the sub menu. Select the desired function (***Enable*** or ***Disable***) then press **OK** to confirm.



If it is enabled, overspeed information will be sent within the TCO1 packets otherwise it will not be sent.

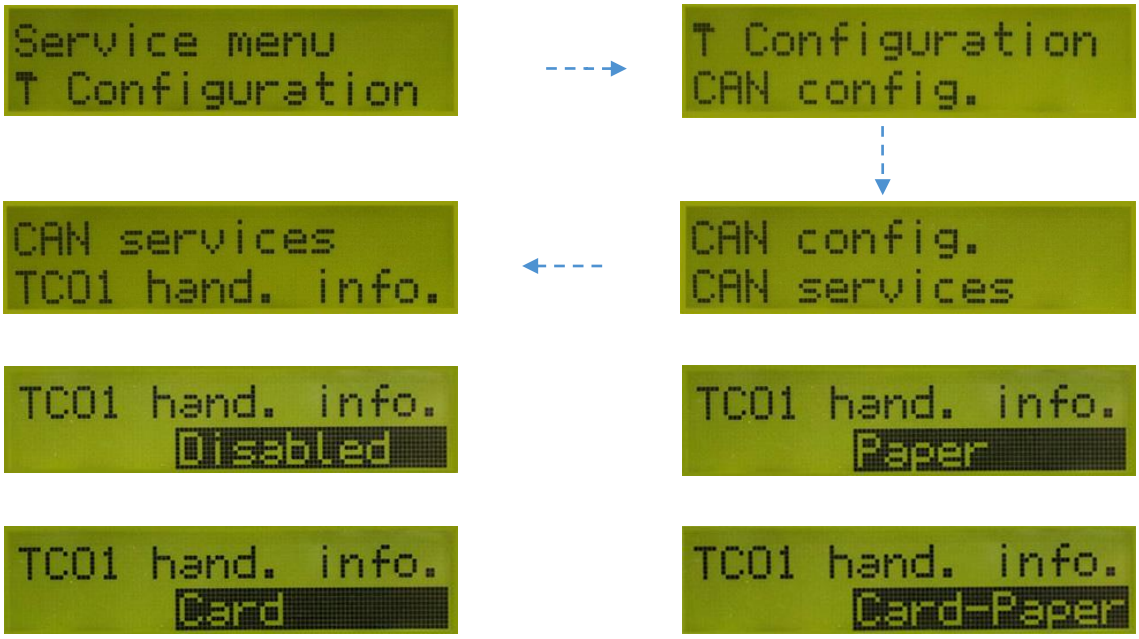
8.2.2.3.5 TCO1 Handling Info

TCO1 Handling info indicates that handling information is present. Information could include “no printer paper”, “no driver card”. TCO1 handling info can be set in this menu.

To open the ***CAN Services > TCO1 hand. info*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***TCO1***

hand. info and press ***OK*** to enter the sub menu. Select the desired function (***Disabled, Paper, Card or Paper-Card***) then press **OK** to confirm.



If it is ***disabled***, then no handling info is included into the TCO1 messages.

If ***paper*** is chosen, then only when the STC-8255 is paper out the handling information is included in the TCO1 messages.

If ***card*** is chosen, then only when there is no personal card (driver or workshop) inserted into slot 1 of the STC-8255, the handling information is included in the TCO1 messages.

If ***paper-card*** is chosen, then any of the circumstances happen (when there is no personal card (driver or workshop) inserted into slot 1 or paper out), the handling information is included in the TCO1 messages.

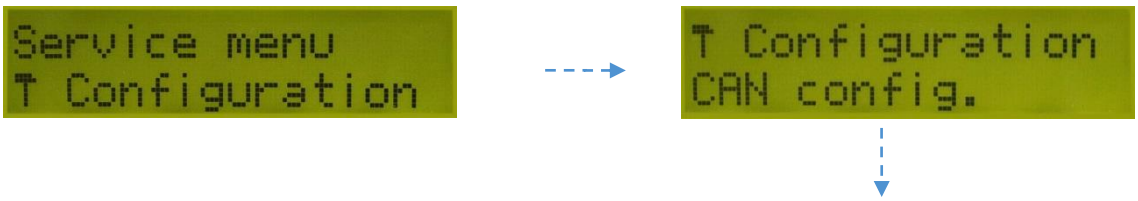
This handling information could flag some alert led(s) on the instrument cluster of the vehicle.

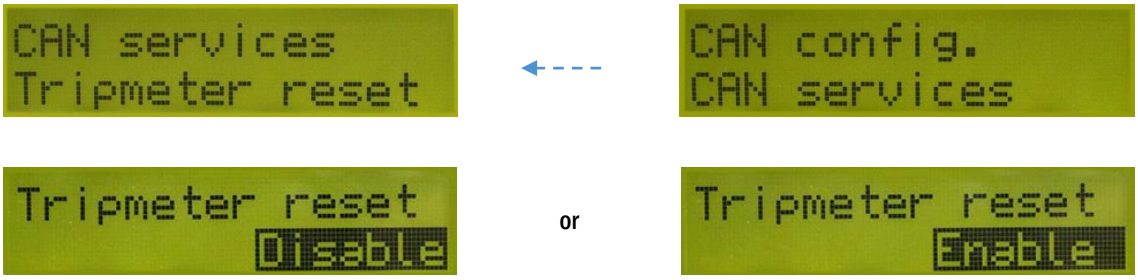
8.2.2.3.6 Tripmeter Reset

The STC-8255 will receive trip-meter reset message to reset the trip distance. Trip-meter reset option is enabled or disabled in this menu.

To open the ***CAN Services > Tripmeter Reset*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Tripmeter Reset*** and press ***OK*** to enter the sub menu. Select the desired function (***Enable*** or ***Disable***) then press **OK** to confirm.



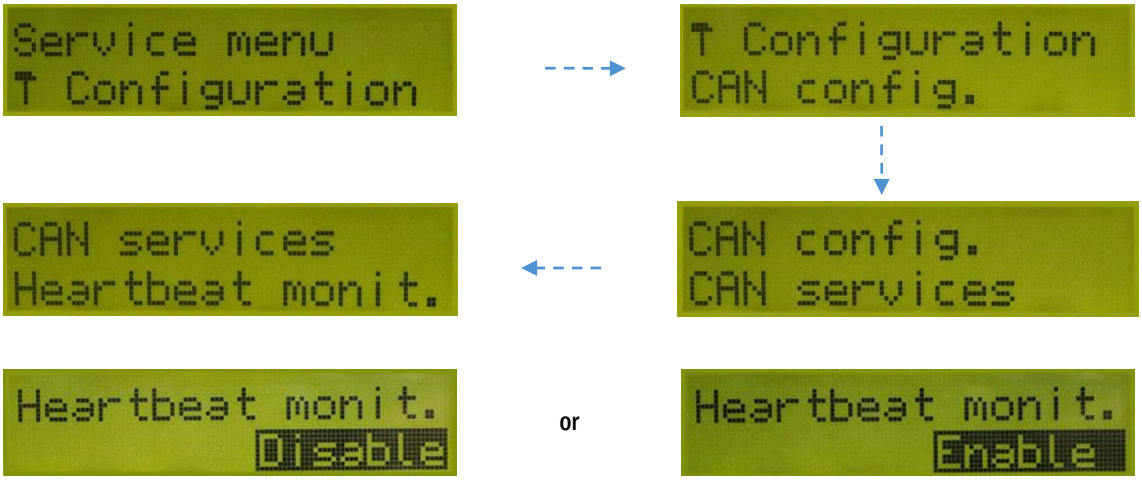


8.2.2.3.7 Heartbeat Monitoring

Heartbeat monitoring message transmitted from the STC-8255 is used for checking the connection between the STC-8255 and the instrument cluster of the vehicle.

To open the ***CAN Services >*** **Hearthbeat Monitoring** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Hearthbeat monit.*** and press ***OK*** to enter the sub menu. Select the desired function (***Enable*** or ***Disable***) then press **OK** to confirm.



If it is enabled, the heartbeat monitoring message is transmitted to the instrument cluster otherwise it is not sent.

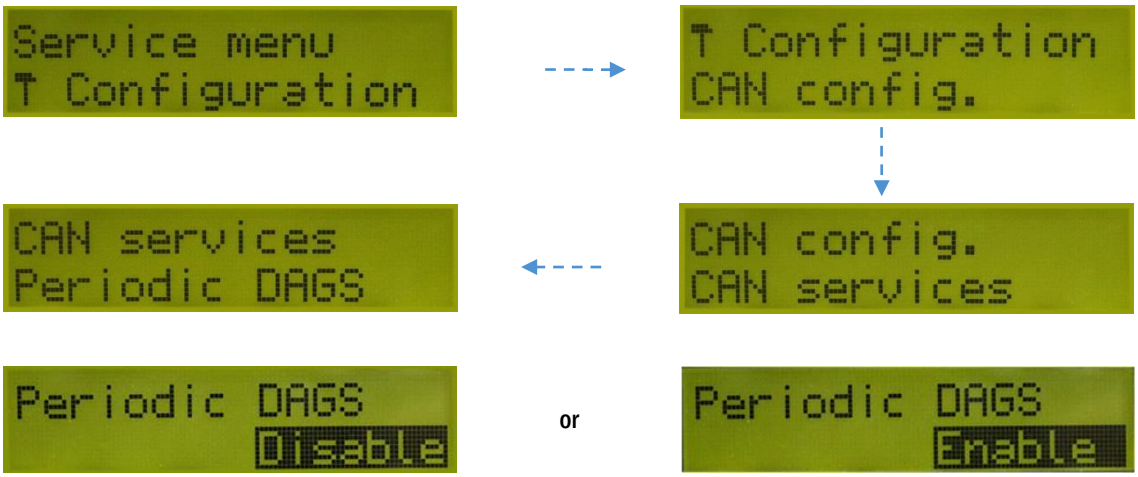
8.2.2.3.8 Periodic DAGS

STC-8255 sends periodical DAGS (UDS) messages related to the driver's activities (current driving remaining time, driver-1 remaining time until next break, drive-1 duration of next break rest, driver-1 remaining time of current break rest, driver-1 time left until next driving period, driver current daily driver time, driver-1 time left until new daily rest period, driver-1 minimum daily rest, driver-1 remaining time of current week, driver-1 time left until new weekly rest period, driver-1 minimum weekly rest, driver-1 additional information, daily driving rights, daily resting rights, unknown period exists, card is ready, driver crew mode, significant time overlap exists) to the CAN-Bus via CAN-A or CAN-C interfaces. These messages are sent in every 10 seconds.

To open the ***CAN Services > Periodic DAGS*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select

CAN services menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***Periodic DAGS*** and press ***OK*** to enter the sub menu. Select the desired function (***Enable*** or ***Disable***) then press **OK** to confirm.

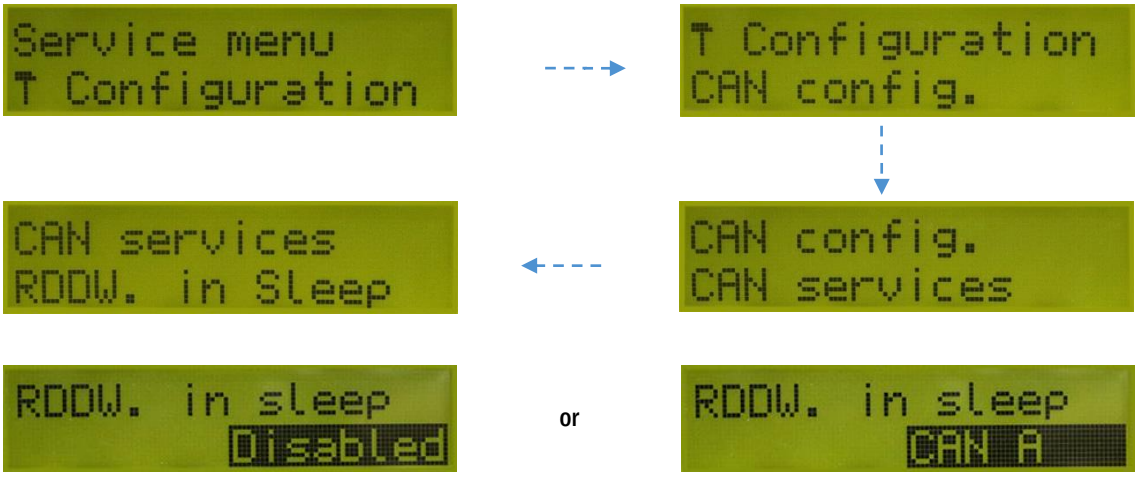


8.2.2.3.9 RDDW. In Sleep

STC-8255's RDD (Remote Data Download) feature while the device is in sleep mode (Ignition off) can be enabled or disabled in this menu.

To open the ***CAN Services > RDDW. in Sleep*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN services*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select **RDDW. In Sleep** and press ***OK*** to enter the sub menu. Select the desired function (***CAN A, CAN C*** or ***Disabled***) then press **OK** to confirm.



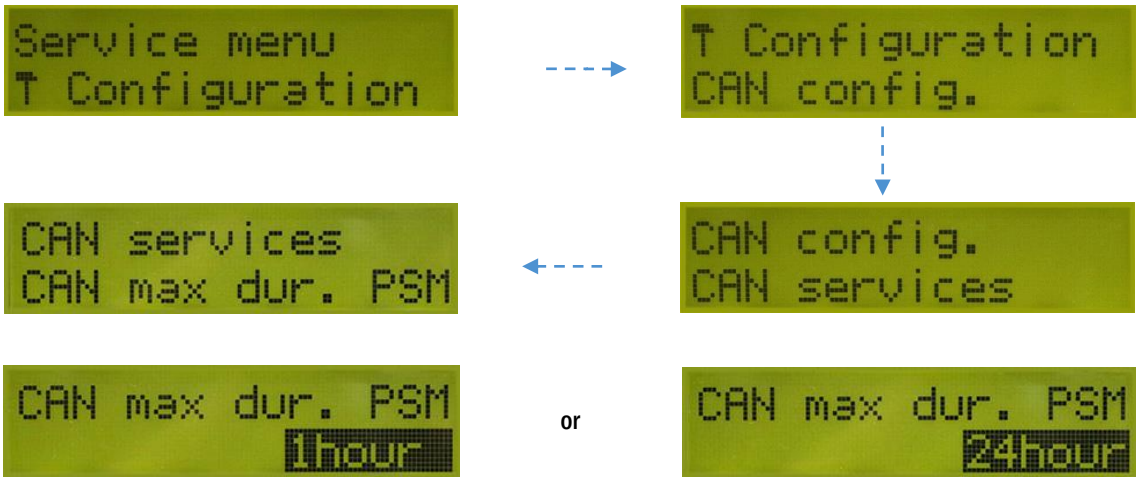
If it is enabled, the STC-8255 allows remote data download while it is in sleep mode, otherwise it is not allowed to download tachograph data remotely in sleep mode. If this mode is enabled, the current consumption in sleep mode increases about 3mA.

8.2.2.3.10 CAN Max. Dur. PSM

This option determines the maximum duration (in hours) that STC-8255 allows remote data download while it is in sleep mode (power saving mode).

To open the ***CAN Services > CAN max dur. PSM*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ◄ / ► keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ◄ / ► keys. Press **OK**, then select ***CAN services*** menu by using the control ◄ / ► keys. Press **OK** to enter the menu. Select ***CAN max dur. PSM*** and press ***OK*** to enter the sub menu. Adjust the maximum allowable duration that you desire, then press **OK** to confirm.



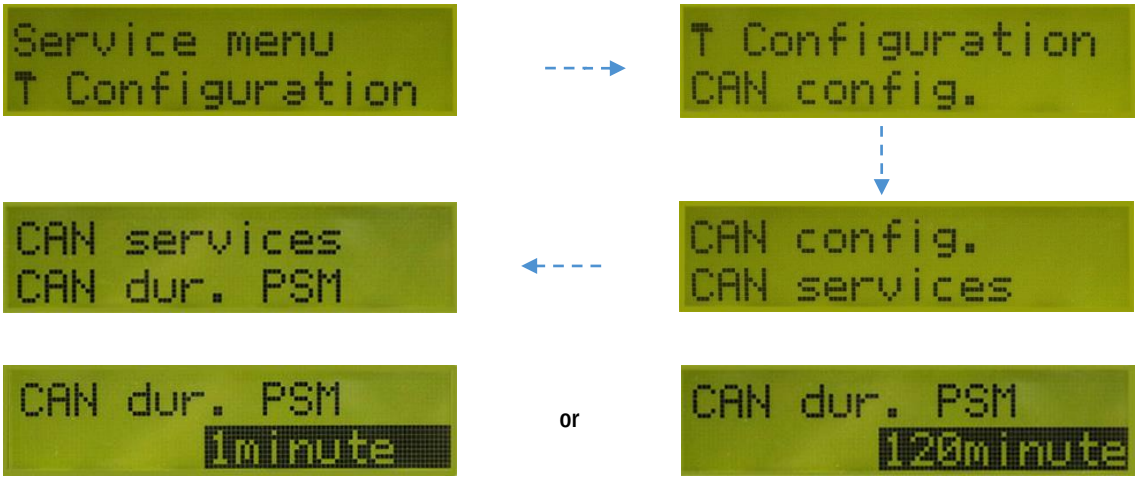
If you adjust this option as 24 hours, then the STC-8255 mass memory can be downloaded remotely within the duration of 24 hour after ignition is switched off. This option requires additional hardware configurations (FMS or RDD devices sold separately on the market) to be connected to the STC-8255 via CAN-Bus.

8.2.2.3.11 CAN dur. PSM

This option determines the duration (in minutes) that STC-8255 goes to active mode after remote data download process is started in power saving mode (sleep mode).

To open the ***CAN Services > CAN dur. PSM*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ◄ / ► keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ◄ / ► keys. Press **OK**, then select ***CAN services*** menu by using the control ◄ / ► keys. Press **OK** to enter the menu. Select ***CAN dur. PSM*** and press **OK** to enter the sub menu. Adjust the allowable duration that you desire, then press **OK** to confirm.



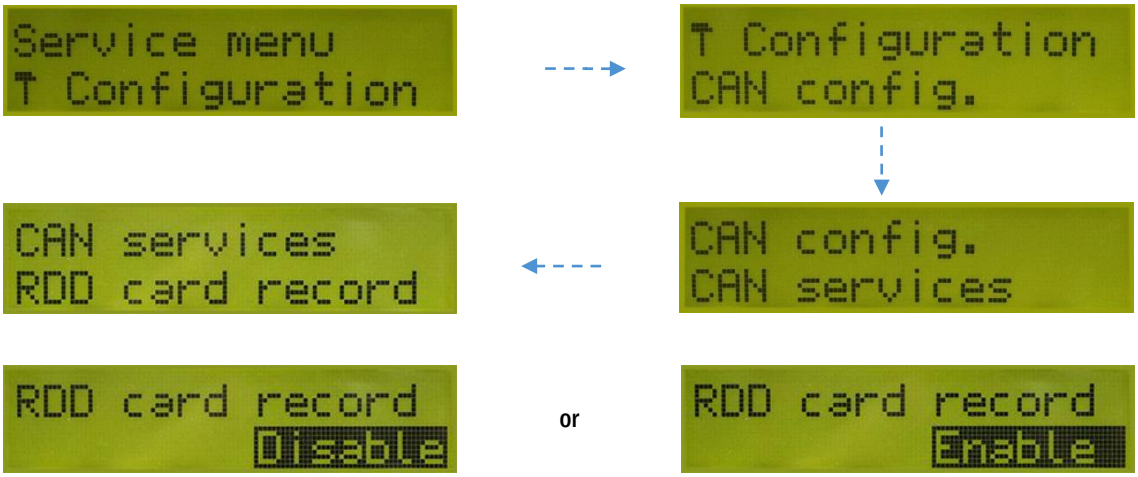
Keeping this duration so long, increases the power consumption while the vehicle ignition is OFF (i.e. tachograph is in sleep mode) and may decrease vehicle's battery charge level.

8.2.2.3.12 RDDW Card Record

This option enables the recording of data download activity to the remotely connected company card which is inserted to the remote data authentication server.

To open the ***CAN Services > RDD card record*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select **Service menu>Configuration** by using control ‹ / › keys. Press **OK** to enter the menu. Select **CAN config.** by using control ‹ / › keys. Press **OK**, then select **CAN services** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select **RDD card record** and press **OK** to enter the sub menu. **Enable** or **Disable** this function, then press **OK** to confirm.

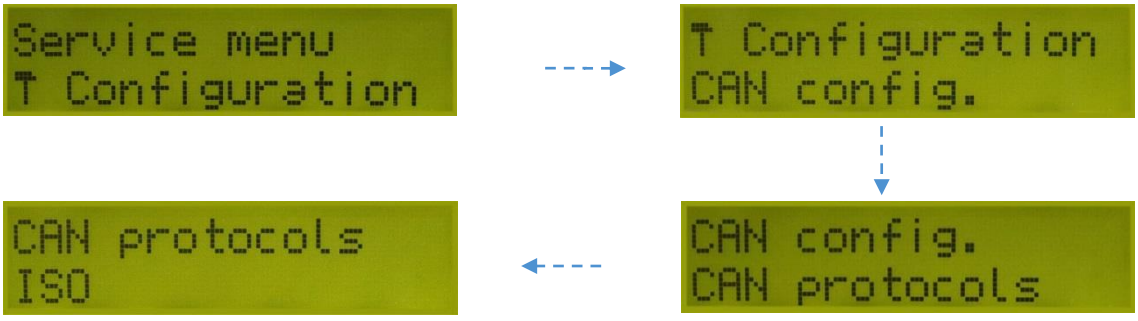


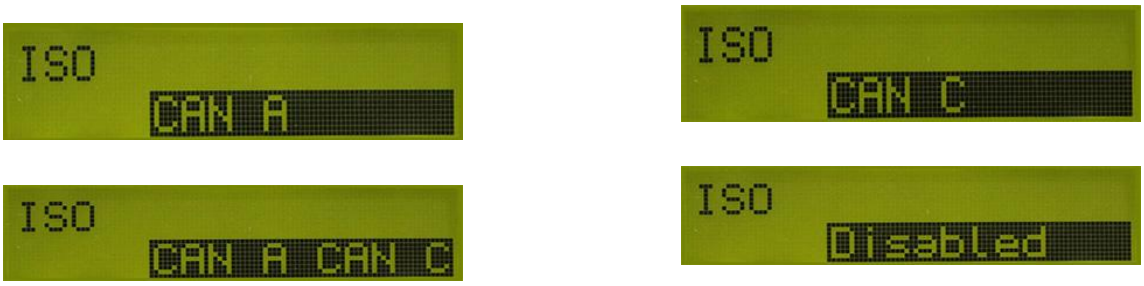
8.2.2.4 CAN Protocols

The protocol that is implemented in the STC-8255 will be selected from this menu. Both ISO (16844) and UDS protocols are supported in STC-8255. For the RDD (Remote Data Download) or FMS protocols UDS protocol should be selected from this menu.

To open the ***CAN Configurations>CAN protocols*** menu, perform as follows:

1. Press **OK** to enter Main Menu. Select ***Service menu>Configuration*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***CAN config.*** by using control ‹ / › keys. Press **OK**, then select ***CAN protocols*** menu by using the control ‹ / › keys. Press **OK** to enter the menu. Select ***ISO*** and press ***OK*** to enter the sub menu. Select the desired CAN output (***Disabled, CAN-A or CAN-C***) then press **OK** to confirm.



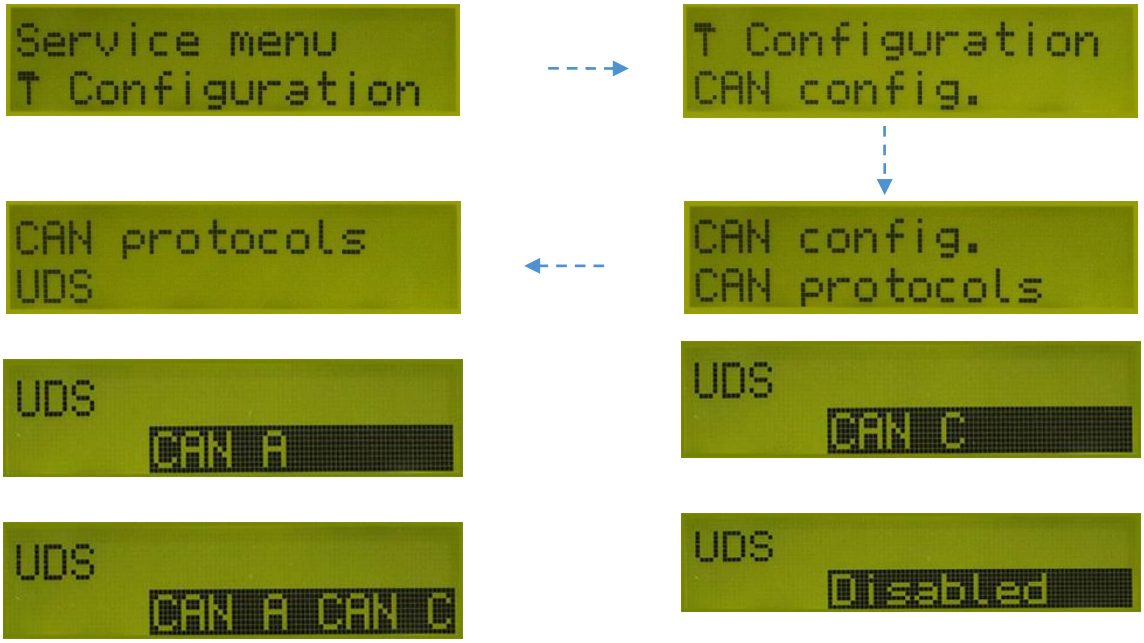


CAN protocol should be selected as ***CAN-A*** or ***CAN-C*** in order to send the TCO1 messages from CAN-A or CAN-C respectively.

If it is ***disabled***, no CAN messages related to the ISO will be transmitted from the CAN interfaces.

CAN UDS protocol should be selected as ***CAN-A*** or ***CAN-C*** in order to send the UDS messages or remote data download message from CAN-A or CAN-C respectively.

If it is ***disabled***, no UDS messages will be transmitted from the CAN interfaces.



9 INSPECTION

Inspection of a digital tachograph system must be carried out within every two years from the last calibration. However, it is necessary to carry out subsequent inspections under the following conditions:

- After every installation.
- After any repairs carried out on the tachograph or the recording equipment system.
- After any change in the characteristic coefficient (w factor) or vehicle constant (k factor) for the vehicle.
- After any change in the effective tyre circumferences of the drive wheels resulting from changing the tyre size.
- If the Vehicle Registration Number (VRN) has changed.
- If the tachograph's UTC time differs from the current UTC time by more than 20 minutes.

9.1 Inspecting Procedure

The most important part of the inspection is to confirm that the functioning of the recording equipment is correct. To perform inspection, do the following procedure:

- Check if the distance travelled and speed have been recorded correctly, completely and displayed by the STC-8255.
- Check if the characteristic coefficient of the vehicle (w) and constant of the vehicle (k) match one another.
- Check the tyre size and the actual circumferences of the drive wheels (l).
- Check the UTC time of the STC-8255 is correct.
- Check the other calibration data.
- Check the battery level, manufacture date.
- Perform the necessary visual inspections.

9.2 Visual Inspection

The following visual inspections must be done during in every inspection procedure.

- Inspect the front of the digital tachograph first to check that if there is a visible damage before removing it.
- Make sure that there is no damage to or drill holes in the entire exterior case of the tachograph so that there is no security breach attempt.
- Check for evidence of tampering with the tachograph seals and labels.
- Check if there is any additional seals or labels that are not belong to the STC-8255, since they might cover drill holes.
- Check the tachograph's type approval mark.
- Check that the seals of the system (digital tachograph, motion sensor and all removable electrical and mechanical connections) are intact.
- Check the conditions of the lines between the tachograph and the motion sensor.
- Check that the components of the digital tachograph are intact (no missing components, no foreign objects, etc.)
- Check that the installation plaque is intact and data on the installation plaque is coherent with the calibration and controls.

➔ Please refer to "section 5.3 Checking before Installation" for details.

9.3 Calibration

During inspection read the calibration data of the installed tachograph and check the data according to the vehicle parameters. If the calibration data needs to be updated then re-calibrate the tachograph according to the new parameters.

➔ Please refer to "section 6.2.5 Calibration", for further information.

9.4 Certification

After each inspection workshop staff should prepare a test certificate in order to be given to the owner of the tachograph. The certificate should consist of the following information:

- Owner of the tachograph,
- Manufacturer, model and serial number of the tachograph,
- The VIN and VRN numbers,
- Results of the inspection,
- Issue date of the certificate.

9.5 Faulty Unit

The repair of the Aselsan STC-8255 digital tachograph is not allowed due to the system security requirements. Therefore, any faulty unit which is failed during inspection must be decommissioned and replaced with new unit.

Please do not forget to download the stored data of the tachograph before decommissioning and store it in a secure data storage media. If the download is not possible from the decommissioned tachograph the workshops must prepare a un-downloadability certificate to be given to the relevant authorities.

9.6 Removal of the Digital Tachograph

Aselsan STC-8255 digital tachograph can be removed from its mounting frame by means of a pair of tachograph extraction tools.

Insert the extracting tools perpendicular into the two pair of holes at the sides of the front plastic cover as shown in the following figure.



Figure 24: Showing the Tachograph Removal Process.

Push the extraction tools sideways towards the sides and at the same time pull the tachograph out of the frame.

Carefully pull the tachograph and disconnect the rear connectors.

9.7 Downloading Vehicle Stored Data

Tachograph data memory records and holds the relevant data (e.g. driver and co-driver activities, country entries for both drivers, special entries, recording control activities, recordings of time adjustments, Lock-in / Lock-out procedures by the company card, identification of the STC-8255, device calibrations, system events & faults data, installation data and speed information over 168 hours and etc.) over a time period of at least 365 calendar days. When the memory of the tachograph is full, the information will be overwritten by new information and thereby it will be lost forever. To avoid this and to secure vehicle data, a frequent download of information is needed. It means that data will be transferred from the tachograph to data storage outside the vehicle.



Downloading will not destroy or erase any data from the memory. Data is only erased from mass storage when it is overwritten by new data.

Tachograph mass storage may be downloaded in the following modes:

1. **Company Mode:** In this mode company card should be in one of the card slots. Driver card or no cards should be in the other slot. If the company is locked-in to the tachograph only that company who has

locked can access to stored data (card insertion/withdrawals, driver activities and places) in company mode. With using different company cards, one cannot access (display, print or download) to these data which is previously locked by another company (card).

2. **Calibration Mode:** In this mode workshop card should be in one of the card slots. Driver card or no cards should be in the other slot. If the company card is locked-in to the tachograph, workshop cards cannot access (display, print or download) to stored data (card insertion/withdrawals, driver activities and places) which is previously locked by any company (card).
3. **Control Mode:** In this mode control card should be in one of the card slots. Driver card or no cards should be in the other slot. Whether it is locked by any company or not, control cards can access (display, print or download) to whole stored data (card insertion/withdrawals, driver activities and places) which is previously locked by any company.

Downloading of mass storage data of the tachograph requires data download equipment (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B Appendix-7).

To explain mass storage download of the STC-8255, perform the following steps:

1. Open the silicon cover in front of the tachograph.
2. Connect the download equipment's 6-pin connector to the tachograph's 6-pin interface behind the silicon cover.
3. Start downloading data according to the instructions on the download equipment.
4. When downloading starts a flashing animation is played on the second line of the display, showing the downloading is in progress.



5. When the downloading is finished animation on the display stops.
6. Get the digitally signed data which identifies the digital tachograph and store it in a secure data storage media.

Do not disconnect the data connection to the digital tachograph before the download is complete.

For detailed information about the data download, please refer to the download equipment's appropriate documentation.

To download card data over tachograph, the STC-8255 should not be in power safe mode. If so, press any button first to wake the STC-8255 up or switch the ignition key ON before the process.

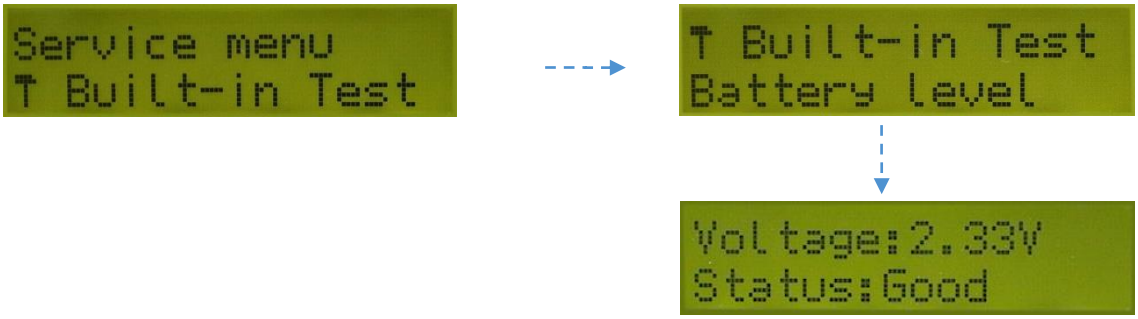
9.8 Battery Re-charging Procedure of the STC-8255

There are two internal batteries in the STC-8255. One of the batteries inside the STC-8255 is rechargeable and the other battery is non-rechargeable. Actually, the STC-8255 digital tachograph is designed so that there is no need to replace the internal batteries for the entire life cycle of the device. In addition, In addition, when the tachograph is installed in the vehicle and connected to the vehicle battery by the backside power connector, the internal back-up batteries will never be drained. However, if the STC-8255 device is held on the storage conditions for a long period of time (approximately 20 years in the non-

activated mode or 10 years in the activated mode) the internal battery reserves may decrease. In these circumstances checking the battery condition is necessary.

To measure the battery level, please perform the ***Built-in test>Battery level*** test which is explained in "section 8.1.5 Battery Level".

If the battery level is good as explained in section 8.1.5 and as shown in the following figure, then there is no need to recharge the battery in that device.



However, if the battery level is 'Low' in contrast to above figure, then connect the STC-8255 to the external power supply (12V or 24V) for at least 12h in order to re-charge the internal re-chargeable battery.

After this 12h re-charging period ends, please disconnect the power supply and let the device be unconnected to the power supply at least 2 minutes. Then, connect the power supply to the STC-8255 and repeat the battery level test again. If the battery level status is still not good, then please consider to send the unit to the manufacturer in order to replace the internal batteries.

If the STC-8255 digital tachograph is stored more than 10 years on the shelf without installing to the vehicle, the battery in the tachograph will be discharged and the Real Time Clock (RTC) data will be erased because of it. In this case, please re-charge the internal battery as mentioned above and set the RTC again.

9.9 Messages, Warnings and Faults

There are four types of messages that can be shown on the display.

- **Messages** – give information on the process or remind the driver. Messages cannot be stored and printed. Press the backspace ⌫ button or **OK** button to clear the message.
- **Pre-warnings** – appear as early reminders to the warnings. Pre-warnings are not stored and cannot be printed. Press **OK** button twice to clear the Pre-warning.
- **Warnings (Events)** – appear in the event of e.g. overspeed or violation of the law or if the tachograph cannot be recording. Warnings are stored and can be printed. Press **OK** button twice to acknowledge the Warning.
- **Faults** – are more critical than warnings and are displayed if there is a fault detected in the tachograph, in the sensor or cards. On the other hand, faults message are shown if the device has detect a tampering. Faults are stored and can be printed. Press **OK** button twice to acknowledge the Fault.

Table 6: List of Message

Pictogram & Message ID	Message	Explanation	User Action
! 1			

Pictogram & Message ID	Message	Explanation	User Action
! 2 1	“non-valid card”	A non-valid or expired card has been inserted to one of the slots.	Acknowledge the message. Eject the non-valid or expired card if it is not ejected automatically.
× 1 × 2 134	“Wrong card type”	Displayed if the inserted card into the tachograph is not a tachograph card. The card inserted is not accepted and ejected automatically.	Please insert a valid tachograph card.
× 1 × 2 135	“PIN is blocked”	Displayed when workshop card is blocked because of wrong card pin entries.	Acknowledge the message. Apply for the new service card to “STAUM”. https://staum.org.tr.
× 1 × 2 136	“Recording Inconsistent”	Displayed if there is an inconsistency with the driver or workshop card records while inserting the cards to any slot.	Acknowledge the message.
× 144	“Upgrade failed”	Displayed when software upgrade is failed.	Acknowledge the message.
 145	“Insert a service card”	Displayed when a card other than Workshop card is inserted to non-activated VU.	Acknowledge the message and insert a valid workshop card.
 149	“Insert a service Card”	Displayed when software upgrade is tried with no Workshop card inserted.	Acknowledge the message and insert a valid workshop card.
  xxhxx	“Drive Cont.”	Displayed if the weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
  xxhxx	“Rest Cont.”	Displayed the remaining time for the upcoming weekly rest period.	Acknowledge the message. Prepare for the rest period.
24h xxhxx	“Drive Cont.”	Displayed if the daily allowable driving time is exceeded.	Acknowledge the message. Give a break.
24h	“Rest Cont.”	Displayed the remaining time for the upcoming daily rest period.	Acknowledge the message. Prepare for the rest period.
xxhxx	xxhxx		
1 2	“Give a break”	Displayed after 4h15m or 4h30m continuous driving time.	Acknowledge the message. Give a break.
xxhxx	xxhxx		
  “xxhxx”	“Drive Cont.”	Displayed if the 2 weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
! 171	“No personalization”	Displayed if the card is inserted into the not personalized tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 172	“Tachograph is not ready”	Displayed if the tachograph is not ready when the card is inserted into the tachograph.	Acknowledge the message. Try again after 1 minute. Have the STC-8255 checked by authorized workshop if this message is displayed frequently.
! 174	“Eject driver card”	Displayed if there is a valid driver card in any slots while the ignition is turning to OFF.	Acknowledge the message. Do not forget to withdraw your driver card from the tachograph.

Pictogram & Message ID	Message	Explanation	User Action
! 174	"Veh. is not rdy to driv."	Displayed if the ignition is ON and there is no valid driver card in the first slot or there is a smart card rather than driver card in the second slot.	Acknowledge the message. Do not forget to insert your driver card to the first slot before you start your trip.
	"Pers. transfer is completed"	Displayed if the personalization file is transferred to the device successfully.	Acknowledge the message.
	"Pers. file is verified"	Displayed if the personalization file is verified successfully.	Acknowledge the message.
	"FUP transfer is completed."	Displayed if the FUP (Firmware Upgrade Package) file is transferred to the device successfully.	Acknowledge the message.
	"Transfer rejected"	Displayed if the FUP file is not verified successfully.	Acknowledge the message.
!	"Power is insufficient"	Displayed if the power is insufficient when the printing process is started or during the printing.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
!	"Card is not ready"	Displayed if the card reading is not finished for both card slots before the printing process is started.	Acknowledge the message. Wait for a while for the card reading process.
	"Check the data"	Displayed after the data download built-in tests which reminds the user to check the downloaded data.	Acknowledge the message. Please check the downloaded data.
	"Not available" "Try again later"	Displayed if necessary calculation has not been finished when starting the data integrity built-in tests.	Acknowledge the message.
	"Buttoncell error"	Displayed if internal battery has not been inserted while performing battery level built-in tests.	Acknowledge the message. Please send the unit to the manufacturer.
	"No response"	Displayed if no response has been received while performing battery level built-in tests.	Acknowledge the message. Try the test again and if the problem still continues then please send the unit to the manufacturer.
	"Successful"	Displayed if the result of the GNSS test in built-in tests is successful.	Acknowledge the message.
	"Failed"	Displayed if the result of the GNSS test in built-in tests is failure.	Acknowledge the message. Try the test again and if the problem still continues then please send the unit to the manufacturer.
	"Stop to print"	Displayed when the printer is busy with printing at the beginning of the printing task.	Stop the vehicle and try to print again when the vehicle is stationary.
	"No paper"	Displayed when the printer is out of paper at the beginning of the printing task.	Load new roll of paper to the printer drawer.

Pictogram & Message ID	Message	Explanation	User Action
	"No data"	Displayed when there is no data for the requested printout type.	Acknowledge the message.
	"Thermal stress"	Displayed if the printer is too hot and the user requests to print any report.	Acknowledge the message. Please wait for a while for the printer head to cool down and try again later.
	"Thermal stress"	Displayed if the printer has continuously printed and the printer head is too hot to continue. Disappears when the user pushes the backspace button or the print head cools down.	Wait for the printer head cools down and printer continues printing job or press backspace to cancel the job. Please do not touch the printer head with your hand when this message appears.
	"Cancel"		
	"Load paper"	Displayed when the printer is out of paper while printing and displayed 45 sec. period until new printer paper roll is inserted.	Insert new printer paper roll.
	-Progress Bar-		
	"Printer is ready"	Displayed after the new roll of paper is loaded to the printer drawer.	Load new roll of paper to the printer drawer.
	-Progress Bar-		
	"Cancel printing"	Displayed when the printer is not loaded with a new roll of paper in a certain period (45 sec).	Press YES to cancel the printing job.
	Yes – No Dialog		
	"not possible"	Displayed when the user is trying to enter the menu while the vehicle is moving.	Stop the vehicle if necessary to navigate in the menus.
	"Printing"	Appears when the printer is busy with printing.	No user action required.
	"Ejection is not possible"	Displayed if the cards inserted to the tachograph slots are requested while the battery supply level is too low.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
1 2	"Ejection is not possible"	Displayed if the cards inserted to the tachograph slots are requested while the vehicle is moving.	Please stop the vehicle and then request the card withdrawal from the tachograph.
	"UTC correct. not possible"	Displayed if you are trying to correct the UTC time which has already been corrected within the last 7 days.	No user action required.
	"Wait for ejection"	Displayed if the cards inserted to the tachograph slots are requested while the inserted activity is recorded to the tachograph cards.	Acknowledge the message. Please try again one minute later.
	"Rebooting..."	Displayed while the tachograph is in rebooting process.	Acknowledge the message.
	"Personalization failure!"	Displayed if the personalization control is fails.	Acknowledge the message. Try the reboot again and if the problem still continues then please send the unit to the manufacturer.

Table 7: List of Pre-Warnings

Pictogram & Message ID	Pre-Warning Messages	Explanation	User Action
>> 129	"Oversp. prev. warning"	The vehicle has exceeded the overspeed limit which is set for that vehicle. If exceeding continues for 30 sec. (or pre-defined value) the overspeed event will be recorded in tachograph memory.	Acknowledge the message. Reduce the vehicle speed.
 130	"Card will expire soon"	Means expiration prewarning day limit of card inserted to the tachograph slot is reached and this card will be expired soon.	Acknowledge the message. Contact the responsible authority to get new one.
 131	"Time for service"	Displayed if the calibration pre-warning date is passed and reminds the driver.	Acknowledge the message. Have the STC-8255 checked and calibrated by authorized workshop.
 146	"Download card"	Displayed if the card download pre-warning date is passed and reminds the driver to download his/her card to external media.	Acknowledge the message. Download the driver card to any external media.
 147	"Download vehicle unit"	Displayed if the tachograph download pre-warning date is passed and reminds the driver to download his/her tachograph to external media.	Acknowledge the message. Download the tachograph mass memory to any external media.

Table 8: List of Warnings

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
! 2	"Card conflict"	Displayed if two tachograph cards incompatible with each other are inserted to the tachograph.	Acknowledge the message. Remove one of the cards (which is useless) from the card slot.
! 3	"Time overlap"	The last withdrawal date/time of the inserted driver card from previous tachograph is later than the date/time of recently inserted tachograph.	Acknowledge the message. Check the date/time of the tachographs used and determine the tachograph with the incorrect UTC time, make the tachograph time to be corrected by authorized workshop. Or, wait for the overlap period to be elapsed.
! 4	"Driving w/o valid card"	Driving started with an inappropriate card or with an inappropriate card combination.	Acknowledge the message. Stop vehicle and insert a valid driver card or remove inappropriate card from co-driver slot.
! 5	"Card insertion while driving"	A driver card was inserted after driving has begun.	Acknowledge the message. No further action required.
! 6	"Card closed incorrectly"	Card was not removed properly from the last tachograph or data could not be saved to the card correctly.	Acknowledge the message. No further action required.
>> 7	"Over speeding"	Driver exceeded the permissible max speed limit for longer than 1 minutes.	Acknowledge the message. Reduce vehicle speed.
! 8	"Power supply interruption"	The power supplied to the STC-8255 is below or above the certain limit more than 200 miliseconds or has been disconnected for more than 200 miliseconds. The event is not triggered in Calibration Mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
! 9	"Motion sensor data error"	The data link failure occurred between the motion sensor and the tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 10	"Motion conflict"	Displayed if the second source of motion data (GNSS or CAN-Bus) is not consistent with the primarily used motion sensor data.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 16	"No further details"	Displayed if an unexpected error occurs while authenticating the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 17	"Motion sensor auth. failure"	Displayed if the tachograph does not authenticate with the motion sensor	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 18	"Card auth. Failure"	Displayed if the tachograph does not authenticate with the smart card.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 19	"Unauthorised change of MS"	Displayed if unpaired motion sensor is linked to the tachograph paired with another motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 20	"Card data integ. err."	Displayed if smart card data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 21	"Data integ. err."	Displayed if the tachograph data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 24	"Hardware sabotage"	Displayed if card(s) ejection by force is detected.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 32	"No further details"	Displayed if a generic error occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 33	"Auth. Failure"	Displayed when error related to authentication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 34	"Data integ. err."	Displayed when error related to non-volatile memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 37	"Hardware sabotage"	Displayed when error related to sensor element or temperature occurs in the motion sensor and the tachograph is not in calibration mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.

Table 9: List of Faults

Pictogram & Message ID	Fault Messages	Explanation	User Action
50	"Printer fault"	Displayed when unexpected interrupt is received from printer.	Acknowledge the message. Repeat print request and if necessary switch off / on the ignition.
51	"Display fault"	Displayed if the display self-test is fails.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
1			Acknowledge the message.

×🖩2 64	"No further details"	Displayed because of generic smart card fault.	Check if the card contact surface is clean. If it is not, clean the contacts and insert it again.
×🖷 157	“Internal fault”	Displayed if there is problem while the tachograph is rebooting.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
×⎍ 158	“Sensor fault”	Displayed when error related to controller ram occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×⎍ 159	“Sensor fault”	Displayed when error related to controller instruction occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×⎍ 160	“Sensor fault”	Displayed when error related communication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×🖷 161	“Internal fault”	Displayed when there is motion sensor authentication failure.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×🖷 162	“Internal fault”	Displayed when the smart card power-up voltage is invalid.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×🖷 163	“Internal fault”	Displayed if the smart card protocol is unsupported.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×🖷 164	“Internal fault”	Displayed if the smart card status is invalid slot.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×🖷 165	“Internal fault”	Displayed if the smart card status is unexpected value.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
×🖷 166	“Internal fault”	Displayed if the user data storage is insufficient.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
×🖷 169	“Internal fault”	Displayed if the internal battery voltage is below critical level.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
×◷ 173	“Clock Failure”	Displayed when there is a fault in the real time clock (RTC) module of the tachograph and recorded to the device as a DTC record.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
×🖷 175	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
×🖷 176	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
×🖷 177	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

9.10 Diagnostic Trouble Codes (DTCs)

The diagnostic trouble codes for the STC-8255 given below is compatible with the ISO16844-7 standard.

Table 10: List of Diagnostic Trouble Codes (DTCs)

Item	Description	HEX				
		DTC Severity	Functional Unit	High Byte	Middle Byte	Low Byte (Failure Type)
	Sensor power supply					
1	above maximum	80	00	00	20	07
2	below minimum	80	00	00	20	03
3	no signal	80	00	00	20	04
	VU power supply					
4	above maximum	80	00	00	00	07
5	below minimum	80	00	00	00	03
6	no signal	80	00	00	00	04
7	Driver Card 1 error	40	00	00	02	00
8	Driver Card 2 error	40	00	00	03	00
9	Printer paper out	80	00	00	06	60
10	No speed signal from sensor	80	00	00	21	80
11	Invalid speed signal or data link error	80	00	00	22	80
12	Speed sensor – Recording Unit: no data link signal	80	00	00	23	80
13	Sensor / Tachograph signature mismatch	80	00	00	24	52
14	Time / Date error	80	00	00	08	00
15	No ignition, but speed pulses present	40	00	00	09	00
16	CAN, internal error	40	00	00	0A	70
17	CAN, bus off	40	00	00	0B	78
18	CAN communication instrument <-> Recording unit No signal	80	00	00	11	77
19	Program memory, checksum error	80	00	00	0C	31
20	Calibration memory, read/ write error	80	00	00	0D	33
21	Calibration error	80	00	00	0D	40
22	Printer error	40	00	00	07	00
23	Card reader 1 error	40	00	00	04	00
24	Card reader 2 error	40	00	00	05	00
25	Buttons / KBD error	40	00	00	0F	00
26	Error in Internal display	40	00	00	10	30
27	Internal error in sensor	80	00	00	25	08
28	Speed pulse output recording unit (B7)	40	00	00	30	00
29	Recording Unit, internal error	40	00	00	01	39
30	Driving without card (Driver card or Workshop card)	80	00	00	12	60
31	GNSS error	80	00	00	FF	00
32	RTP temperature out of range	80	00	00	FF	01
33	SP temperature out of range	80	00	00	FF	02
34	SP voltage failure	80	00	00	FF	03
35	SP front side physical violation	80	00	00	FF	04
36	SP mirroring violation	80	00	00	FF	05
37	SP LSD violation	80	00	00	FF	06

9.11 Care and Maintenance

9.11.1 Care of the Tachograph

To increase the life-time of the tachograph please be aware of the following:

- Try to keep the tachograph and the environment where it is installed clean and dust-free.
- Clean the plastic box, display and the function keys with a slightly moistened cloth.
- Do not use any alcohol or chemical cleaning solvents to clean any parts of the tachograph.
- Do not insert objects or other cards (rather than tachograph cards, e.g. credit cards) to the card slots otherwise the cards and the tachograph may be damaged.
- Protect the tachograph from direct sunlight (do not install it on the vehicle dashboard).

9.11.2 Care of the Cards

Please take care of the cards and keep in mind the following:

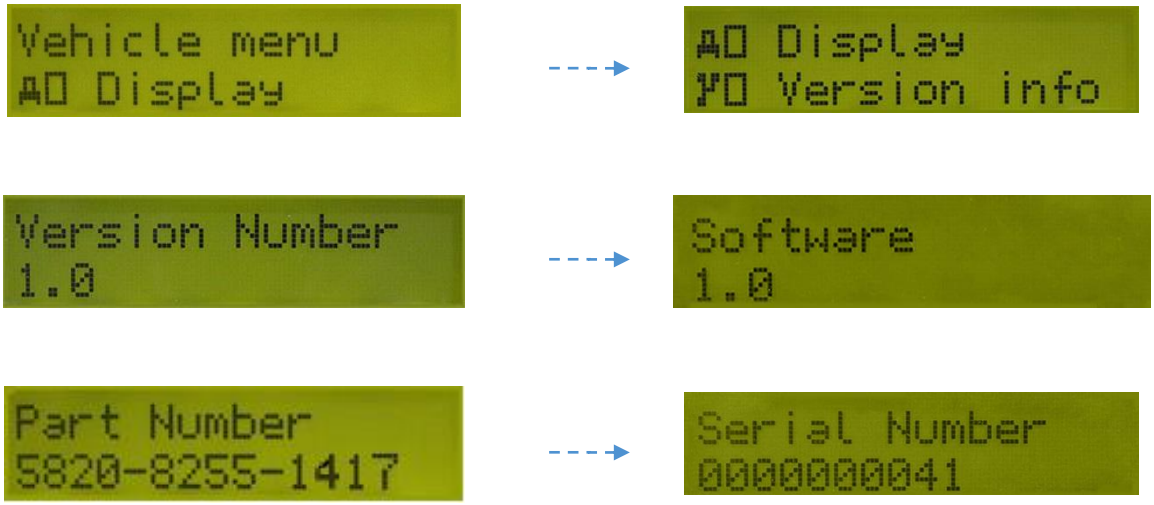
- Try to keep the card contacts free from dirt and dust.
- Clean the cards with a slightly moistened soft cleaning cloth if the contacts have grease or oil.
- Do not bend or fold the cards and do not use it for anything other than their primary usage.
- Protect it from damage and do not use damaged cards in the tachograph.
- Do not let the cards to be under strong electromagnetic field.

9.12 Software & Hardware Information

You can display the software & VU version numbers, serial & part number and sealing label info defined on the STC-8255.

To display software and VU version, follow the steps:

1. Press **OK** to enter main menu. Select **Vehicle menu>Display** by using control ‹ / › keys. Press **OK** to enter the menu. Select **Version info** by using control ‹ / › keys. Press **OK** to display the version info. Use control ‹ / › keys to see the software, hardware versions, part number, serial number and seal number. Press **OK** or **Backspace** ⌫ to navigate one level back.





9.13 Upgrading the STC-8255 Operating Software

If the software update of the STC-8255 digital tachograph is released, it is strongly recommended that you should update the operating software of the STC-8255 digital tachograph before installing it into a vehicle.

The STC-8255 updated software must be approved by the manufacturer (Aselsan Inc.) and certified by the official EU approval bodies. The certified software versions are available at the following link (https://www.aselsan.com.tr/en/capabilities/public-safety-communicatio-systems/digital-tachograph-systems).

In order to upgrade the operating software of the STC-8255 digital tachograph, perform the following steps:

1. Check the above internet site and make sure that new certified software release exists.
2. Obtain the latest release of STC-8255 software and the "Workshop Software Loader" software upgrade program either by downloading from the Aselsan's website or by requesting a CD including these data from Aselsan Inc.
3. Obtain a USB to 6-Pin serial converter cable or RS-232 serial to 6-Pin cable either from "*LISLE Design*" or from "*ASELSAN Inc.*" (sold separately).
4. Before starting the software upgrade please install the "Workshop Software Loader" software upgrade program into the computer you will use.
   ➔ Please refer to the, "Workshop Software Loader User Manual" for installation process.
5. Please connect the serial cable or USB cable into one of the empty ports of the computer depending on the cable you choose and connect the 6-Pin connector into the tachograph's 6-Pin interface at front panel.
6. If you choose USB to 6-Pin cable then install the cable's driver (USB to UART converter) into the PC.
   ➔ Please refer to the, cable manufacturer's (LISLE Design) relevant documentation.
7. After, connecting and installing the driver of the 6-Pin cable, please note the COM Port parameters from the PC Settings (Under the device manager>Ports menu).
8. Insert your workshop card into the first slot of the STC-8255 and enter its respective PIN.
9. Run the "Workshop Software Loader" program in the PC then you will see the following GUI on the PC screen as shown in the figure below.

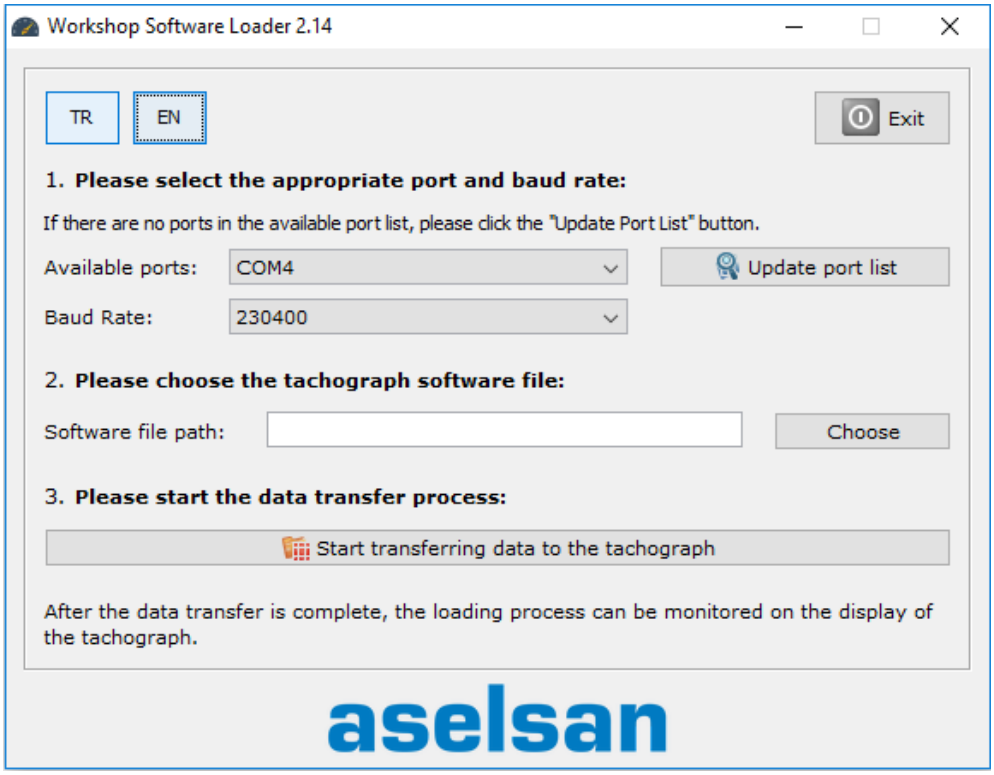


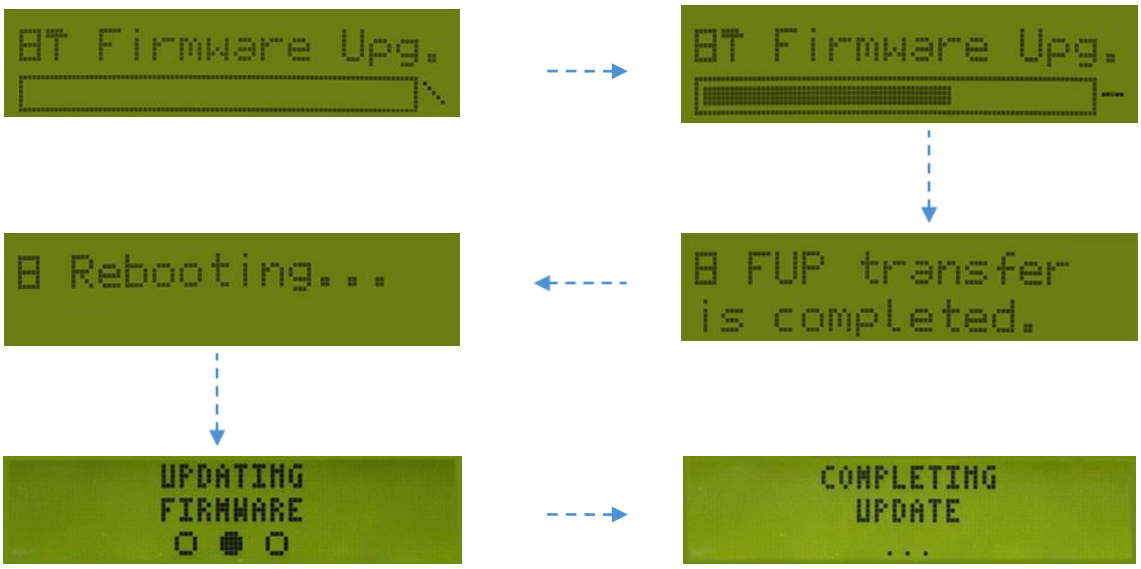
Figure 25: Workshop Software Loader Program Graphical User Interface (GUI)

10. Select the port that is assigned to your USB to 6-Pin cable as you noted at step 7. If the port is not seen on the available ports drop down list then update the port list and choose your desired port.
11. Finally, choose your updated software get from the CD or downloaded to the computer from the internet.

    Please note that: The software update file extension should be as "*.fup"

12. Press ***"Start transferring data to the digital tachograph"*** button after choosing the software to be uploaded. You will see the following screens on the tachograph's display.

    Please note that: Do not disconnect the 6-Pin connector from the tachograph's 6-Pin interface until you see the operation successful message on the PC and on the tachograph's display.



Then press **OK** to confirm. After confirming, the STC-8255 will reboot and start with the new operating software.

13. Make sure that STC-8255 software is upgraded either from the ***Vehicle Menu>Display>Version Info*** or from the ***technical printout***.

9.14 Technical Data

The following table summarizes the STC-8255 digital tachograph's most important technical data.

Table 11: Technical Data

Technical Data	
Nominal power supply	12 V or 24V
Power supply range	9V to 32V
Power consumption in operation 12V (with motion sensor) 24V (with motion sensor)	 70mA (approx., max illumination) 40mA (approx., max illumination))
Power consumption when printing 12V 24V	 max. 4.2A max. 1.8A
Power consumption in standby (w/o motion sensor) 12V 24V	 ~10 mA (for standard conf.) ~7 mA (for standard conf.)
Display	Dot-Matrix LCD (99 x 21 Dots), 2 lines 16 char. each. (backlight color optional)
Printer	24 characters per line. Paper width 56-57 mm, paper roll diameter max. 25 mm,
Protection class (device front after installation)	IP40
Protection class (device rear)	IP40
Operating temperature	-25°C to +80°C
Printing temperature	-15°C to +50°C
Storage temperature	-40°C to +85°C
Long-term storage temperature	0°C to +50°C
Speed measurement range	0 to 220 km/h
Frequency ranges Speed-impulse in/output (B3/ B6 & B7)	 Max. 1.6 kHz
Data interfaces (front)	RS-232 download interface K-Line calibration interface
Data interfaces (rear)	Motion sensor data signal (B4) CAN-A (A4, A8) CAN-C (C5, C7) K-Line calibration interface (D7) Information Data-Out Interface (D8)
Inputs	1x Motion Sensor Real Time Speed and Data inputs (B3 & B4) 1 x Illumination input (A2) 2 x Status inputs (D1, D2) (opt.) 1 x Motor revolution input (C3) 1 x CAN wake-up input (D3)
Outputs	3 x v-pulse output (B6, B7, D6)

Technical Data	
	1 x 4 imp/m output (B8) 1 x warning output (D4)
Dimensions without front panel (W x H x D)	179.9mm x 50.4mm x 148.5mm
Dimensions of front panel (WxHxD)	187.5mm x 58mm x 14.8mm
Weight	Less than 690g (including paper roll)
Installation	Standard 1-DIN radio slot (ISO-7736)

9.15 National / Regional Abbreviations and the European Time Zones

The following table provides an overview of national and regional abbreviations, as well as time zones:

Table 12: Country Codes

Nation	Abbreviation	Time zone offset
Austria	A	+01:00h
Albania	AL	+01:00h
Andorra	AND	+01:00h
Armenia	ARM	+04:00h
Azerbaijan	AZ	+03:00h
Belgium	B	+01:00h
Bulgaria	BG	+02:00h
Bosnia and Herzegovina	BIH	+01:00h
Belarus	BY	+02:00h
Croatia	HR	+01:00h
Cyprus	CY	+01:00h
Czech Republic	CZ	+01:00h
Denmark	DK	+01:00h
Estonia	EST	+02:00h
Faeroe Islands	FR	+00:00h
Finland	FIN	+02:00h
France	F	+01:00h
Georgia	GE	+02:00h
Germany	D	+01:00h
Greece	GR	+02:00h
Hungary	H	+01:00h
Iceland	IS	+00:00h
Ireland	IRL	+00:00h
Italy	I	+01:00h
Kazakhstan	KZ	+04:00h to +06:00h
Latvia	LV	+02:00h
Lichtenstein	FL	+01:00h
Lithuania	LT	+02:00h
Luxembourg	L	+01:00h
Malta	M	+01:00h

Nation	Abbreviation	Time zone offset
Macedonia	MK	+01:00h
Monaco	MC	+01:00h
Montenegro	MNE	+01:00h
Netherlands	NL	+01:00h
Norway	N	+01:00h
Poland	PL	+01:00h
Portugal	P	+00:00h
Republic of Moldova	MD	+02:00h
Romania	RO	+02:00h
Russian Federation	RUS	+03:00h
San Marino	RSM	+01:00h
Serbia	SRB	+01:00h
Slovakia	SK	+01:00h
Slovenia	SLO	+01:00h
Spain	E	+01:00h
Sweden	S	+01:00h
Switzerland	CH	+01:00h
Tajikistan	TJ	+05:00h
Turkey	TR	+02:00h
Turkmenistan	TM	+05:00h
Ukraine	UA	+02:00h
United Kingdom	UK	+00:00h
Uzbekistan	UZ	+05:00h
Vatican City	V	+01:00h
Yugoslavia	YU	+01:00h
European Community	EC	
Rest of Europe	EUR	
Rest of World	WLD	

9.16 Region Abbreviations of Spain

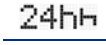
Table 13: Regions of Spain

Region	Abbreviation
Andalusia	AN
Aragon	AR
Asturias	AST
Cantabria	C
Catalonia	CAT
Castil-Leon	CL
Castile-La Mancha	CM
Valencia	CV
Extremadura	EXT
Galicia	G
Balearic Islands	IB
Canary Islands	IC
La Rioja	LR
Madrid	M
Murcia	MU
Navarra	NA
Basque Community	PV

9.17 Pictogram Combinations

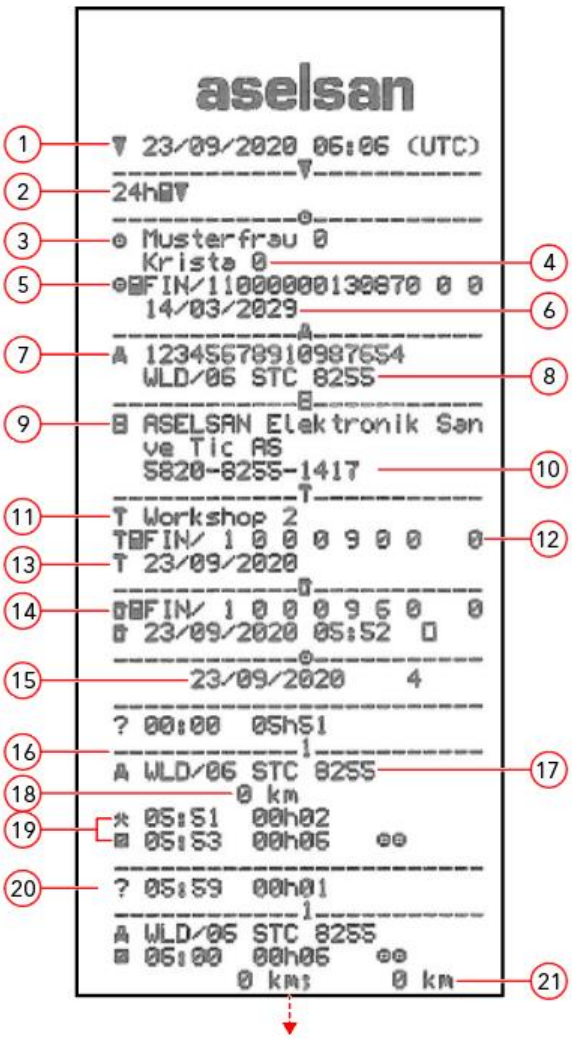
Table 14: Pictogram Combinations

Symbol	Meaning / Description
	Control Place
	Location start of daily work period
	Location end of daily work period
	From time
	To time
	From vehicle
OUT	Out of scope begin
OUT	Out of scope end
	Driver card
	Company card
	Control card
	Workshop card
---	No card
	Crew driving
I	Driving time for one week
II	Driving time for two weeks
24h	Daily driver activity printout from smartcard
24h	Daily driver activity printout from tachograph memory
!	Printouts of events and faults from smartcard
!	Printouts of events and faults from tachograph memory
	Technical data printout
>>	Over speeding (speed infringement) printout
!	Insertion of a non-valid smartcard
!	Card conflict
!	Time overlap
!	Driving without appropriate card
!	Card insertion while driving
!	Last card session not correctly closed
>>	Over speeding
!	Power supply interruption
!	Motion data error
!	Security breach
!	Time adjustment (by workshop)
>	Over speeding control
!	Vehicle motion conflict
1	Card fault (driver slot)
2	Card fault (co-driver slot)
	Display fault

Symbol	Meaning / Description
	Downloading fault
	Printer fault
	Sensor fault
	Vu internal fault
	Still the same daily work period?
	End of previous work period?
	Confirm or enter location of end of work period
	Enter start time
	Enter location of start of work period
	Card existence at ignition state warning
	Clock failure warning
	Company entry
	Display GNSS
	Print D1/D2
	Display D1/D2
	Print V-profile
	Display V-profile
	Print N-profile
	Display N-profile
	Print vehicle V-graph
	Print card activity graph
	Daily rest counter warning
	Two week driving warning
	Display version info
	GNSS report

10 SAMPLE PRINTOUTS

10.1 24h Driver Card Printout



1 Date and time of printout (UTC).

2 Type of printout (24h, card).

3 Card holder's surname.

4 Card holder's first name.

5 Country and card identification number.

6 Expiry date of the card inserted.

7 Vehicle identification, VIN.

8 Registering member state and Vehicle Registration Number (VRN).

9 Tachograph manufacturer's name.

10 Tachograph part number.

11 Workshop name where the last calibration is made.

12 Workshop card identification number.

13 Date of the last calibration.

14 Last control the inspected driver has been subjected to.

15 Enquiry date and daily card presence counter.

16 Slot where card was inserted.

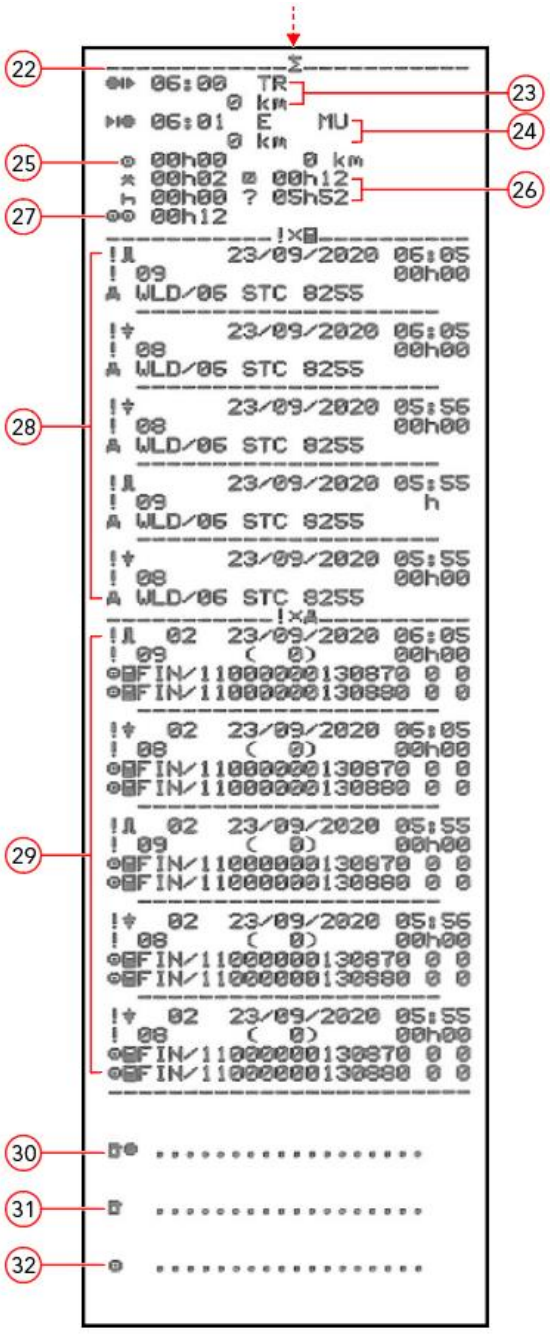
17 Vehicle Registration Number (VRN), for the vehicle where the driver card was inserted.

18 Vehicle odometer at card insertion.

19 Activities while driver card inserted, start and duration time, crew status.

20 Activity while driver card not inserted to any tachograph, start and duration time.

21 Card withdrawal: Vehicle odometer and distance travelled since last insertion for which odometer is known.



22 Daily summary of activities.

23 Begin country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

24 End country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

25 Driving activity duration and trip distance.

26 Other activities' durations.

27 Driving crew status duration.

28 Last five events and faults stored in the driver card.

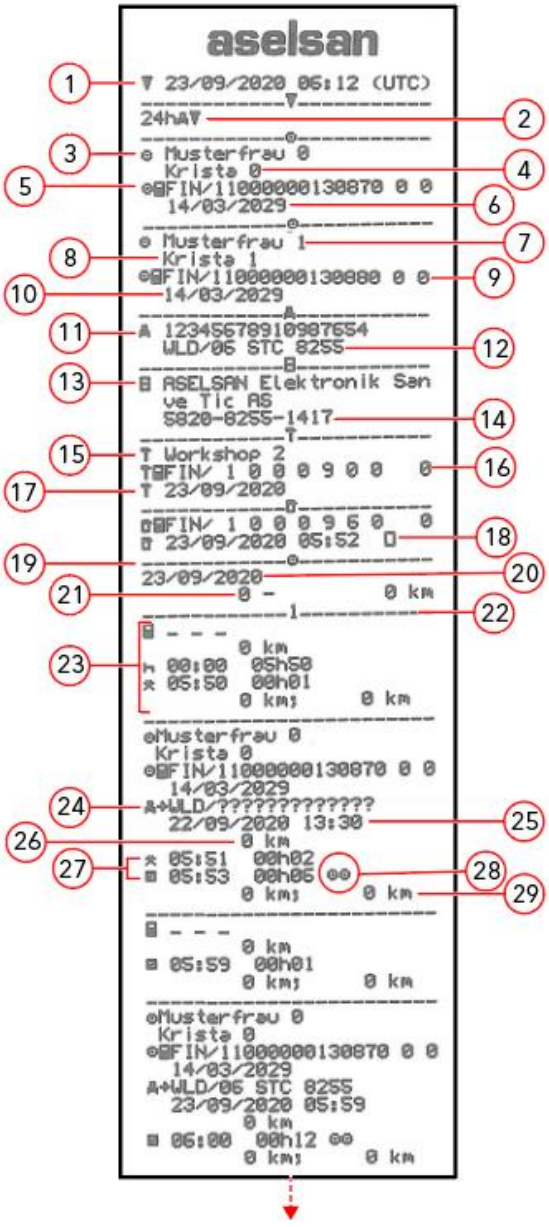
29 Last five events and faults stored in the vehicle unit (VU).

30 Control place.

31 Controller´s signature.

32 Driver's signature.

10.2 24h Vehicle Unit Printout



1 Date and time of printout (UTC).

2 Type of printout (24h, VU).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

7 Card holder's surname (co-driver).

8 Card holder's first name (co-driver).

9 Country and card identification number (co driver).

10 Expiry date of the co-driver card.

11 Vehicle identification, VIN.

12 Registering member state and Vehicle Registration Number (VRN).

13 Tachograph manufacturer.

14 Tachograph part number.

15 Workshop name where the last calibration is made.

16 Workshop card identification number.

17 Date of the last calibration.

18 Date and time, and type of the last control.

19 Drivers' activities stored in the VU per slot in chronological order.

20 Enquiry date.

21 Vehicle odometer at 00:00 and 23:59 (if the printout refers records of present day, right hand odometer value is for present time).

22 Driver slot indicator.

23 Driver's activities when there is no card in driver slot.

24 Registration member state and vehicle registration number of previous vehicle used.

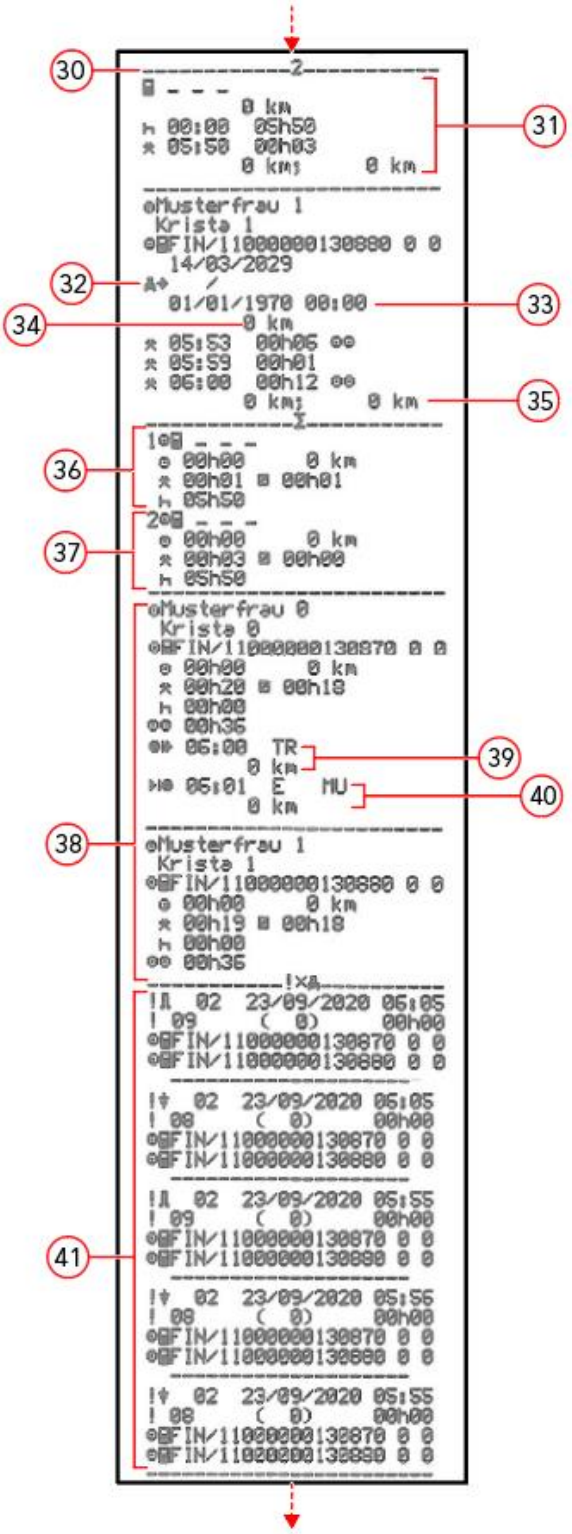
25 Date and time of card withdrawal from previous vehicle.

26 Vehicle odometer at card insertion.

27 Activities with start and duration time.

28 Activity status as crew.

29 Vehicle odometer at card withdrawal and trip distance during card is inserted in the slot.



30 Co-driver slot indicator.

31 Co-driver's activities when there is no card in co-driver slot.

32 Registration member state and vehicle registration number of previous vehicle used.

33 Date and time of card withdrawal from previous vehicle.

34 Vehicle odometer at card insertion.

35 Vehicle odometer at card withdrawal and trip distance during card is inserted in the slot.

36 Summary of periods without card in driver slot.

37 Summary of periods without card in co-driver slot.

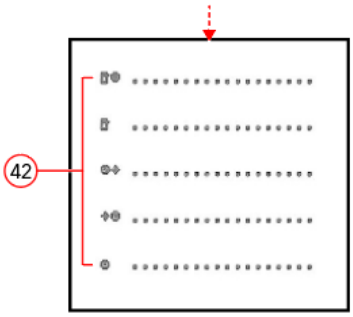
38 Daily summary of activities with card(s) is inserted.

39 Begin country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

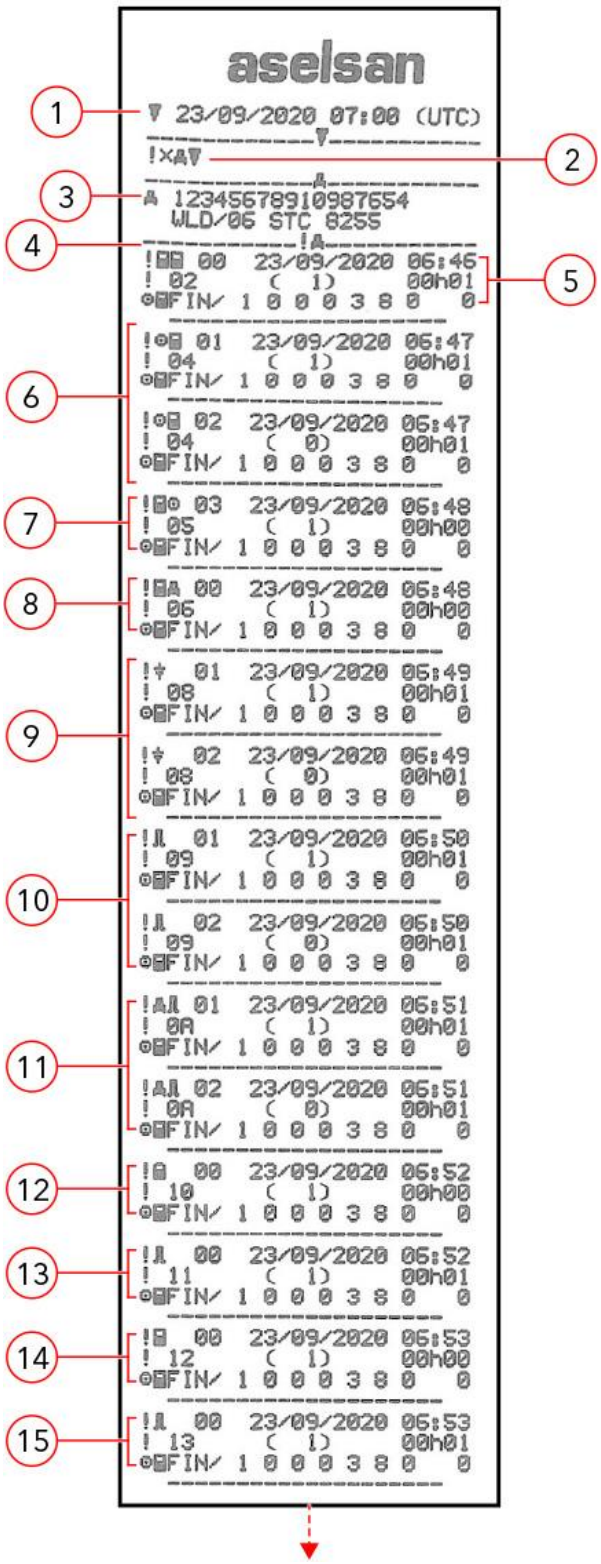
40 End country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

41 Last five events and faults stored in the vehicle unit (VU).

42 Hand-written and signature part.



10.3 Events & Faults Vehicle Unit Printout



1 Date and time of printout (UTC).

2 Type of printout (Events, VU).

3 Vehicle Identification Number (VIN)
Vehicle Registration Number (VRN) and
Registering Member State.

4 Block identifier of events.

5 Card conflict event.

6 Not appropriate card event.

7 Card inserted while driving event.

8 Card session close event.

9 Power supply interruption event.

10 Motion data error event.

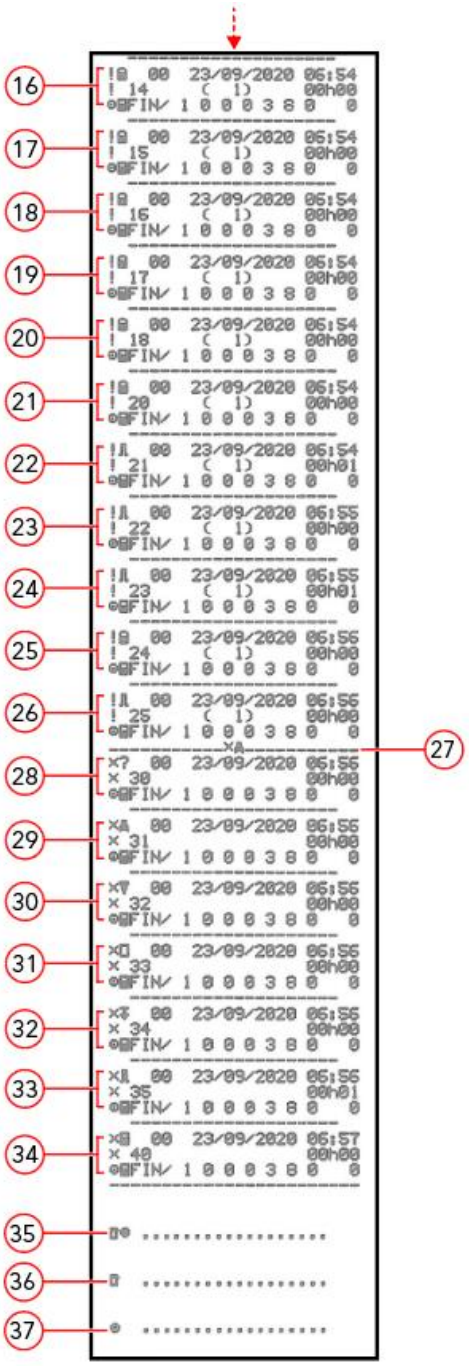
11 Vehicle motion conflict event.

12 No further details event.

13 VU MS auth. failure event.

14 Card auth. failure event.

15 Unauth. MS change event.



16 Card integrity error event.

17 VU integrity error event.

18 VU internal transfer error event.

19 VU unauth case opening event.

20 VU hardware sabotage event.

21 MS general event.

22 MS auth failure event.

23 MS integrity error event.

24 MS internal transfer error event.

25 MS unauth case opening event.

26 MS hardware sabotage event.

27 Block identifier of faults.

28 VU general fault.

29 VU internal fault.

30 Printer fault.

31 Display fault.

32 Downloading fault.

33 Sensor fault.

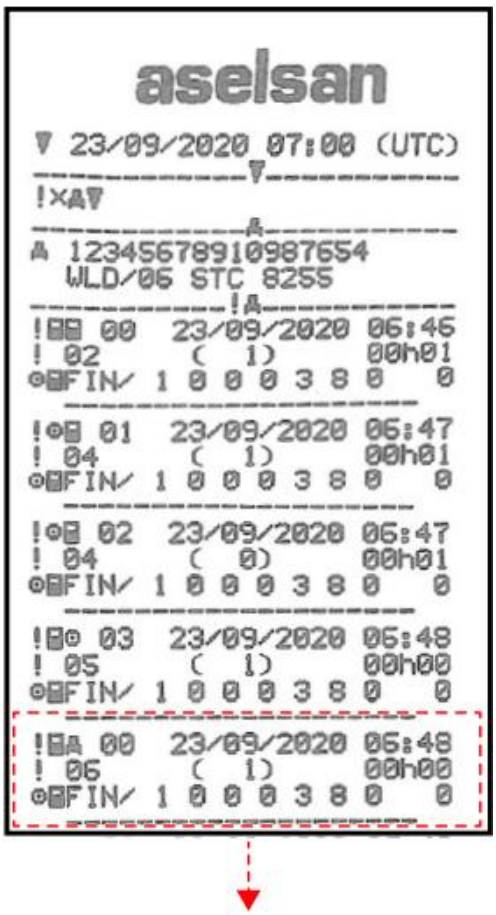
34 Card general fault.

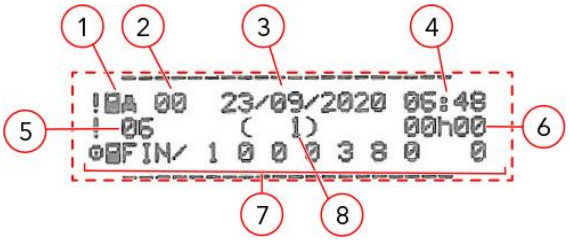
35 Control place.

36 Controller's signature.

37 Driver's signature.

10.4 Detailed View of Events & Faults Vehicle Unit Printout





1 Type of event or fault.

2 Event or fault record purpose:

00 - one of the 10 most recent (or last) events or faults.

01 - the longest event for one of the last 10 days of occurrence.

02 - one of the 5 longest events over the last 365 days.

03 - the last event for one of the last 10 days of occurrence.

04 - the most serious event for one of the last 10 days of occurrence.

05 - one of the 5 most serious events over the last 365 days.

06 - the first event or fault having occurred after the last calibration.

07 - an active/on-going event or fault.

3 Date of event or fault.

4 Start time of event or fault.

5 Event or fault code.

6 Duration of event or fault.

7 Card inserted in slot 1 at the beginning of event or fault (Driver card).

8 Number of events of the same type during the day.

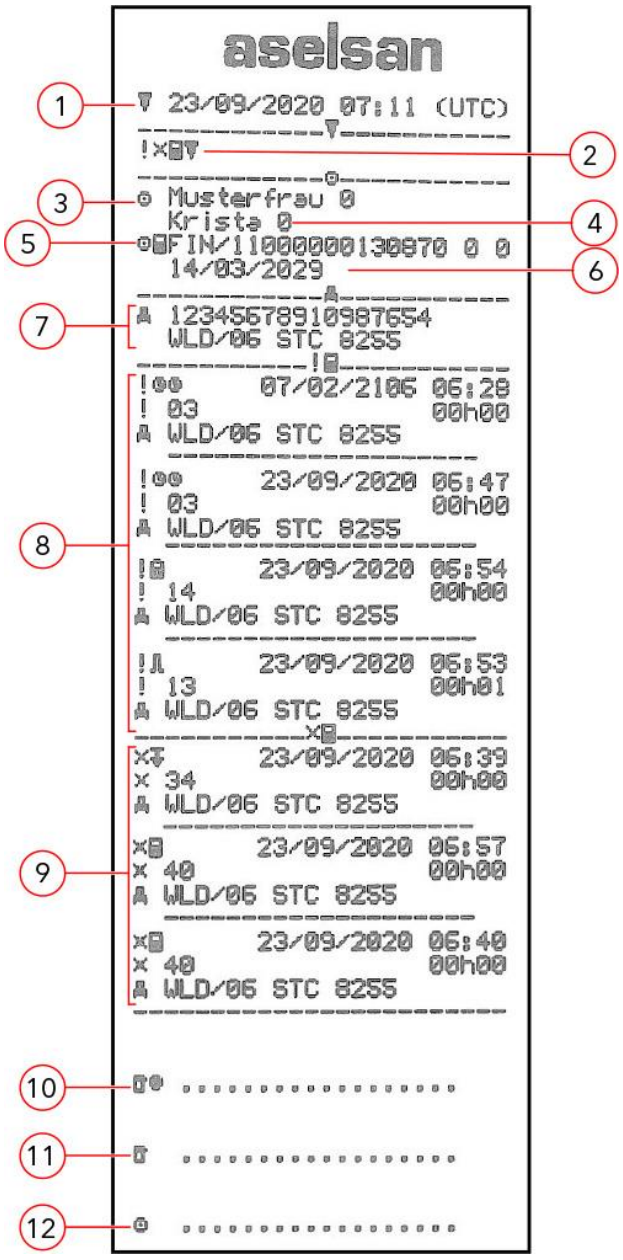
0 – For this event, it is not necessary to save "Number of similar events"

1 – One event of this type occurred on this day.

2 – Two event of this type occurred on this day, but only the one was saved.

n – "n" events of this type occurred on this day, but only the one was saved.

10.5 Events & Faults Card Printout



1 Date and time of printout (UTC).

2 Type of printout (Events, card).

3 Card holder's surname.

4 Card holder's first name.

5 Country and card identification number.

6 Expiry date of the card.

7 Vehicle identification. VIN, registering member state and VRN.

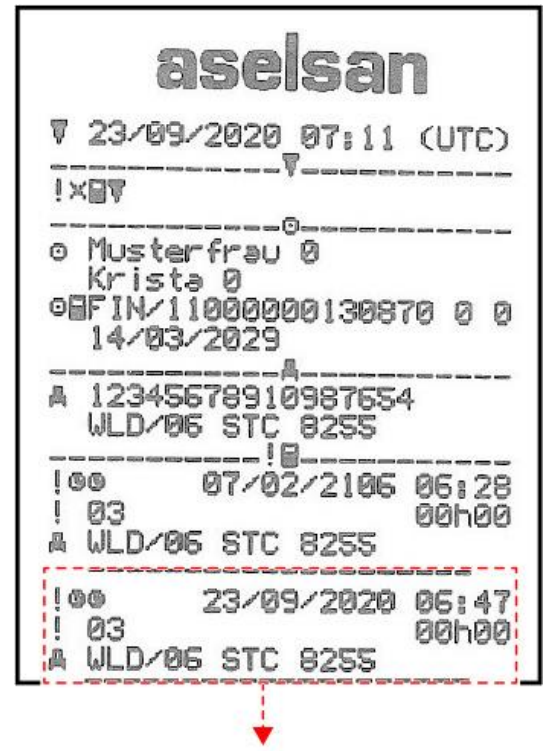
8 List of all events stored on the card.

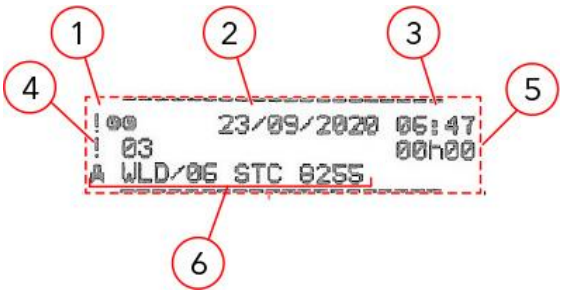
9 List of all faults stored on the card.

10 Control place.

11 Controller´s signature.

12 Driver´s signature.





1 Type of event or fault.

2 Date of event or fault.

3 Start time of event or fault.

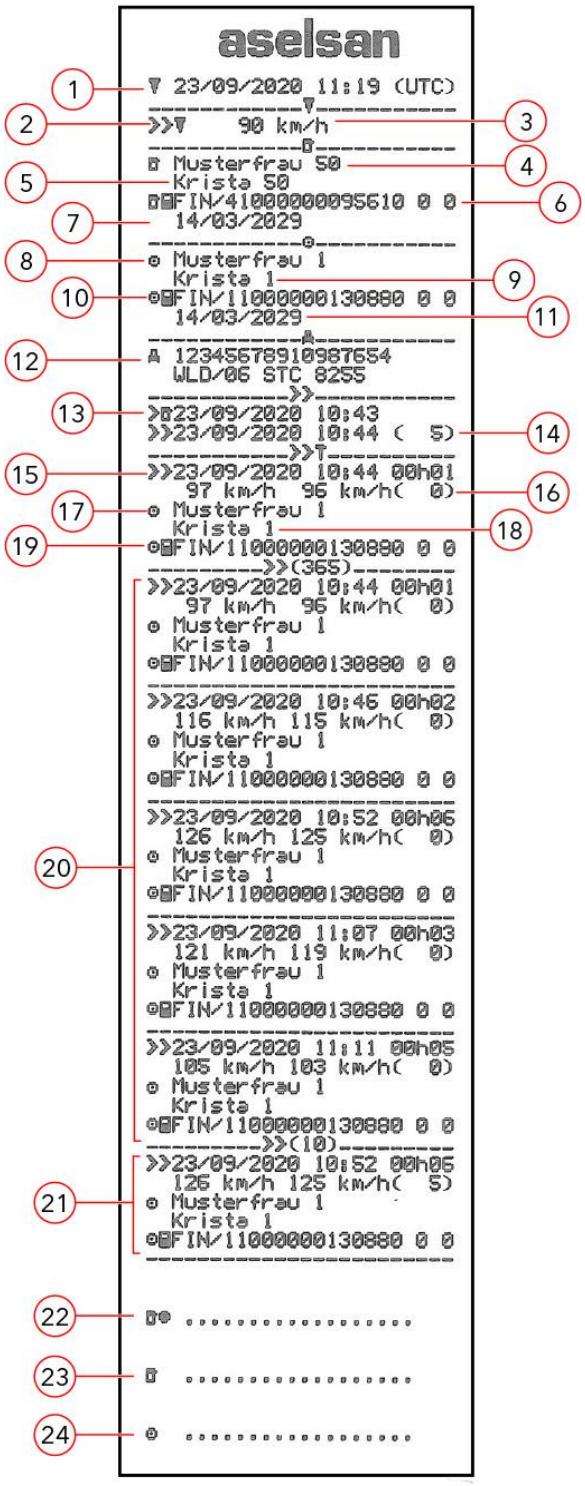
4 Event or fault code.

5 Duration of event or fault.

6 Inserted Vehicle Registration Number.

(VRN) and registering Member State (RMS).

10.6 Overspeeding Printout



1 Date and time of printout (UTC).

2 Type of printout (overspeeding).

3 Speed limiting setting.

4 Card holder's surname (driver).

5 Card holder's first name (driver).

6 Country and card identification number.

7 Expiry date of the driver card.

8 Card holder's surname (co-driver).

9 Card holder's first name (co-driver).

10 Card and country identification number.

11 Expiry date of the co-driver card.

12 Vehicle Identification Number (VIN)
Vehicle Registration Number (VRN) and
Registering Member State (RMS).

13 Date and time of first overspeeding control.

14 Date and time of the last overspeeding control
and number of overspeeding events.

15 First overspeeding after the last calibration, date
time and duration.

16 Max and average speed.

17 Driver card holder's surname.

18 Driver card holder's name.

19 Card and country identification number.

20 Five most serious overspeeding over the last
365 days: Date time and duration, max and
average speed, driver and drivers card identification.

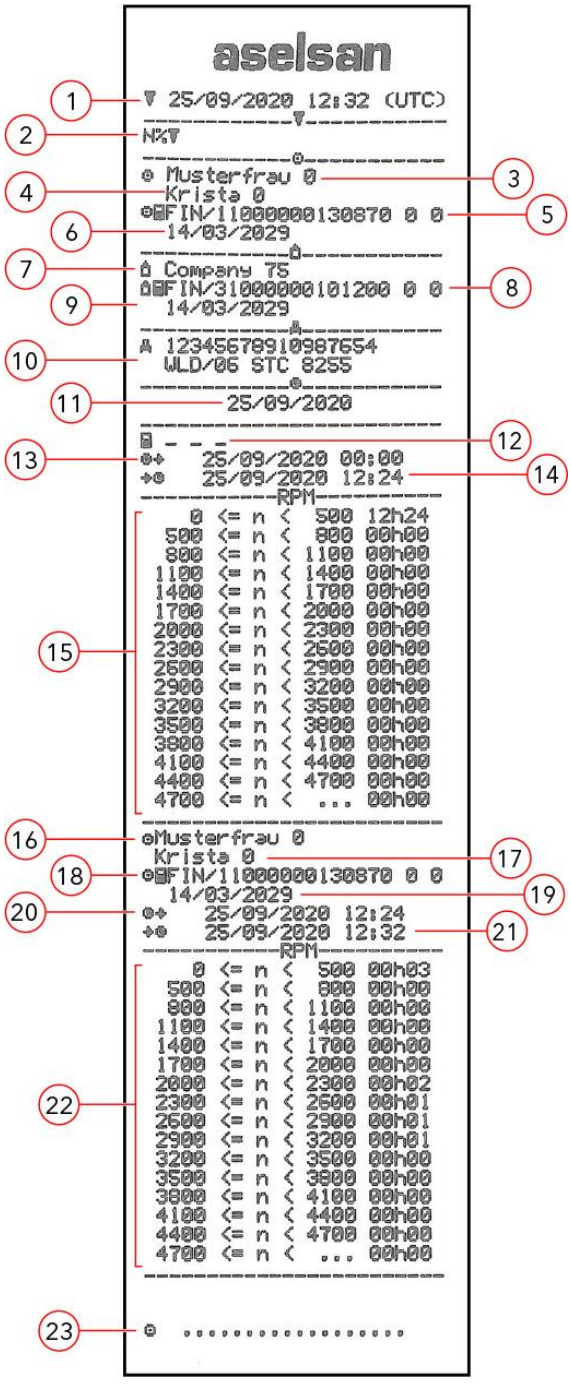
21 Five most serious overspeeding over the last ten
days: Date time and duration, max and average speed
driver and drivers card identification.

22 Control place.

23 Controller´s signature.

24 Driver´s signature.

10.7 N-Profile Printout



1 Date and time of printout (UTC).

2 Type of printout (N-profile).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

7 Card holder's surname (co-driver).

8 Card holder's first name (co-driver).

9 Expiry date of the co-driver card.

10 Vehicle Identification Number (VIN) Vehicle Registration Number (VRN) and Registering Member State (RMS).

11 Enquiry date.

12 First card slot info during profile recording (there is no card during recording, this area is shown as dashed line).

13 Begin date and time of the recording of rpm profile.

14 End date and time of the recording of rpm profile.

15 List of defined engine revolution levels and durations.

16 Card holder's surname (driver).

17 Card holder's first name (driver).

18 Card and country identification number.

19 Expiry date of the driver card.

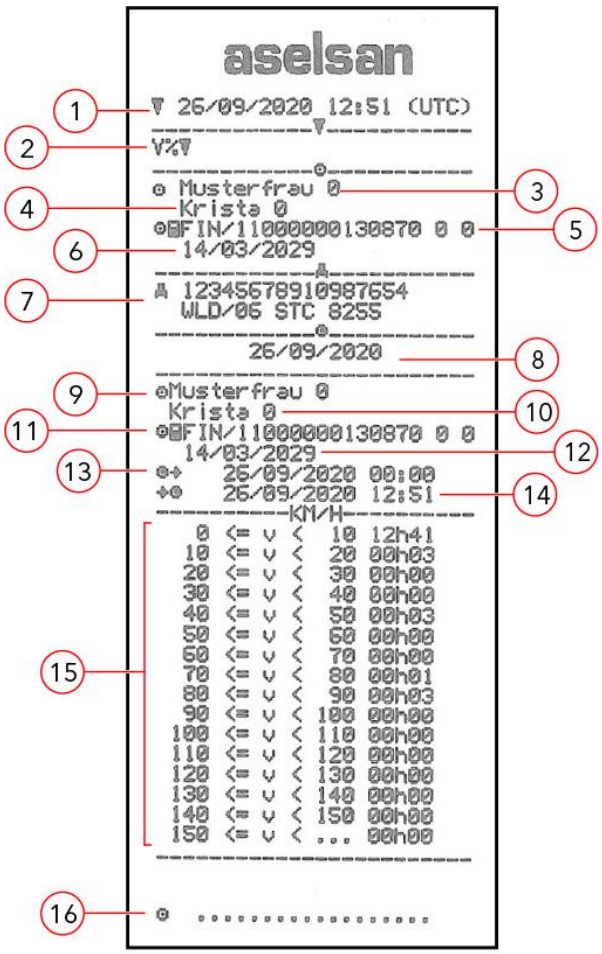
20 Begin date and time of the recording of rpm profile.

21 End date and time of the recording of rpm profile.

22 List of defined engine revolution levels and durations.

23 Driver´s signature.

10.8 V-Profile Printout



1 Date and time of printout (UTC).

2 Type of printout (V-profile).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

7 Vehicle Identification Number (VIN),
Vehicle Registration Number (VRN)
and Registering Member State (RMS).

8 Enquiry date.

9 Card holder's surname (driver).

10 Card holder's first name (driver).

11 Card and country identification number.

12 Expiry date of the driver card.

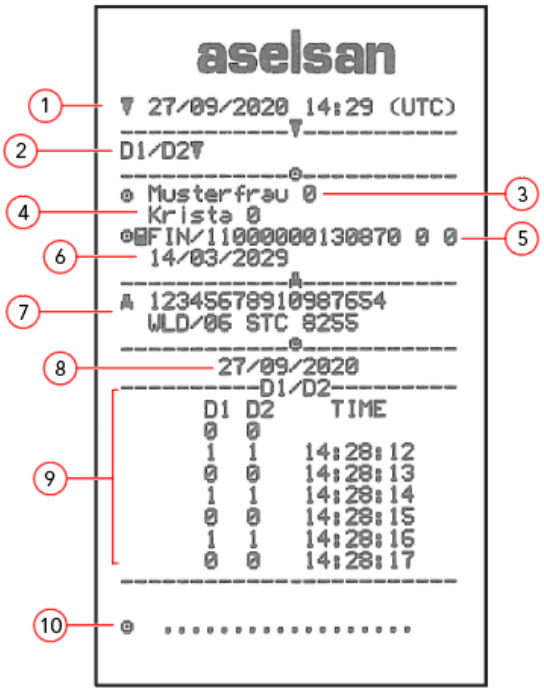
13 Begin date and time of the recording of speed profile.

14 End date and time of the recording of speed profile.

15 List of defined speed ranges and duration.

16 Driver´s signature.

10.9 Status D1-D2 Printout



1 Date and time of printout (UTC).

2 Type of printout (Status D1-D2).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

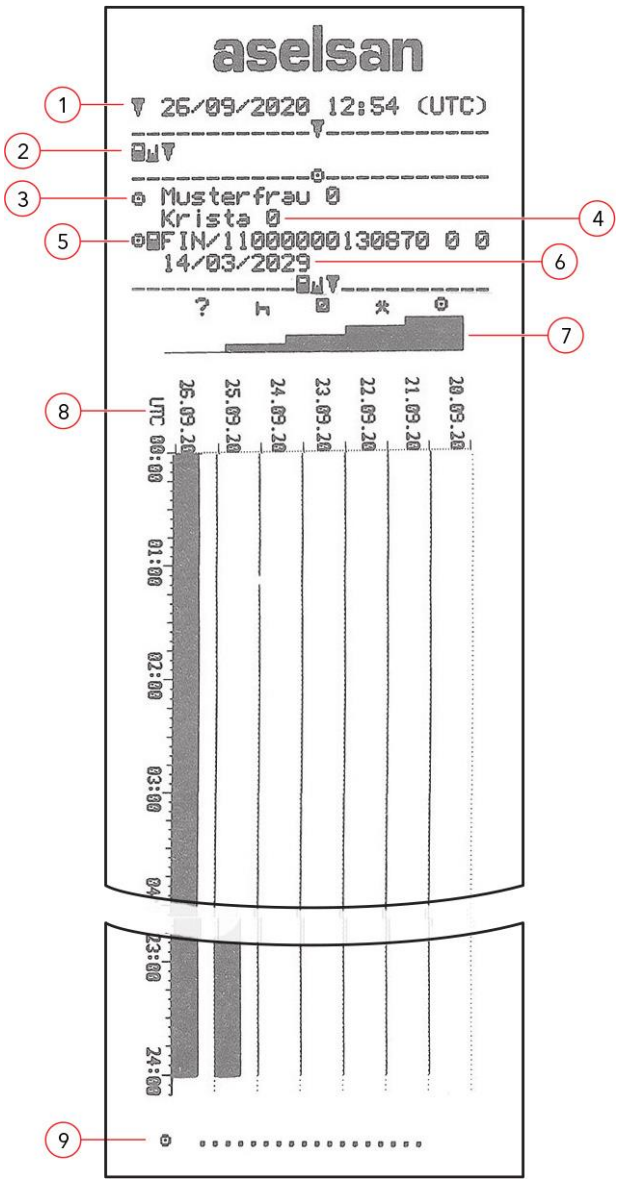
7 Vehicle Identification Number (VIN),
Vehicle Registration Number (VRN)
and Registering Member State (RMS).

8 Enquiry date.

9 List of status D1-D2.

10 Driver´s signature.

10.10 Activity Graph Printout



1 Date and time of printout (UTC).

2 Type of printout (Activities).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

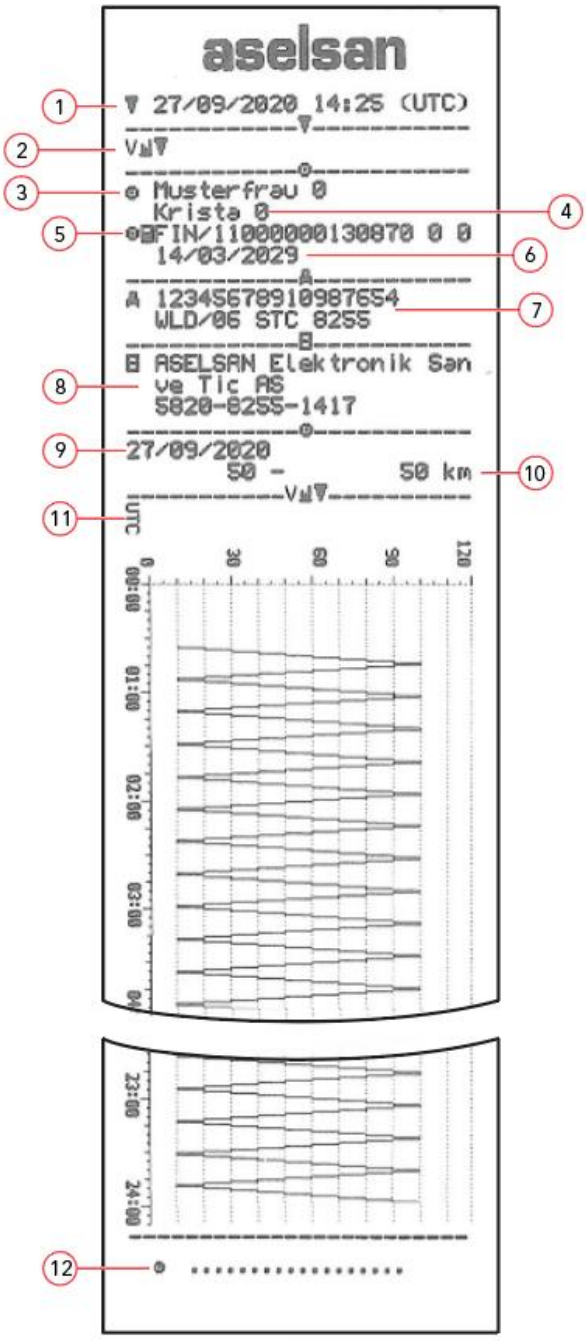
6 Expiry date of the driver card.

7 Legend of the activity symbols.

8 Activities diagram of the last seven calendar days from the selected day.

9 Driver´s signature.

10.11 V-Graph Printout



1 Date and time of printout (UTC).

2 Type of printout (V-graph).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

7 Vehicle Identification Number (VIN),
Vehicle Registration Number (VRN)
and Registering Member State.

8 Tachograph manufacturer's name and
tachograph part number.

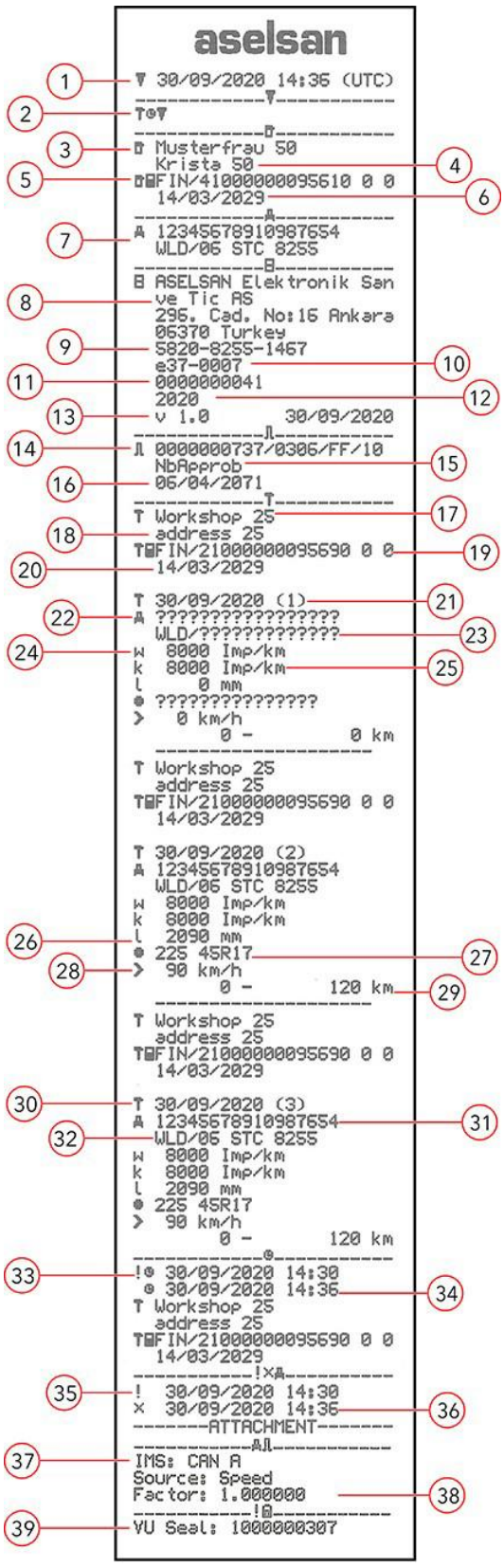
9 Selected day.

10 beginning (00:00) and end (23:59) of the day
(if the printout refers records of present day,
right-hand odometer value is for present time).

11 Recording of speed profile of the selected day.

12 Driver´s signature.

10.12 Technical Data Printout



1 Date and time of printout (UTC).

2 Type of printout (technical data).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

7 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State (RMS).

8 Tachograph manufacturer.

9 Tachograph part number.

10 Tachograph approval number.

11 Tachograph serial number.

12 Year of manufacture.

13 Software version and installation date.

14 Motion sensor serial number.

15 Motion sensor approval number.

16 Motion sensor first installation date.

17 Workshop name having performed the calibration.

18 Workshop address.

19 Workshop card identification.

20 Workshop card expiry date.

21 Calibration date and purpose.

22 VIN (shown as question marks before registered).

23 VRN (shown as question marks before registered).

24 Characteristic coefficient of vehicle.

25 Constant of the recording equipment.

26 Effective circumference of wheel tyres.

27 Size of tyres.

28 Authorized speed setting.

29 Old and new odometer values.

30 Calibration date and purpose.

31 VIN.

32 VRN and country of registration.

33 Old date and time.

34 New date and time.

35 Most recent event date time.

36 Most recent fault date time.

37 IMS source.

38 IMS factor.

39 Seal number.

➔ *Please refer to "Section 6.2.6* Checking Calibration Data *" on page 37.*

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Warranty Information

The warranty period is two years and years in repair period.

- System / Device / Material / Software,
- Packaging / Shipping / Delivery,
- Marketing / Sales,
- Acceptance Tests,
- Assembly / Facility / Testing,
- Operation / Maintenance / Repair,
- Operation Manual / Document,
- Training,
- Payment / Invoice

You can reach for your problems / requests on the above listed items to the following contact information.

About Producer

Title	:	ASELSAN Inc.
Cage Code	:	T4238
Address	:	Mehmet Akif Ersoy Mahallesi, İstiklal Marşı Caddesi No: 16 Yenimahalle 06200 ANKARA / TURKEY
Telephone	:	+90 312 5921000
Fax	:	+90 312 3541302
Call Center	:	+90 850 3601030
Web	:	www.aselsan.com.tr